Behavioral Research Through Interpretable, Dimensionality-reduced Generative AI Embeddings (BRIDGE): A Method to Incorporate Real-World Stimuli in Consumer Experiments

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Statements and Declarations

Ethical considerations

All studies involving human participants were reviewed by the Singapore Management University Institutional Review Board. The coffee certification experiments (Experiments 1 and 2) were conducted under IRB-25-013-E002-M1(1025); the wine choice study and the validation study (Web Appendix §G) were conducted under IRB-19-100-E036(1019).

Consent to participate

Informed consent to participate was obtained electronically from all participants.

Consent for publication

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Declaration of conflicting interest

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Data availability statement

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Abstract

Traditional experiments often rely on a few, stylized stimuli, which can limit realism and undermine generalizability beyond the sampled stimuli—known as the stimulus-sampling problem. To address this challenge, this paper introduces BRIDGE, a novel analytical method that enables the use of many unaltered real-world descriptions as experimental stimuli. Leveraging foundational generative AI embeddings, BRIDGE develops (1) structured, low-dimensional, and interpretable representations of focal constructs and (2) statistical controls for non-focal nuisance variations, facilitating causal inference. Extensive Monte Carlo simulations, two coffee certification experiments, and a large-scale choice experiment (plus its validation study) show that BRIDGE recovers true parameters even when textual stimuli contain unobserved nuisance variations, and can effectively account for different sources of confounding. In the choice experiment, 1,000 participants evaluated approximately 50,000 unique product descriptions randomly sampled from a corpus of nearly 120,000. Results show that entirely incidental initial products can shape participants' subsequent preferences. By incorporating many unaltered product texts into experiments, BRIDGE enhances the realism, generalizability, and practical relevance of consumer research in information-rich environments. A detailed researcher's guide and Python package `bridge` are provided.

Keywords: Research Design, Interpretable Artificial Intelligence, Generative AI, Consumer Behavior, Marketing.

Code, data, and documentation are available at https://doi.org/10.17605/OSF.IO/5D6KX.

Real-world product descriptions can be complex, elaborate, and diverse. For instance, when choosing a vacation destination, a description of Bali (Indonesia) might highlight its stunning beaches, ancient temples, and a diverse range of activities like surfing and yoga retreats. Nairobi (Kenya) may be portrayed through its lively city-meets-safari appeal, with bustling Maasai craft markets and the opportunity to spot lions, giraffes, and other wildlife in the nearby national park. Consumers may be presented with many such product descriptions in free-form, unstructured texts. That richness helps consumers anticipate experiences and evaluate products but poses a challenge for researchers: How much of this real-world complexity should experimental stimuli retain?

Using stimuli with rich detail can help activate the same psychological processes that affect real-world consumer choices, enhancing both experimental and mundane realism (Camerer 1997; Morales, Amir, and Lee 2017; Wilson, Aronson, and Carlsmith 2010).[1] However, the inherent variability of real-world descriptions can make it difficult to ensure comparability across stimuli. A single category may have hundreds or thousands of product descriptions, which may differ in countless ways, from what is said ("content," e.g., attributes, claims, information) to how it is said ("form," e.g., words used, style, length). Yet, causal inference requires isolating the effect of a focal element embedded in the unstructured text (e.g., surfing, cultural heritage, wildlife) while holding constant the many other features that covary with it.

Some researchers condense, abbreviate, and stylize real-world product descriptions. For example, previous studies presented vacation experiences to participants using stylized displays (e.g., "A = (average décor, $120 per night)"; Frederick, Lee, and Baskin (2014), Studies 1a–1s) or abbreviated descriptions (e.g., holiday destinations described by name only; Sharot, Velasquez, and Dolan (2010), Study 1). While this approach may exclude content integral to the original allure of the products, it allows researchers to isolate the causal influence of a focal aspect of product descriptions on consumer response by affording greater control over nuisance variables (Calder,

[1]Experimental realism is the degree to which the experiment engages participants psychologically; mundane realism is the extent to which a study mirrors real-life situations, tasks, and environments (Wilson, Aronson, and Carlsmith 2010).

Phillips, and Tybout 1981; Camerer 1997; Wilson, Aronson, and Carlsmith 2010). However, this control comes at a cost: with only a handful of select stimuli per condition, the observed effects may be due to idiosyncrasies of the specific stimuli rather than the intended construct (Clark 1973; Pham 2013; Wells and Windschitl 1999). This is the stimulus-sampling problem, and variation in unstructured texts tends to amplify it: the potential stimulus space of unstructured texts is far larger than that of structured stimuli, making it difficult to sample representatively.

The stimulus-sampling problem undermines a study's internal validity (Simonsohn, Montealegre, and Evangelidis 2025), as the sample may be biased by uncontrolled confounds, inflating Type I error (Judd, Westfall, and Kenny 2012; Wickens and Keppel 1983). It threatens construct validity[2] (Wells and Windschitl 1999), as specific features of a sampled stimulus may introduce alternative construct interpretations, and external validity, as findings may not generalize beyond the stimuli used (e.g., Baribault et al. 2018; Judd, Westfall, and Kenny 2012). Scholars have speculated that past failures to replicate experimental results may stem from the stimulus-sampling problem (e.g., Westfall, Judd, and Kenny 2015).

In this paper, we propose BRIDGE (Behavioral Research through Interpretable, Dimensionality-reduced Generative AI Embeddings), a novel data analysis methodology that enables a fundamentally different approach to stimulus sampling. Under this approach, participants in a single experiment can be exposed to distinct stimuli randomly selected from a corpus of real-world product descriptions, resembling a series of micro-experiments.

BRIDGE facilitates the analysis of participants' responses to diverse, unstructured stimuli while maintaining control across varying stimulus characteristics. It (1) uses a large language model (LLM) to generate high-dimensional, numerical embeddings of the unstructured, non-numerical stimuli and (2) transforms these embeddings into lower-dimensional, interpretable "knowledge representations" of the product attributes described in the stimuli, serving as numerical representations that are amenable to computational reasoning (Carvalho, Pereira, and Cardoso

[2]As Wells and Windschitl (1999) point out, "the failure to sample stimuli can threaten construct validity…when 'the operations which are meant to represent a cause or effect can be construed in terms of more than one construct' (Cook and Campbell 1979, p. 59)" (p. 1116).

2019; Levesque 1986; Tenenbaum et al. 2011). The lower dimensionality and interpretability of these knowledge representations make them practical for use in theory-driven hypothesis testing with key, defined product attributes, as they can be incorporated in statistical models of participant behavior. BRIDGE also enables the development of statistical controls for nuisance variables to ensure stimulus comparability, extending conventional ANOVA/ANCOVA-style testing to settings with many varied real-world stimuli, while encompassing conventional experimental designs as special cases. In doing so, it facilitates broader stimulus sampling, increases statistical power through multiple distinct product presentations per participant, and improves generalizability through greater coverage of product offerings in the market.

We organize our paper as follows. We develop BRIDGE, describing the data structures it addresses. We present a synthetic-data study identifying key boundary conditions. We demonstrate our methodology using multiple experiments. The first two experiments use controlled stimuli and conventional designs to provide evidence that the inference obtained from BRIDGE (without explicit confound specification) is comparable to an "unconfounded" benchmark that uses direct knowledge of the confound structure to yield consistent and efficient estimates. In the third experiment, we illustrate the use of BRIDGE in addressing a novel consumer research question that would be challenging to investigate using conventional experimental designs. This experiment involves 1,000 participants who are each presented with the real-world descriptions of 32 pairs of wines randomly sampled from nearly 120,000 available on the market, each wine description averaging 40 words (SD = 11.2) and the experiment encompassing almost 50,000 unique wines. The inclusion of such rich and diverse stimuli would be impracticable using traditional experimental designs with the same number of participants. We present validation studies comprising both perturbation analyses and a follow-up experiment to further support our findings. We conclude with a discussion of the potential applications of BRIDGE methodology, its limitations, and possible directions for future research.

BRIDGE: Method Development

The traditional experimental approach—using a few, simplified, and stylized stimuli—is subject to the stimulus-sampling problem. First identified as the "language-as-fixed-effect fallacy" (Clark 1973) and later framed as the stimulus-sampling problem (Wells and Windschitl 1999), this issue arises when idiosyncratic features of the selected stimuli are confounded with the intended construct. It threatens a study's *internal validity* when the selected stimuli produce effects for reasons other than the hypothesized one, such as through uncontrolled confounds (Simonsohn, Montealegre, and Evangelidis 2025). It undermines *construct validity* when the specific features of the stimuli offer alternative construct explanations for an effect, making it unclear whether the operationalized variable accurately reflects the intended theoretical construct (Cook and Campbell 1979; Wells and Windschitl 1999). It challenges *external validity*, as the findings may not generalize to the larger population of stimuli in the category (Baribault et al. 2018; Pham 2013; Westfall, Judd, and Kenny 2015). In addition to these validity concerns, treating stimuli as fixed rather than random when they are sampled from a broader population inflates Type I error rates (Judd, Westfall, and Kenny 2012). These issues are amplified by the file drawer problem (Rosenthal 1979): if stimuli that yield significant results are more likely to be reported, they are disproportionately likely to be adopted in future studies. Selective reuse truncates the distribution from which stimuli are drawn, further inflating the probability of false conclusions and skewed inference. Together, these mechanisms may explain why scholars have attributed past failures to replicate experimental results, in part, to the stimulus-sampling problem (e.g., Westfall, Judd, and Kenny 2015).

Recent work proposes managing the stimulus-sampling problem *ex ante* at the design stage. The "Mix-and-Match" framework (Simonsohn, Montealegre, and Evangelidis 2025) provides a systematic procedure for stratified sampling of diverse stimuli within each condition and matching them across conditions to manage confounds. Its accompanying "Stimulus Plots" enable exploratory assessment of stimulus variation by comparing observed heterogeneity in results to that expected under homogeneity.

Several features of the questions and contexts that BRIDGE targets, however, present challenges for such design-based approaches. When stimuli are unstructured real-world descriptions, the nuisance dimensions (non-focal variation) along which they vary can be numerous, latent, and difficult to identify *a priori*, let alone measure and balance across conditions. In addition, when stimuli vary simultaneously on multiple dimensions (e.g., attributes), the number of potential strata grows multiplicatively with each added dimension. While Simonsohn, Montealegre, and Evangelidis (2025) recommend a small number of categories (e.g., five) rather than exhaustive crossing, selecting which dimensions to stratify on requires knowing which dimensions matter, which may be difficult with unstructured stimuli. Stimulus Plots provide a valuable diagnostic for *detecting* stimulus variation, but their diagnostic value also depends on having sufficient observations per stimulus, a requirement that becomes harder to meet as the number of stimuli in a study grows.

BRIDGE complements design-based approaches by addressing these challenges *ex post* at the analysis stage, in contexts where researchers aim to test hypotheses using many unstructured textual stimuli (e.g., real-world wine tasting notes, financial product summaries, and sustainability initiative narratives). BRIDGE offers a unified model that pools information across stimuli while accounting for non-focal variation. Rather than requiring the careful curation of select stimuli before an experiment, it enables researchers to sample at scale from large, real-world corpora. It does so by using an interpretable AI model to decompose each stimulus into (1) structured, low-dimensional representations of the focal attributes and (2) statistical controls for latent nuisance variables. In short, where Mix-and-Match manages confounds by carefully curating *what* is shown, BRIDGE algorithmically controls for variation in *what was* shown.

To understand BRIDGE, consider the standard analytical framework that it builds upon. In a traditional experiment, responses are analyzed using a linear model:

$$y_i = \alpha + \beta D_c(i) + \gamma \text{covariates}(i) + \delta \text{attributes}(i) + \epsilon_i, \tag{1}$$

where y_i represents the response of participant i; α is the intercept; β is the hypothesized effect; γ are coefficients for covariates describing systematic factors; and δ are coefficients for stimulus attributes. $D_c(i)$ is a dummy variable indicating whether participant i is in condition c, covariates(i) are participant covariates (e.g., participant characteristics), and attributes(i) are attributes of interest (e.g., the emotional tone in which a description is written); the latter two pre-defined, based on theory. For simplicity, we omit a task index t, though this framework can accommodate studies with multiple tasks, task-varying effects, and task-specific stimuli (i.e., task-varying β and γ), as well as interaction terms between the condition dummies and the covariates of interest.

In many experiments covariates$(i) = 0$ and attributes$(i) = 0$, and hypothesis testing on β is equivalent to ANOVA. If covariates$(i) \neq 0$, hypothesis testing on β is equivalent to ANCOVA, which assesses if the group means differ while accounting for the covariates. If attributes$(i) \neq 0$, it is typical to employ a regression equation and use Wald tests to examine if β, γ, or $\delta = 0$. This reduces to a t-test on β when considering only the treatment effect.

The stimulus-sampling problem arises directly from a key assumption in this approach: any systematic difference in responses between participants in different conditions must be attributable solely to the experimental manipulation $D_c(i)$. When participants are presented with diverse, unstructured real-world stimuli, the stimuli inevitably vary along numerous additional dimensions beyond the included variables (the manipulation, covariates, and attributes). These extraneous variations are *nuisance variables*: undesired, non-focal sources of variation that can affect the outcome (Kirk 2013). A nuisance variable that covaries with the manipulation (treatment) is a *confound*—"an extraneous variable that covaries with the variable of interest" (Shadish, Cook, and Campbell 2002, p. 506). Confounds violate the assumption directly: unless they are represented and adjusted for in the model, they supply alternative explanations for the systematic difference in responses between conditions (i.e., for the observed effect); statistically, this manifests as biased estimates. The component of the nuisance variation that is orthogonal to the included variables does not bias the estimates but inflates the error variance, reducing precision and therefore statistical power (Kirk 2013). In sum, unmodeled nuisance variation harms inference in two ways:

its confounding component biases estimates, while the orthogonal remainder erodes precision.

Consequently, experimenters often choose a few, simplified, and/or stylized stimuli. This approach improves comparability across conditions by minimizing nuisance variation, with the remaining idiosyncratic (non-systematic) variation absorbed by the error term, ϵ_i. However, in prioritizing experimental control to protect internal validity, this approach often simplifies stimuli and variety, diminishing construct and external validity. Paradoxically, it also undermines the internal validity that it seeks to protect. Failing to account for the random variability introduced by the stimulus sample leads to underestimated error terms and, consequently, inflated "false positive" rates—increasing the likelihood of concluding an effect exists when it does not (Judd, Westfall, and Kenny 2012; Westfall, Judd, and Kenny 2015; Wickens and Keppel 1983).

One strategy to address these issues is to introduce many real-world descriptions as stimuli and manually code the nuisance variables, translating them into control variables. However, this strategy faces several difficulties. First, the relevant nuisance variables may not be known *a priori*—the many nuanced dimensions along which real-world descriptions differ (e.g., tone, style, voice, formality) are often hard to identify and isolate (Clark 1973). For instance, one product description might use vivid and emotive language to create an immersive experience, while another might rely on technical jargon or minimalist phrasing to convey sophistication. Such characteristics may be irrelevant to the research question yet still influence participant responses. Second, even if identifiable, nuisance dimensions can be numerous, and including a control for each can quickly exhaust the information in the data. Consider, for example, the number of possible brands in the market for a typical consumer product such as running shoes; controlling for brand may require many fixed effects in a conventional model. Third, the sheer scale of coding may be prohibitive. Our wine study, for example, uses approximately 50,000 distinct stimuli selected from a set of nearly 120,000 descriptions—a volume that would be difficult to code manually.

BRIDGE employs an alternative strategy: it uses AI to systematically extract knowledge representations for key attributes and statistical controls for the nuisance variables and incorporate them into the analysis. This enables the specification of models of the form:

$$y_i = \alpha + \beta D_c(i) + \gamma \text{covariates}(i) + \delta \text{attributes}(i) + \zeta \text{controls}(i) + \epsilon_i, \tag{2}$$

where ζ represents coefficients for the statistical controls of nuisance variables, and controls(i) are the controls, which are algorithmically extracted from the stimuli. controls(i) are distinct from attributes(i) in that, whereas attributes(i) capture observed focal differences of interest that are often manipulated or controlled in traditional experiments, controls(i) represent variables that are algorithmically extracted to absorb the latent, unobserved nuisance variation — adjusting for its confounding component while also absorbing idiosyncratic variation that would otherwise inflate error variance.

To illustrate, consider an experiment where the stimuli vary in color. If color were known, a researcher could control for it by including it in the model, thereby ruling it out as a confounding factor when estimating β. Now, suppose color were unknown *a priori*. A strategy might be to use an algorithm to automatically code color, thereby enabling a similar analytical approach. This is analogous to BRIDGE but differs in scale and complexity: rather than a single observable attribute like color, BRIDGE is designed to identify and control for multiple, latent nuisance dimensions—tone, style, formality, and more—simultaneously.

If stimuli of many colors are introduced, adding a coefficient for each color leads to the inclusion of many coefficients. A fixed-length numerical representation (for color, RGB—a three-dimensional representation) leads to a more efficient econometric model. As BRIDGE is designed for real-world descriptions that can vary in many ways, it similarly develops low-dimensional representations of the potentially high-dimensional nuisance variables in stimuli as statistical controls.

Section Roadmap. We outline the general step-by-step procedure for conducting a study using BRIDGE and discuss how to validate BRIDGE in a new application. To facilitate adoption, we provide a Web Appendix comprising a roadmap (Web Appendix §A), a detailed researcher's guide introducing a Python package that includes all analytical components (§B), perturbation analyses (§C), the Monte Carlo programming framework and complete results (§D), supplementary materials for the coffee certification experiments (§E), supplementary analyses for the wine choice experiment (§F), and a validation study with controlled stimuli (§G). Last, we discuss how BRIDGE's structured, algorithmic approach enhances research objectivity and reproducibility, facilitating research practices like preregistration.

BRIDGE: An Interpretable AI Model for Theory Testing

BRIDGE uses an LLM to encode each stimulus as a *semantic embedding*: a numerical representation that captures the meaning of the text. BRIDGE is developed to address two challenges in the use of these embeddings for hypothesis testing. First, semantic embeddings are inherently unstructured—they lack explicit organization. Consequently, they lack interpretability: individual dimensions of the encoding do not correspond to specific, understandable features, making it challenging to relate these dimensions to theoretical constructs of interest or to link them directly to participant responses. Second, they are high-dimensional, often consisting of thousands of dimensions, which can pose computational challenges and reduce statistical power—an issue that is particularly crucial when working with typical sample sizes in laboratory studies.

Given a set of theory-defined focal attributes, including those that express the manipulation, BRIDGE uses a fully connected feedforward network to compress a high-dimensional, unstructured semantic embedding into a lower-dimensional intermediate representation (Johnson and Lindenstrauss 1984).[3] As illustrated in Figure 1, the compression network consists of two stages: a projection layer and a reduction layer. The projection layer initially reduces the dimensionality of the embedding by exploiting the structure of the encoding; in the case of Matryoshka representations (Kusupati et al. 2022), this is simply a masking layer, as the embeddings have a nesting structure that permits truncation with minimal information loss; in other architectures, a learnable

[3]BRIDGE is embedding-agnostic: any semantic embedding can serve as input. However, the quality of the input embeddings affects the precision of the extracted representations. For instance, using BERT embeddings (768 dimensions) in place of OpenAI text-embedding-3-large (3,072 dimensions) in our wine study yields anchoring coefficients (reported below) in the expected direction but attenuated and nonsignificant ($\hat{\beta} = 0.011$–0.015 vs. 0.038–0.039). See Web Appendix §F.6 for details.

projection to reduce dimensionality may be useful. The reduction layer, a feedforward network whose width is discoverable through automatic tuning with tools such as Optuna (Akiba et al. 2019), further compresses the projected embedding to produce a low-dimensional intermediate representation. A partitioned feedforward network maps this intermediate representation to an interpretable layer that is partitioned into components, each of which is dedicated to predicting a distinct attribute during training.[4] The network learns to further compress the intermediate representation and extract attribute-relevant information (Bishop 1995; Hornik, Stinchcombe, and White 1989), embedding it in an attribute-specific vector space. The output from each component of the interpretable layer serves as the corresponding attribute's representation.

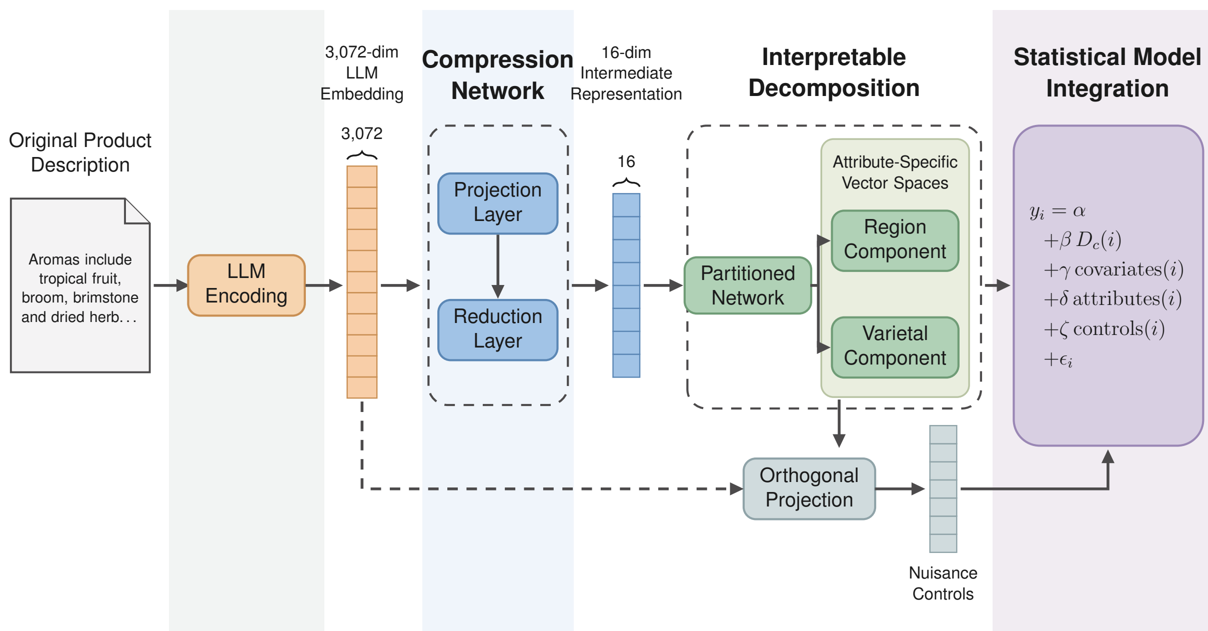


Figure 1: Schematic of BRIDGE Architecture

Note: An LLM maps an unstructured product description to a 3,072-dimensional embedding. A compression network—consisting of a projection layer and a reduction layer—compresses this embedding to produce a 16-dimensional intermediate representation. A partitioned network maps the intermediate representation to attribute-specific vector spaces (e.g., Region and Varietal components). The original LLM embedding is projected onto the orthogonal complement of the attribute-specific subspace to derive nuisance controls. Both the attribute representations and nuisance controls feed into the statistical model.

To ensure interpretability in the attribute-specific components, BRIDGE is trained using a

[4]This approach is different from (standard) feedforward networks where the complete output of a preceding layer is used to predict an attribute (Caruana 1997).

multi-term loss function:

$$\mathcal{L} = \underbrace{\sum_{k=1}^{K} \mathcal{L}_{\text{CE}}^{(k)}}_{\text{classification}} + \lambda \underbrace{\frac{1}{N} \sum_{i=1}^{N} \sum_{k=1}^{K} \log\left(1 + \sum_{j \in \mathcal{M}_i \setminus \{i\}} \exp\left(\frac{\cos(\mathbf{e}_{k,i},\ \mathbf{e}_{k,j})}{\tau}\right)\right)}_{\text{contrastive}} \tag{3}$$

where K is the number of focal attributes; $\mathcal{L}_{\text{CE}}^{(k)}$ is the cross-entropy classification loss for attribute k; $\mathbf{e}_{k,i}$ is the ℓ_2-normalized representation for attribute k of stimulus i; $\mathcal{M}_i$ denotes the mini-batch containing stimulus i; τ is a temperature parameter; and λ is the contrastive weight (see Web Appendix §F.2).

The classification term focuses on accurately predicting the attributes (e.g., region, varietal) associated with each output node. The contrastive term serves two complementary purposes: it encourages the representations of distinct levels within each attribute to be well-separated (within-attribute expansiveness) and, indirectly, encourages the attribute-specific vector spaces to remain distinct from one another (between-attribute distinctiveness). The loss function balances interpretability and performance, as the classification term benefits from the interpretable layer being only sufficiently large to express the key attribute-specific information in the product descriptions—a larger layer can lead to less precise training and worse validation loss—whereas the contrastive term benefits from larger dimensional spaces, as these facilitate orthogonality in the attribute- and attribute-level-specific representations.

Statistical controls for the nuisance variables are developed by projecting the original LLM representations onto the orthogonal complement of the subspace spanned by the attribute-specific representations. The orthogonal complement of any subspace W is the subspace of vectors orthogonal to every vector in W. This ensures that the nuisance controls are, by construction, orthogonal to the attribute representations.

Step-by-Step Procedure for Implementing the Research Design

Conducting a study using BRIDGE involves five key steps. As depicted in Figure 2 (for a detailed discussion of each step, see Web Appendix §B), *Step 1* is collecting product descriptions. These

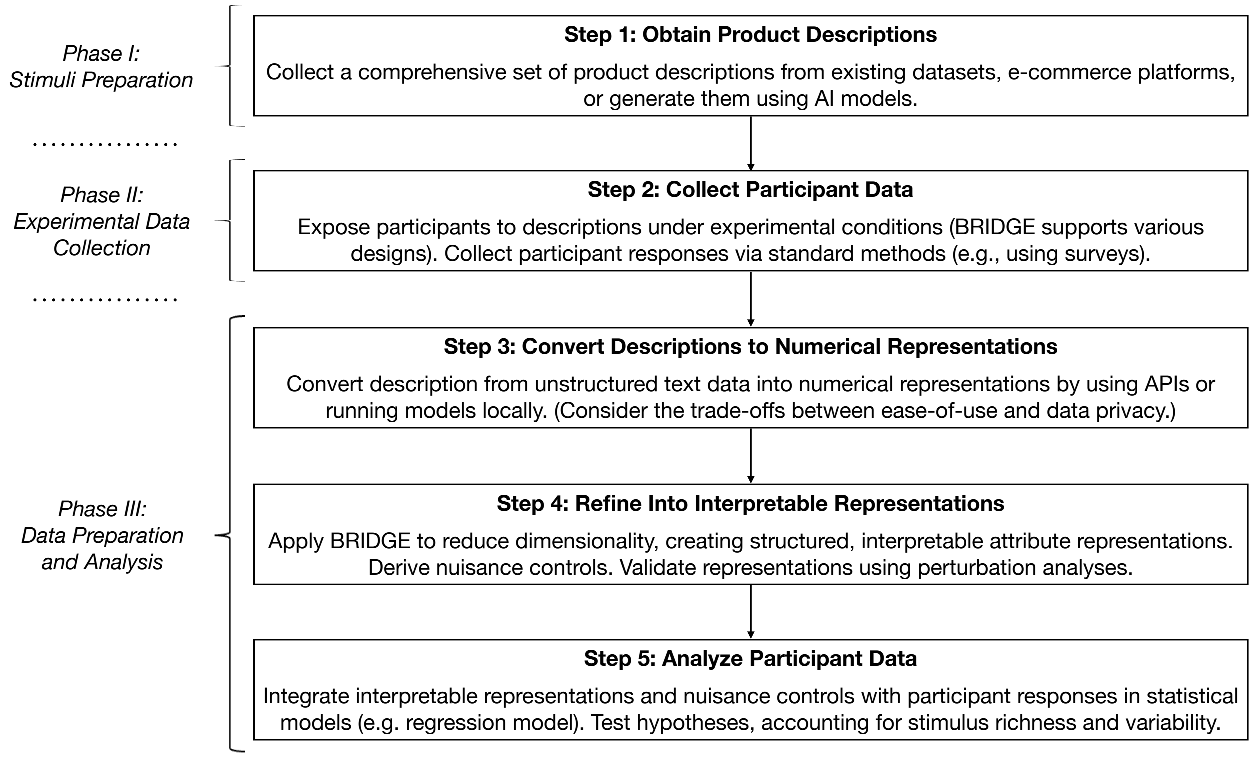


Figure 2: Procedure of a Study Using BRIDGE

can be sourced from existing datasets, collected from e-commerce platforms, or generated using AI models. *Step 2* involves participant data collection. BRIDGE supports within-subjects, between-subjects, and mixed experimental designs. It (a) converts the unstructured product descriptions to high-dimensional numerical representations ("embeddings") in *Step 3* and (b) refines these representations through a partitioned neural network, which generates structured, interpretable, and low-dimensional representations in *Step 4*. BRIDGE maps each attribute of interest to a distinct numerical component, facilitating clear comparisons across diverse stimuli.[5] BRIDGE also derives statistical controls for nuisance variables by orthogonalizing the original LLM embedding, absorbing extraneous variations.[6] In *Step 5*, researchers leverage the refined attribute representations

[5]Representations derived from BRIDGE (e.g., for region and varietal in our wine study) can be viewed as pre-trained embeddings. Researchers working with the same stimuli could potentially leverage these existing representations directly, bypassing the need for retraining.

[6]BRIDGE thus produces two outputs: (i) low-dimensional, attribute-specific representations that capture how focal constructs are conveyed in text, and (ii) orthogonal nuisance controls that absorb residual, non-focal textual variation. These outputs serve different inferential roles downstream—nuisance controls adjust for unobserved textual confounding and absorb idiosyncratic, treatment-unrelated variation, while the attribute representations capture within-attribute variation in how an attribute is expressed. Before proceeding to analysis, researchers should

and nuisance controls to analyze participant responses.

Validation

Applying BRIDGE in a new setting begins where any experiment does — by establishing that the focal attributes are present and measurable in the stimuli. When an attribute of the stimulus has been recorded — a product's origin or category, for instance — labels are already available, and BRIDGE's role is to compress them and to control the accompanying non-focal variation. When an attribute has not been recorded, such as whether a description is humorous or persuasive, it must first be measured on a labeled subset to conventional standards before BRIDGE can learn it. For instance, it may be coded by raters drawn from the same population as the participants, with agreement reported. In neither case does BRIDGE remove the obligation to measure the construct; it scales that measurement from a labeled sample to a large corpus.

Even with the attribute properly measured, validation does not end with prediction. The held-out classification accuracy and weighted F1 reported in *Step 4* are necessary evidence that the attribute is recoverable from the text, but they are not sufficient: high accuracy can ride a superficial cue that merely co-occurs with the attribute, and it speaks to neither whether the representation matches participants' perception nor whether the nuisance controls absorb the confounds that threaten the inference. Researchers should add checks matched to the failure they most suspect — comparing representations built from partial versus full stimuli when an explicit cue may be doing the work, re-estimating on subsamples where that cue is absent, substituting an alternative embedding to confirm conclusions do not hinge on one representation, and human-coding a subset to confirm the representation tracks independent judgment; the perturbation analysis below is one such check.

Two limits bound these efforts: no data-reduction control can be guaranteed to absorb every confound, and because the controls absorb only what the embedding encodes, confound control

validate BRIDGE by reporting classification accuracy and weighted F1 for each focal attribute on a held-out test set, confirming that the learned representations preserve the intended attribute distinctions. We discuss how to interpret these metrics — and what they do and do not establish for a new application — in the *Validation* subsection that follows.

is embedding-dependent even though the method is embedding-agnostic. Validation is thus necessarily *local*: predictive accuracy is a precondition for *considering* BRIDGE in a new application, not proof that its estimates are unconfounded there; the researcher's guide (Web Appendix §B) details a richer protocol.

Perturbation analyses. To evaluate whether the extracted representations in *Step 4*—and the downstream statistical conclusions in *Step 5*—are robust to nuisance variation, BRIDGE introduces a validation technique termed *perturbation analysis*. The objective is twofold: to ensure that the model's focal attribute representations remain stable when non-focal variation is introduced, and to verify that its nuisance controls effectively absorb any introduced noise.

The procedure involves modifying (perturbing) the stimuli to alter non-focal elements without changing focal attributes. For example, this might involve appending phrases (e.g., "New Design!", "Limited Stock!") to existing product descriptions to deliberately introduce nuisance variation that can confound a naïve analysis. These perturbed stimuli are then processed by the trained BRIDGE model and used in the downstream econometric model.

Evaluation proceeds in two steps. First, representational stability is assessed by calculating similarity metrics between the original and perturbed outputs — for example, the RV coefficient, a multivariate generalization of the squared Pearson correlation that indexes the similarity of two data configurations on a 0–1 scale, with values near 1 indicating that the representations are essentially unchanged by the perturbation. Robustness is indicated by high similarity across the attribute representations, coupled with low similarity across the nuisance controls (confirming the controls captured the injected noise). Second, the final statistical analysis is replicated using the perturbed representations. If the resulting coefficient estimates, significance levels, and model fit align with the original findings, it confirms that the observed effects stem from genuine theoretical constructs, as the deliberately introduced nuisance variation—and the alternative explanations it presents—is effectively absorbed by the model. Full implementation details and code are provided in Web Appendix §C.

Enhancing Research Transparency, Objectivity, and Reproducibility

BRIDGE enhances research transparency, objectivity, and reproducibility by mapping the research process into a structured, *end-to-end algorithmic pipeline* that can be precisely pre-specified and preregistered before data collection and analysis. Traditional experimental designs often require numerous subjective decisions about stimulus selection, simplification, and data analysis—decisions that can introduce researcher degrees of freedom, unintentionally compromising reproducibility and credibility (the "garden of forking paths" problem, Gelman and Loken 2013). BRIDGE allows researchers to commit *a priori* to: (1) a *real-world dataset* and a *stimulus sampling strategy* (e.g., random sampling) to reduce stimulus-selection bias, (2) an *embedding method* (e.g., an open-source model such as NV-Embed-v2) to encode the stimuli, (3) a *BRIDGE architecture* or a *systematic procedure* for determining it (e.g., using automated tools like Optuna within defined parameters, as shown in Web Appendix §F.2), and (4) a complete *analytical plan*, including independent and dependent variables, derived representations, nuisance controls, and models for hypothesis testing. The researcher's guide and accompanying code (Web Appendix §B) detail such a transparent and reproducible workflow, and a fully reproducible example based on our Coffee Certification Experiments—including pipeline code, experiment data, and pre-computed outputs—is provided alongside the `bridge` package.

Monte Carlo Studies Evaluating BRIDGE's Performance

We first apply BRIDGE to synthetic data under controlled conditions to validate its inferential performance. The complete study design and programming framework are presented in Web Appendix §D; below, we provide an overview and key results.

Simulation Study Design

Consider an experiment investigating whether exposure to one environmental concern (factory-farmed meat) shifts consumer preferences toward another environmental feature (organic farming). Participants are randomly assigned to a treatment condition (information about the detrimental effects of factory farming) or a control condition (no such information). They then evaluate 24 packaged salads. Each salad's description conveys its attributes—organic status, size, type, and weight—in natural language. The researcher's goal is to estimate the causal effect of the information about factory farming on preferences for a meat-free meal option, and whether it spills over to the organic premium.

Crucially, the study aims to use real-world descriptions to address the stimulus-sampling problem and boost generalizability. However, the real-world product descriptions obtained by the researcher vary in elaboration: some are terse and factual, others are vivid and creative. This variation can create a confound. For instance, if the treatment-condition participants were to encounter more elaborately written descriptions, a naïve analysis would not be able to discriminate between the treatment effect and the utility/disutility from elaboration. In addition, elaboration may amplify how much consumers value the organic attribute: a vividly described organic salad may command a larger premium than a tersely described one. Consequently, not only may the analysis yield a positive bias on the treatment effect (and therefore a false positive, a Type I error) but also a positive bias on organic interacted with the treatment (also a false positive, another Type I error). The objective is to recover the true treatment effect and the true interaction effect without observing the confound (elaboration), using only the stimuli themselves.

Data-Generating Process (DGP)

We simulate data for $N = 1{,}000$ participants, randomly assigned to a treatment or control condition, each completing 24 evaluation tasks on a 13-point Likert scale. To introduce a confound, we treat product descriptions as differing in elaboration across three styles—Factual (least elaborate), Engaging (moderately elaborate), and Creative (most elaborate). We present participants in the treatment condition with a greater proportion of Engaging and Creative descriptions (8 Engaging, 12 Creative, and 4 Factual per participant), and the control group with more Factual descriptions (12 Factual, 8 Engaging, and 4 Creative per participant); these differences in style allocation are unobserved.

Formally, product j is characterized by the tuple $\left(\text{Organic}_j, \text{Size}_j, \text{Type}_j, \text{Weight}_j, \text{Treatment}_j, \text{Style}_j\right)$, where Organic_j and Treatment_j are binary indicators, Size_j and Type_j are categorical, Weight_j is continuous, and $\text{Style}_j \in \{\text{Factual}, \text{Engaging}, \text{Creative}\}$. $\text{Intensity}_j \in \{0, 1, 2\}$ maps the three styles to increasing levels of descriptive elaboration. Participant i's latent utility for product j is:

$$\begin{aligned}\text{preference}_{ij} = \alpha + \beta_1 \text{Treatment}_j + \beta_2 \left(1 + \lambda \cdot \text{Intensity}_j\right) \cdot \text{Organic}_j + \beta_3 (\text{Treatment}_j \times \text{Organic}_j) \\ + \beta_4 \text{Size}_j + \beta_5 \text{Type}_j + \beta_6 \text{Weight}_j + \theta \cdot (\text{Intensity}_j - 1) + \varepsilon_{ij},\end{aligned}$$

where $\beta_1 = 0.25$, $\beta_2 = 0.50$, and $\beta_3 = 0.50$ capture the treatment effect, organic premium, and their interaction (spillover), respectively; β_4, β_5, and β_6 represent size, type, and weight effects (with $\beta_6 = 0.10$); and $\varepsilon_{ij} \sim \mathcal{N}(0, 1)$.

Confounds can threaten inference in two ways. First, when the confound covaries with the treatment and directly shifts preferences (Kirk 2013), it creates an omitted variable bias on the main effect of the treatment. We term this channel *main effect confounding*. Second, when it changes the *effective strength* of a focal attribute, it acts as an unobserved moderator (Blackwell and Olson 2022). In this case, it creates an omitted variable bias on the interaction of the treatment with the attribute. We term this channel *interaction effect confounding*.

The parameter θ controls the *additive* effect of style on preferences (main effect confounding), shifting utility by $-\theta$, 0, and $+\theta$ for Factual, Engaging, and Creative descriptions, respectively. The parameter λ controls the *moderating effect* of description intensity on the organic premium (interaction effect confounding). When $\lambda > 0$, the organic effect becomes style-specific: β_2 for Factual, $\beta_2(1 + \lambda)$ for Engaging, and $\beta_2(1 + 2\lambda)$ for Creative.

The implications for theory development are complex as these channels can elevate both Type I and Type II errors. If there is no treatment effect, a bias can lead to a false positive (a Type I error). If the bias attenuates the estimate, it can lead to a false negative (a Type II error).

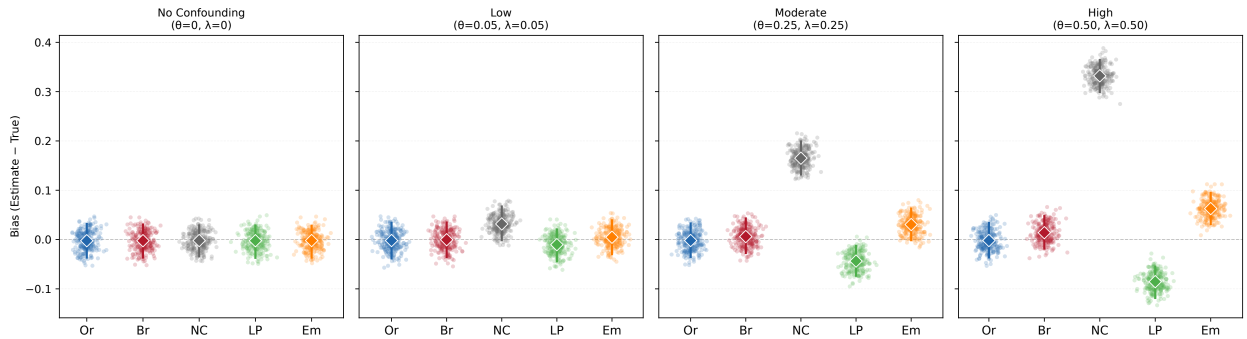
Analytical Approaches

We evaluate five estimators. In every estimator, the observed product attributes (Size, Type, and Weight) enter directly as controls; the estimators differ only in how they handle the *unobserved* style confound, whose allocation covaries with the treatment. The Oracle conditions on the true style assignments—information unavailable in practice—and serves as the efficient but infeasible benchmark; the remaining four estimators are feasible, using only the observed information (product descriptions, product attributes). The question is whether BRIDGE, which learns nuisance controls and attribute representations from the descriptions, can match the infeasible Oracle.

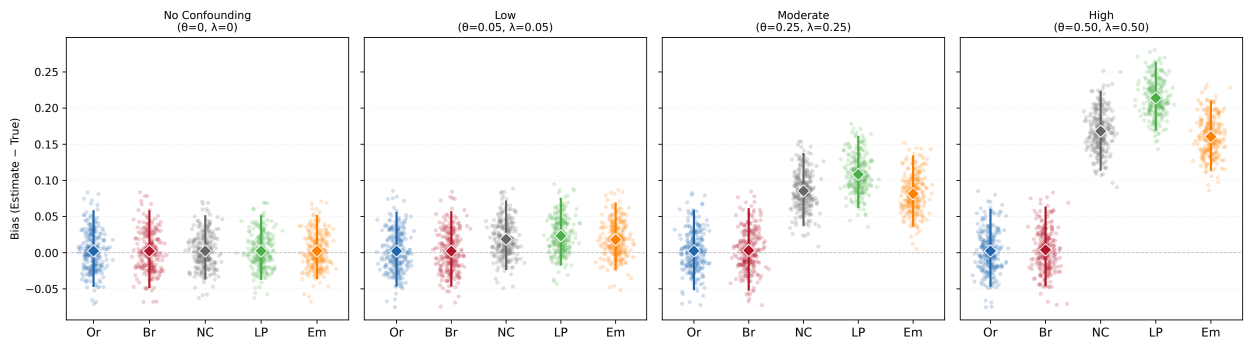
1. **Oracle (including Style Dummies):** An infeasible estimator that includes dummies describing the unobserved confound structure alongside style $\times$ organic interaction terms.
2. **BRIDGE:** Interacts BRIDGE's orthogonal nuisance controls and attribute representations with the organic indicator to capture both main effect and interaction effect confounding, without observing style.
3. **No Controls:** Omits text controls entirely; assumes there is no confound.
4. **Linear Projection:** Includes nuisance controls derived by using Ridge regression on the attribute labels and then applying singular value decomposition (SVD) to the residuals—a linear decomposition of the original LLM embedding.
5. **Empath Controls:** Includes control variables derived by applying SVD to variables from the Empath library (Fast, Chen, and Bernstein 2016), which uses deep learning (neural embeddings trained on 1.8 billion words) to generate pre-validated psycholinguistic measures.

Results

We examine parameter vectors corresponding to four scenarios (no confounding: $\theta = 0$, $\lambda = 0$; low confounding: $\theta = 0.05$, $\lambda = 0.05$; moderate confounding: $\theta = 0.25$, $\lambda = 0.25$; high confounding: $\theta = 0.50$, $\lambda = 0.50$). Figure 3 describes the bias of the estimated treatment effect and interaction effect (treatment $\times$ organic) for 250 different simulations (where we vary the seed).



(a) Treatment effect ($\beta_1 = 0.25$)



(b) Treatment $\times$ Organic interaction ($\beta_3 = 0.50$)

Figure 3: Estimation Bias Across Four Confounding Levels

Note: Bias is the estimate minus the true value. Each dot represents one of 250 random seeds; diamonds indicate the mean bias; vertical bars span the 2.5th–97.5th percentile interval; the dashed line at zero indicates no bias. Or = Oracle (infeasible), Br = BRIDGE, NC = No Controls, LP = Linear Projection, Em = Empath Controls.

BRIDGE closely approximates the Oracle—the efficient but infeasible estimator—across all four scenarios (Figure 3). The BRIDGE and Oracle bias distributions are centered on zero and virtually indistinguishable, regardless of confounding intensity, for both the treatment effect and the treatment $\times$ organic interaction. BRIDGE and Oracle are also virtually identical across all confounding levels for the three style-specific organic effects (see Web Appendix §D for the corresponding figures).

In contrast, the alternatives align with the true values only in the absence of confounding ($\theta = 0$, $\lambda = 0$). Even at low confounding ($\theta = 0.05$, $\lambda = 0.05$), the No Controls estimator shows substantial treatment effect bias. At moderate confounding ($\theta = 0.25$, $\lambda = 0.25$), all three alternatives are severely biased on the interaction and, to varying degrees, on the treatment. These biases are reflected in the coverage rates: an accurate estimator's 95% confidence interval should contain the true value in 95% of the simulations (237.5 of 250). With moderate confounding, coverage of the treatment effect drops to 0, 99, and 179 for No Controls, Linear Projection, and Empath, respectively, while coverage of the interaction effect drops to 26, 4, and 30. BRIDGE and Oracle closely match the expected rate (treatment: 240 and 244, interaction: 236 and 237).

Table 1: Monte Carlo Simulation Results ($\theta = 0.25$, $\lambda = 0.25$)

	Oracle	**BRIDGE**	**No Controls**	**Linear Projection**	**Empath Controls**
Treatment ($\beta_1 = 0.25$)	0.240 (0.021)	0.248 (0.021)	0.428 (0.019)	0.208 (0.020)	0.292 (0.020)
	[0.199, 0.280]	[0.207, 0.288]	[0.390, 0.466]	[0.169, 0.248]	[0.253, 0.330]
	✓	✓	✗	✗	✗
Treatment × Organic ($\beta_3 = 0.50$)	0.512 (0.029)	0.512 (0.029)	0.567 (0.027)	0.593 (0.027)	0.563 (0.027)
	[0.454, 0.569]	[0.455, 0.569]	[0.513, 0.621]	[0.540, 0.645]	[0.510, 0.616]
	✓	✓	✗	✗	✗
Organic, Factual ($\beta_2 = 0.50$)	0.518 (0.024)	0.518 (0.024)	0.577 (0.019)†	0.565 (0.019)†	0.588 (0.019)†
	[0.470, 0.566]	[0.470, 0.565]	[0.539, 0.615]	[0.528, 0.602]	[0.550, 0.626]
	✓	✓	✗†	✗†	✗†
Organic, Engaging (0.625)	0.611 (0.027)	0.608 (0.027)	–	–	–
	[0.557, 0.664]	[0.556, 0.661]			
	✓	✓			
Organic, Creative (0.75)	0.688 (0.032)	0.689 (0.031)	–	–	–
	[0.626, 0.751]	[0.628, 0.750]			
	✓	✓			
R-squared	0.481	0.481	0.452	0.479	0.470

Note. Standard errors in parentheses, 95% confidence intervals in brackets. ✓ indicates ground truth falls within the 95% CI, ✗ indicates it does not. †Pooled organic coefficient; these estimators cannot decompose by style. All models include size and type dummies. Style-specific organic effects for BRIDGE and Oracle are computed via the delta method: for each style s, $\hat{\beta}_2$ is projected to the conditional mean nuisance and attribute representation profile for style s, with standard errors computed from the gradient and the relevant covariance submatrix. The full coefficient vector is reported in Web Appendix Table W1.

Table 1 presents coefficient estimates and 95% confidence intervals for all estimators in the moderate confounding condition ($\theta = 0.25$, $\lambda = 0.25$). The Oracle, which observes the unobserved style assignments, eliminates confound-related uncertainty and accurately recovers all five focal coefficients: treatment ($\hat{\beta}_1 = 0.240$), the treatment × organic interaction ($\hat{\beta}_3 = 0.512$), and style-specific organic effects of 0.518, 0.611, and 0.688 for Factual, Engaging, and Creative (true values: 0.50, 0.625, 0.75). The privileged information on which it is based (style assignments), however, is unobserved in practice. Therefore, the Oracle serves only as a benchmark for the best achievable inference.

BRIDGE, without observing style, approaches the Oracle. The treatment effect ($\hat{\beta}_1 = 0.248$)

and the treatment $\times$ organic interaction effect ($\hat{\beta}_3 = 0.512$) are both consistent with ground truth. The style-specific organic effects—0.518, 0.608, and 0.689—are virtually indistinguishable from the Oracle. The mechanism is the interaction of BRIDGE's learned representations with the organic indicator: the nuisance controls and the attribute representations carry style-related information that differs systematically between organic and conventional descriptions. Both serve as continuous proxies for the unobserved style $\times$ attribute interaction.

The absence of a comparable control mechanism leads to severe bias. No Controls overestimates both the treatment effect ($\hat{\beta}_1 = 0.428$ versus the true 0.25), indicating main effect confounding, and the treatment $\times$ organic interaction ($\hat{\beta}_3 = 0.567$ versus the true 0.50), reflecting interaction effect confounding. Linear Projection underestimates the treatment ($\hat{\beta}_1 = 0.208$) and fails to accurately estimate the interaction ($\hat{\beta}_3 = 0.593$); the linear decomposition absorbs some style variance but cannot fully capture the relationship between style and attribute effects. The model with Empath lexical features overestimates both the treatment ($\hat{\beta}_1 = 0.292$) and the interaction ($\hat{\beta}_3 = 0.563$). Furthermore, all three alternatives yield a single pooled organic coefficient (0.57–0.59) that collapses style-varying effects; only the Oracle and BRIDGE capture both confound channels and separately recover the style-specific effects.

Discussion

BRIDGE recovers all five structural parameters—treatment, treatment $\times$ organic interaction, and style-specific organic effects—matching the infeasible Oracle. The alternative estimators fail differently, illustrating the distinct confound channels. Under main effect confounding alone, lexical controls such as Empath may plausibly absorb the dominant style axis and recover the treatment effect. But when textual features (such as extent of elaboration or elaboration style) also moderate attribute strength (the λ channel), these methods collapse the attribute effect to a single average, yielding biased estimates of the interaction effect.

These considerations delineate a practical complexity boundary. When product descriptions are simple and few, conventional designs with carefully controlled stimuli likely suffice and algorithmic nuisance controls add little incremental value. BRIDGE becomes necessary in product contexts where many varied descriptions are the norm in the real-world marketplace.

Specifically, when the attribute space is large and semantically structured—for example, our wine study involves 426 regions and 708 varietals—consumer decisions likely exhibit interaction effect confounding: a terse mention of "Burgundy" and an evocative paragraph about Burgundian terroir both indicate the same region but likely activate consumer associations and preferences differently. Because the confound structure is also unknown *ex ante*, BRIDGE is the robust default: it matches simpler methods when they suffice—the statistical controls are orthogonal to the product's attributes by design, so including them even when nuisance variables play a minimal role does not affect consistency (Figure 3)—while remaining the only feasible and accurate approach when both confound channels are active.

Coffee Certification Experiments

Having established BRIDGE's performance in controlled Monte Carlo simulations with specified confound structure, we test BRIDGE in experiments with human participants. Researchers often do not observe whether (and which) non-focal features of the stimuli covary with the focal treatment—the confound structure may not be known in advance. Even when a specific confound is suspected, accounting for it typically requires creating additional stimulus versions, running follow-up studies, or collecting more measures (for covariate adjustments). BRIDGE addresses potential confounds by extracting nuisance controls from the stimuli without explicit confound specification.

To examine the efficacy of BRIDGE, we situate our experiments in coffee certification framing: fair trade in Experiment 1 and organic in Experiment 2. Product texts that include certification information can vary in length, sometimes in systematic ways. For example, a fair trade or organic disclosure may be conveyed succinctly (e.g., a certification label only) or elaborately (e.g., sourcing and standards information) compared to products without the certification disclosure. Such variation can introduce a text-length confound that obscures the true treatment effect.

We deliberately introduce a text-length confound to assess BRIDGE's performance. Participants evaluate a pair of coffee descriptions and are randomly assigned to conditions in which the treatment description is shorter than, matched to, or longer than the control description. The matched condition is the standard stimulus-control design in which the stimuli differ only on the focal dimension, approximating an "unconfounded" treatment effect; in the remaining conditions, the treatment is confounded by description length. We then examine whether BRIDGE's nuisance controls recover, in the confounded conditions, treatment-effect estimates comparable to the matched benchmark, and compare BRIDGE with conventional estimators.

Method

Participants and design

Participants were recruited from the Prolific U.S. panel and completed one of the two (mutually exclusive) experiments for small monetary compensation; no participant took part in both. Experiment 1 recruited 353 participants (43.9% women; 55.5% men; 0.6% non-binary; M_{age} = 44.9; https://aspredicted.org/7dt3ev.pdf). Experiment 2 recruited 352 participants (49.7% women; 48.9% men; 1.1% non-binary; 0.3% prefer not to say; M_{age} = 44.7; https://aspredicted.org/fz6y8h.pdf). In each experiment, participants were randomly assigned to one of the three between-subjects conditions (treatment framing length: shorter-than-control, length-matched, longer-than-control).

Procedure and measures

Participants evaluated a pair of coffee descriptions—one with certified framing (treatment; Experiment 1: fair trade, Experiment 2: organic) and one without (control)—and indicated their relative preference on a 7-point scale (1 = strongly prefer Option A, 7 = strongly prefer Option B). All participants were presented with the same two underlying coffee profiles (nutty/balanced; chocolatey/full-bodied). For each participant, certified framing was randomly assigned to one profile, and the left/right placement of the options was randomized. We varied the treatment description length (short, medium, or long); the control description was always medium length.

This design yields a length-matched benchmark condition (medium treatment, medium control), and provides a controlled setting to assess how experimentally induced nuisance variation in text length (shorter-than-control, length-matched, longer-than-control) can affect inference about the focal treatment effect. Finally, participants provided basic background information (gender, age). Detailed stimulus materials and methods are available in Web Appendix §E.

Model specifications

The key dependent variable is participants' 7-point recoded preference, ranging from -3 (strongly prefer the Base Option [control option]) to 3 (strongly prefer the Comparison Option [certification-framed option]). We fit two Bayesian linear regressions: the Naïve and BRIDGE models. The models are designed to test, using different approaches, the extent to which participants' preferences are influenced by the certified frame (treatment) when its length is also varied (confound).

Specification 1: naïve model. As in our preregistrations, the first model estimates the effect of treatment frame on recoded preference using a Bayesian linear regression with condition indicators. The treatment/medium-length condition served as the reference category.

In Experiment 1, the utility that participant i assigned to the Comparison Option was represented as:

$$u_i = \beta_0 + \beta_1 \mathbb{I}(\text{FairTrade+Short}_i) + \beta_2 \mathbb{I}(\text{FairTrade+Long}_i) + \varepsilon_i, \tag{4}$$

where u_i represents the utility of participant i for the Comparison Option; β_0 is the intercept, representing the effect of the fair-trade/medium-length option; FairTrade+Short_i and FairTrade+Long_i are the dummy variables for the corresponding conditions; β_1 and β_2 are the coefficients capturing the influence of shorter or longer description length on participants' preference; and ε_i is an i.i.d. Gaussian error term. A parallel utility function was specified for certified organic framing in Experiment 2; we estimated each experiment separately. A significant intercept β_0 indicates that the fair-trade/medium frame differs significantly from the non-fair-trade/medium control. Significant β_1 or β_2 coefficients indicate additional effects of the shorter or longer fair-trade frame relative to the fair-trade/medium frame.

Specification 2: BRIDGE. This model adds a vector of nuisance controls, extracted by BRIDGE from the orthogonalized residuals of the LLM embeddings relative to the learned focal attribute representations, to estimate the utility that participant i assigned to the Comparison Option. The estimator is both feasible and flexible, as it requires neither (1) knowledge of the experimental conditions nor (2) specification of what the confound is; BRIDGE learns the nuisance structure directly from the text.

Specifically, we represent utility as:

$$u_i = \beta_0 + \mathbf{z}_i^\top \boldsymbol{\gamma} + \varepsilon_i, \tag{5}$$

where u_i represents the utility of participant i for the Comparison Option; β_0 is the intercept, representing the effect of the certified/medium-length option; $\mathbf{z}_i$ is a $K \times 1$ vector of nuisance controls for participant i, constructed using the BRIDGE method; $\boldsymbol{\gamma}$ is a $K \times 1$ vector of coefficients corresponding to the nuisance controls; and ε_i is an i.i.d. Gaussian error term.

Additional specification: word-count model. This model tests the extent to which adding the treatment–control word-count difference as a covariate controls for the specific confound (length) in measuring β_0 (the effect of treatment frame).

Results: Relative Preference for Certified Coffee

All completed observations were included in our analyses. Experiments 1 and 2's key findings are described in Table 2.

Naïve model

Table 2 reports the treatment effects measured by the canonical estimator (naïve to the confound) when fit separately to the data from each version of the treatment text (shorter, length-matched, longer). When description length is matched (both treatment and control stimuli are medium-length), the treatment framing increases the 7-point recoded preference by +1.32 in Experiment 1 (fair trade) and +0.97 in Experiment 2 (organic). These estimates serve as our key benchmark:

Table 2: Coffee Certification Experiments

Estimator	Experiment 1: Fair Trade		Experiment 2: Organic	
	Estimate	**95% CI**	**Estimate**	**95% CI**
Naïve (matched)	1.32	[0.96, 1.69]	0.97	[0.56, 1.38]
Naïve (shorter)	0.16	[−0.22, 0.52]	0.38	[−0.02, 0.78]
Naïve (longer)	1.29	[0.88, 1.69]	1.26	[0.89, 1.61]
Naïve (pooled)	0.92	[0.70, 1.14]	0.88	[0.66, 1.11]
Word Count	0.80	[0.57, 1.04]	0.76	[0.51, 0.99]
BRIDGE	1.24	[1.00, 1.49]	1.02	[0.51, 1.54]

Note. Naïve (matched/shorter/longer) estimates are derived from a single Bayesian linear regression with condition indicators (medium-length as reference): matched $= \hat{\beta}_0$, shorter $= \hat{\beta}_0 + \hat{\beta}_1$, longer $= \hat{\beta}_0 + \hat{\beta}_2$. Naïve (pooled) is the weighted average of the condition-specific effects, with weights proportional to sample sizes. Word Count controls for the word-count difference Δ_{wc} between treatment and control descriptions. BRIDGE uses one nuisance control extracted from AI text embeddings; results with two nuisance controls are similar. All models use Gaussian errors and are estimated separately by experiment. All 95% CIs are posterior credible intervals.

holding constant the confound (description length) corresponds to the canonical estimator in the standard stimulus-matched experimental design.

However, when the treatment text is shorter than the control text, the estimated effect of treatment framing reduces to +0.16 in Experiment 1 and +0.38 in Experiment 2; in each experiment, the estimate from the shorter-than-control data is substantially smaller than from the length-matched data and not within its 95% CI. By contrast, when the treatment text is longer than the control text, the estimated effect is +1.29 in Experiment 1 and +1.26 in Experiment 2, both comparable to the length-matched benchmark (within its 95% CI). When the confound is mixed and unobserved (such as would result if a researcher were unaware of the confound and pooled the data across scenarios), the estimate is +0.92 and +0.88 in Experiments 1 and 2, respectively. Thus, in Experiment 1, uncontrolled variation in description length attenuates the measured effect of fair-trade framing; in Experiment 2, the estimate is comparable to the benchmark from the matched data. These scenarios illustrate that uncontrolled nuisance variations can lead to materially different conclusions across experiments (attenuating the treatment effect in Experiment 1, risking a false negative, while leaving inference in Experiment 2 unaffected), depending on the nature and extent of the confound (differences in treatment versus control text length).

BRIDGE

In both experiments, BRIDGE effectively recovers the "unconfounded" treatment effect without knowledge of the experimental conditions. In Experiment 1, BRIDGE ($\hat{\beta}_0 = 1.24$, 95% CI [1.00, 1.49]) recovers a treatment effect comparable to the length-matched benchmark ($\hat{\beta}_0 = 1.32$; posterior difference $\delta = -0.081$, 95% CI $[-0.528, 0.361]$). In Experiment 2, BRIDGE ($\hat{\beta}_0 = 1.02$, 95% CI [0.51, 1.54]) similarly recovers the benchmark ($\hat{\beta}_0 = 0.97$; $\delta = 0.051$, 95% CI $[-0.626, 0.707]$).

Word-count model

If a researcher is aware of the text-length confound, one possibility is to treat it as a covariate and attempt to statistically control for it. We fit a linear model controlling for the word-count difference between the treatment and control descriptions. In Experiment 1, this yields $\hat{\beta}_0 = 0.796$ with a slope of 0.021 per word of difference, producing condition-specific estimates of 0.564 for the shorter condition (treatment text: 10 words), 0.796 for the matched condition (21 words), and 1.450 for the longer condition (52 words). In Experiment 2, the corresponding model yields $\hat{\beta}_0 = 0.756$ with a slope of 0.018, producing estimates of 0.508 (shorter, 10 words), 0.756 (matched, 24 words), and 1.306 (longer, 55 words). Across both experiments, the word-count model overcorrects at the matched condition; in Experiment 1, this difference (-0.528) is significant (95% CI $[-0.971, -0.096]$ does not contain 0). Thus, a standard word-count covariate does not adequately capture the relationship between description length and preference, mischaracterizing the treatment effect across conditions.

Coffee Certification Experiments Discussion

Experiments 1 and 2 show that confounding variation in text length can distort the estimated effect of certified framing under conventional analyses. Conceptually, our design parallels the Monte Carlo study, now with real participants. The pooled naïve estimates are consistent with main effect confounding: uncontrolled length variation attenuates the treatment effect relative to the length-matched benchmark (markedly so in Experiment 1). The inclusion of word count does not reliably recover the benchmark, suggesting standard covariate adjustments might not be adequate for this confounding. In contrast, even when the confound was not explicitly specified, BRIDGE recovers treatment effects comparable to the length-matched benchmark in the respective experiments. This motivates the use of BRIDGE's nuisance controls, as they help isolate the treatment effect while flexibly accounting for nuisance variation in text. Broadly, the experiments show that BRIDGE can complement careful stimulus design by providing a safeguard when residual or unanticipated confounding variation remains.

Preference Dynamics in Unstructured Real-World Product Descriptions

Next, we demonstrate how BRIDGE can (1) enable the sampling of myriad unaltered real-world product texts as study stimuli, addressing the stimulus-sampling problem, and (2) apply to hypothesis testing in information-rich environments. Whereas the simulations and the Coffee Certification Experiments provide direct evidence that BRIDGE corrects text-based confounding in settings where the confound structure is known or induced, the wine study makes a complementary contribution: it enables a large-scale, naturalistic choice experiment that conventional designs cannot accommodate.

Normative theories view consumer preferences as stable and merely revealed during decision making (Rabin 1998). Mounting evidence, however, shows that preferences are often constructed based on how options are presented (Payne et al. 2000; Slovic 1995). Making repeated choices can stabilize preferences as consumers learn the attribute trade-offs underlying their choices (Amir and Levav 2008; Hoeffler and Ariely 1999). Individuals can also anchor on accessible information as a reference point, with subsequent judgments biased toward or away from that anchor (Ariely, Loewenstein, and Prelec 2003; Spicer et al. 2022). Yet much of this evidence relies on numerical stimuli, with some extending to semantic categories (Chernev 2011). Previous studies on preference dynamics employ structured choice contexts in which options are defined by a few simpler, aligned attributes (Amir and Levav 2008; Donkers et al. 2020).

In contrast, consumers in real-world settings often encounter options described in unstructured text—wine tasting notes and coffee flavor narratives—rich with subjective features that may not align across options. Learning from earlier choices is more difficult in such environments because (1) the consequences of a choice can be hard to trace back to specific antecedents (Einhorn and Hogarth 1981), and (2) the concreteness of the information can affect its meaning and implications for decision making (Ebbesen and Konecni 1980), and processing it often requires more cognitive effort (Martin, Seta, and Crelia 1990; Stone and Schkade 1991). Some scholars have argued that judgment phenomena observed with simpler, stylized stimuli may not occur in information-rich environments (Ebbesen and Konecni 1980; Einhorn and Hogarth 1981; Gigerenzer 1991). Thus, it remains unclear whether unstructured product descriptions encountered at the outset affect subsequent decisions.

We examine how consumers develop preferences when making sequential choices based on real-world, unstructured product descriptions. We posit that preferences for complex, verbally described products are initially constructed but become largely invariant, exhibiting consistency in subsequent choices. When consumers encounter products described in prose, they need to devote effort to assess this information. Product options encountered at the outset can activate the accessibility of select semantic knowledge about the options (Strack and Mussweiler 1997) and set standards of comparison (Mussweiler 2003). Because anchors exert influence by increasing the availability of anchor-consistent information (Chapman and Johnson 1999), the initial products provide a reference frame for consumers to discover their preference structure (i.e., attribute trade-off weights), thereby aligning subsequent preferences with the initial, incidental product anchors.

To test this hypothesis, we present each participant with a sequence of choice tasks involving pairs of randomly selected, real-world product descriptions in prose. In two experiments, participants are asked to choose their preferred option (our wine study) or to indicate a relative preference (our coffee anchoring study). For each participant, each product in the sequence is drawn randomly (without replacement) from 119,955 wines or 36 coffees. We design the experiments so that (1) the initial products (anchors) are randomly chosen and therefore incidental; (2) subsequent products are also randomly chosen, ensuring exogenous variation in the distance between the anchors and these subsequent products; and (3) the real-world product descriptions are unaltered (unabridged and non-stylized) and their characteristics do not always align.

We use BRIDGE to analyze participants' decisions. Such an analysis would be intractable using conventional methods like ANOVA because no two participants face the exact same choice task (effectively placing each participant in a unique condition) and because of the nuisance variables inherent in real-world descriptions. To provide converging evidence, we then conduct a follow-up experiment (Web Appendix §G) using a limited number of controlled stimuli. Although this approach forgoes the benefits of broad stimulus sampling, it yields data that are amenable to traditional analysis techniques, ensuring our results reflect the phenomenon itself rather than method artifacts.

Context and Product Descriptions

Wine descriptions in the real-world marketplace are rich in sensory detail and nuance—the complexity makes wine an ideal context for examining our hypothesis of anchoring. First (Step 1 in Figure 2), we obtain a comprehensive dataset containing the verbal descriptions (tasting notes) of 119,955 wines from Wine Enthusiast, a globally recognized publication. These tasting notes are developed through extensive blind taste tests, where professional tasters describe each wine's character and consumption experience without access to identifying labels such as the producer, name, varietal, or region. They are essential for communicating the nuanced profiles of wines to consumers and are often adapted by retailers into point-of-sale marketing materials.

The activation of salient concepts in initial products can serve as a semantic reference frame (Mussweiler 2003; Strack and Mussweiler 1997). A wine's taste, cultural roots, and geographical-climatic provenance are shaped by its region of origin (terroir) and grape varietal (MacNeil 2015). Retailers often use these attributes to organize and display their products (e.g., Wine.com, a large

US-based online wine shop, as well as many online or brick-and-mortar stores). As participants are likely to draw on taste expectations associated with region and varietal when forming preferences (Latour and Latour 2010; Rocklage, Rucker, and Nordgren 2021), we focus our analysis on these attributes.

Our data comprise 119,955 wines from 426 wine-growing regions and 708 wine-grape varietals—a vast product space. Each record includes a wine's name, region, varietal, and tasting note (averaging 40 words in length; SD = 11.2 words). BRIDGE fits the data well: it achieves 98.2% classification accuracy for region (426 classes) and 98.0% for varietal (708 classes), far exceeding chance baselines of 0.23% and 0.14%, respectively.[7] Detailed classification metrics including weighted F1 scores are reported in Web Appendix §F.2. A sensitivity analysis examining the impact of training data size on classification performance is reported in Web Appendix §F.3.

Experimental Design and Participant Data

We collected participant data in collaboration with Qualtrics, a global leader in market research (Step 2 in Figure 2). We engaged 1,000 consumers from Australia (250 participants), New Zealand (200 participants), and the United States (550 participants), ranging in age from 25 to 89, with 50.5% women. Participants met three criteria: a minimum age of 25 years (set for ethical reasons), current employment, and reported wine consumption of at least one glass in the prior 28 days. The majority (86.5%) reported consuming at least one glass of wine per week. We instructed our service provider to ensure that the participants' demographics were representative of the wine-consuming population in their respective countries.

Each participant completed 32 sequential tasks. In each task, participants saw a pair of wines—each selected randomly without replacement from our corpus of 119,955 real-world product descriptions—and were asked to pick their preferred option. Each wine was described by its name

[7]The displayed names themselves often contain the focal attributes: 64.7% of names include the wine's varietal, 26.4% its region, and 17.4% both, with near-identical fractions among the wines shown to participants — these metrics reflect how faithfully the compressed representations preserve the attributes present in the record, not the inference of hidden labels from prose. Moreover, we test directly whether the anchoring results depend on this (see the results below and Web Appendix §F.7).

and tasting note. This shows BRIDGE's flexibility, as it (a) allows participants to encounter many different options and (b) presents wines in prose, resembling how consumers encounter these products in the real world.

We opted for a relatively long sequence of 32 decision tasks as a more stringent test of the hypothesized anchoring effect. The extended sequence also allows us to examine candidate anchors in different positions in the sequence.

Variable Operationalization and Model

Two options were shown in each choice task. Our dependent variable was 1 if the option on the right ("right-option") was chosen and 0 if the option on the left ("left-option") was chosen. To examine anchoring, we differenced the similarity of the right-option to a *candidate anchor* and the similarity of the left-option to the same anchor. A positive (and significant) coefficient on this "right minus left" variable would indicate that participants were more likely to choose the option on the right when it was more similar to the anchor. Based on this operationalization, we construct four independent variables examining the effect of different anchor positions:

1. *Similarity to Options in the First Task (Sim^{First}):* the anchor is the pair of options in the first task; its coefficient tests the hypothesized anchoring effect.

2. *Similarity to Options in the Previous Task (Sim^{Prev}):* the anchor is the pair of options in the immediately preceding task; its coefficient tests for the recency effect.

3. *Similarity to Options in the Next Task (Sim^{Next}):* the anchor is the pair of options in the next task. As the options in the next task are unknown at the time of decision making, its coefficient serves as a placebo test. We expect a nonsignificant coefficient, ruling out alternative explanations such as pre-existing preferences (revealed preferences).

4. *Similarity to Options in the Final Task (Sim^{Final}):* the anchor is the pair of options in the final task. Like Sim^{Next}, its coefficient serves as a placebo test.

To rule out alternative explanations, we also calculated similarity to the second task ($\text{Sim}^{\text{Second}}$), two tasks before the current task ($\text{Sim}^{\text{Two Before}}$), two tasks after the current task ($\text{Sim}^{\text{Two After}}$), and the second-to-final task ($\text{Sim}^{\text{Second-to-Final}}$).

We fit a hierarchical Bayesian model where the utility that participant i assigns to option k in task t is:

$$
\begin{aligned}
u_{ikt} = \beta_{0i} &+ \beta_{1i}\text{Sim}^{\text{First}}_{ikt} + \beta_{2i}\text{Sim}^{\text{Prev}}_{ikt} + \beta_{3i}\text{Sim}^{\text{Next}}_{ikt} + \beta_{4i}\text{Sim}^{\text{Final}}_{ikt} \\
&+ \beta_{5i}\text{Sim}^{\text{Second}}_{ikt} + \beta_{6i}\text{Sim}^{\text{Two Before}}_{ikt} + \beta_{7i}\text{Sim}^{\text{Two After}}_{ikt} + \beta_{8i}\text{Sim}^{\text{Second-to-Final}}_{ikt} \\
&+ \sum_{q=1}^{N} \delta_{qi}\text{Nuisance}_{qikt} + \varepsilon_{ikt}.
\end{aligned}
$$

ε_{ikt} is i.i.d. Gumbel, yielding a logit choice model. $N = 5$, chosen based on the elbow of the scree plot of the nuisance variables.

We estimated a sequence of nested fixed-effects models. *Model 1* includes the four focal variables ($\text{Sim}^{\text{First}}$, Sim^{Prev}, Sim^{Next}, $\text{Sim}^{\text{Final}}$). *Model 2* incorporates the four additional variables ($\text{Sim}^{\text{Second}}$, $\text{Sim}^{\text{Two Before}}$, $\text{Sim}^{\text{Two After}}$, $\text{Sim}^{\text{Second-to-Final}}$). *Model 3* augments Model 1 by adding the BRIDGE-derived nuisance controls (Nuisance_q). *Model 4* includes all eight variables and the nuisance controls. In addition, to account for individual-level differences in sensitivity to these variables, we introduce participant heterogeneity through random effects. *Model 5* builds upon Model 3, allowing for heterogeneity in the intercept, the four focal similarity variables ($\text{Sim}^{\text{First}}$, Sim^{Prev}, Sim^{Next}, $\text{Sim}^{\text{Final}}$), and the nuisance controls (δ_{qi}). *Model 6* introduces participant-level heterogeneity in *all* variables—the intercept, all eight similarity measures, and the nuisance controls.

We developed measures based on the cosine similarity of the BRIDGE-derived attribute representations of the options. Specifically, we used the sum of the cosine similarities between an option's attribute representations and the attribute representations of the two options shown in an anchor task.[8] Higher scores indicate greater similarity between a presented option and the

[8]The BRIDGE architecture used to generate these representations was determined using hyperparameter optimization to ensure objectivity and reproducibility (see Web Appendix §F.2 for details). Post-hoc analysis confirmed a substantial

options in the anchor task.

Each option was selected at random (without replacement from the corpus) for each participant. Participants encounter exogenous variation in both the anchors and the options presented in the tasks. Thus, the *similarity* measures vary randomly across both participants and tasks, providing exogenous variation for estimating the anchoring effect and examining alternative explanations. This stimulus-randomization procedure ensures that no single wine (or its varietal or region) could drive the focal variable, reducing potential bias from a specific stimulus.

Results

The data comprise 32,000 choice tasks involving 49,548 unique wines (41.3% of available wines) from 367 wine-growing regions and 566 grape varietals. This broad sampling ensured substantial individual exposure to diversity: on average, each participant encountered 28.6 (SD = 2.99) unique wine regions and 31.6 (SD = 3.02) unique varietals across their 32 tasks. All observations were included in our analyses (i.e., no data exclusions); the models are estimated on the 28,000 observations (tasks 3–30 for each participant) for which all eight similarity measures are defined.

Table 3 indicates participants anchored their preferences to the wines presented in the initial task, favoring wines that were similar to these initial wines in subsequent choices.[9] The coefficient for "Similarity to First Task" was positive and different from zero across all seven models with this predictor, including these six models (posterior means ranging from 0.038 to 0.042, ps from .002 to .014), and the corresponding single-predictor model (see Web Appendix §F.4.1). This finding indicates that once the product options in the initial task established a reference frame, participants' subsequent preferences remained largely invariant and consistent with that initial

degree of practical orthogonality between the resulting 8-dimensional region and varietal representations. The median absolute pairwise cosine similarity between average attribute vectors was 0.233 (IQR: 0.083–0.485), with over half (57.9%) of pairs exhibiting similarity below 0.3 (and 28.9% below 0.1), indicating they capture distinct information.

[9] Full results for single-predictor models are provided in Table W8 in Web Appendix §F.4.1, fixed-effects models (M1–M4) with nuisance control estimates in Table W9 in Web Appendix §F.4.2, and the fully heterogeneous models (M5 and M6) including all fixed and random effects in Table W10 in Web Appendix §F.4.3. We report exact posterior p-values computed from the 4,000 MCMC draws: $p_{\text{exact}} = 2 \times (1 - p_d)$, where p_d is the proportion of posterior draws on the same side as the point estimate; no distributional assumptions are required.

anchor, even though the anchor was incidentally encountered.

In contrast, the coefficients for similarity to other candidate anchors (e.g., wines shown in the previous, next, or final tasks) and the additional similarity terms were indistinguishable from zero across all models. For "Similarity to Previous Task," the 95% credible interval includes zero, which does not support recency accounts. The same holds for "Similarity to Next Task" and "Similarity to Final Task," indicating that the observed effect of the initial (incidental) anchor is not an artifact of revealed preferences.

Model fit comparisons using the Leave-One-Out Information Criterion (LOOIC) reveal that accounting for individual differences improves explanatory power. Models 5 and 6, which incorporate participant-level heterogeneity, both outperform their fixed-effects counterparts (Models 3 and 4), with Model 6 achieving the best fit overall (LOOIC = 38,293 vs. 38,320 for Model 5).

We also estimated models incorporating the position of each task in the sequence and its interaction with the similarity to the options in the first task. The interaction term's 95% credible interval included zero, suggesting that the anchoring effect does not diminish over the experiment, ruling out fatigue as an alternative explanation. Such temporal stability is consistent with comparison-based accounts of anchoring, in which the accessibility changes a comparison induces are long-lasting rather than transient (Mussweiler 2003). These results are detailed in Web Appendix §F.4.4.

We decomposed our similarity-to-first-task measure into two components: similarity to the wine that the participant (i) *chose* in the first task ($\text{Sim}^{\text{Chosen First}}$) and (ii) did *not* choose ($\text{Sim}^{\text{Unchosen First}}$), re-estimating each of the six model specifications with the two components in place of $\text{Sim}^{\text{First}}$. If the anchoring effect reflected stable preferences, participants should prefer wines resembling their initial choice, and $\text{Sim}^{\text{Chosen First}}$ should drive the effect. Instead, across all six specifications, only similarity to the *unchosen* wine was significant ($\hat{\beta}$ = 0.072 to 0.085, all $p < .001$); similarity to the chosen wine was not ($\hat{\beta}$ = -0.002 to 0.006, all $p > .76$), and the difference between the two components was itself significant in every specification ($p \leq .019$). Full results are reported in Web Appendix §F.4.5.

Table 3: Key Results: Wine Study

Effect	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Intercept	0.084	0.084	0.083	0.083	0.089	0.089
	(0.012)	(0.012)	(0.012)	(0.012)	(0.012)	(0.013)
Similarity to First Task	0.039	0.041	0.038	0.039	0.038	0.042
	(0.014)	(0.014)	(0.014)	(0.014)	(0.015)	(0.016)
Similarity to Previous Task	0.017	0.017	0.015	0.016	0.015	0.016
	(0.014)	(0.014)	(0.013)	(0.014)	(0.015)	(0.015)
Similarity to Next Task	-0.007	-0.007	-0.009	-0.009	-0.010	-0.009
	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)
Similarity to Final Task	-0.003	-0.003	-0.005	-0.004	-0.004	-0.002
	(0.014)	(0.014)	(0.014)	(0.014)	(0.015)	(0.015)
Similarity to Second Task	—	0.001	—	-0.001	—	-0.003
	—	(0.014)	—	(0.014)	—	(0.016)
Similarity to Two Tasks Before	—	-0.011	—	-0.012	—	-0.011
	—	(0.014)	—	(0.014)	—	(0.015)
Similarity to Two Tasks After	—	-0.009	—	-0.009	—	-0.013
	—	(0.014)	—	(0.013)	—	(0.015)
Similarity to Second-to-Final Task	—	0.013	—	0.012	—	0.010
	—	(0.014)	—	(0.014)	—	(0.016)
LOOIC	38766	38772	38694	38700	38320	38293
Nuisance Controls Included	No	No	Yes	Yes	Yes	Yes
Full Heterogeneity Included?	No	No	No	No	Yes	Yes

Note. Fixed-effects estimates (posterior means). Standard errors reported in parentheses. Dashes (—) indicate the variable is not included. Nuisance control fixed-effect estimates and heterogeneity estimates (standard deviations of participant-level random effects for M5 and M6) suppressed for brevity and reported in Web Appendix §F.4.

The conceptual account underlying our framework suggests that comparisons are directional: a target is evaluated against a standard, and the standard organizes the judgment (Mussweiler 2003). The asymmetry is consistent with the possibility that the wine a participant came to favor in the first task served as the target, whereas the unchosen wine served as the comparison standard, carrying the initial reference frame into later tasks.

Comparing models with and without nuisance controls (e.g., Model 1 vs. Model 3) shows that adding these controls does not substantially shift the anchoring coefficient (0.039 vs. 0.038). This stability reflects a key feature of BRIDGE's architecture: because the similarity measures are computed from representations that are already partitioned to isolate attribute signal from nuisance variation, the nuisance controls are largely redundant. In contrast, when similarity is computed from unpartitioned full embeddings—which conflate focal attributes with non-focal variation (e.g., writing style)—the nuisance controls become essential for detecting the effect (see Benchmark Comparisons in Web Appendix §F.5). Moreover, as the Monte Carlo simulations and coffee certification experiments demonstrate, BRIDGE's nuisance controls provide meaningful corrections when confounding variations are present. Thus, BRIDGE presents a two-layer safeguard for inference—on the one hand, its attribute representations are orthogonal to the nuisance variation and enable the specification of lower-dimensional and tractable measurement strategies; on the other hand, it develops nuisance controls that adjust for unobserved textual confounding.

As noted previously, the wine names often display the focal attributes (64.7% show the varietal, 26.4% the region), raising the question of whether the anchoring effect depends on the attributes being visible in the names. Restricting the sample to cue-free tasks would discard the most informative variation and leave as few as 115 observations. Instead, we retain the full sample and interact the anchoring term with four indicators for whether the varietal or region was visible in the first (anchor) task and in the current task. As the options shown in each task are randomized, the four indicators are nearly orthogonal and each interaction is separately identified. The average anchoring slope remains 0.035–0.039 (exact $p = .005–.020$), no interaction is significant in any of Models 1–6, and a joint test cannot reject that all four interactions are simultaneously zero ($p = .49–.76$); the only term approaching significance is a *positive* coefficient on the current task's varietal visibility ($p = .07–.08$ in Models 5 and 6); full results are reported in Web Appendix §F.7. The anchoring effect thus does not depend on whether the structured attributes happen to appear in the free-form names.

Robustness Check: Perturbation Analysis Results

We applied the perturbation analysis technique described previously. OpenAI's GPT-4.1 was prompted to slightly expand each description while maintaining tone, style, and core attributes (e.g., region, varietal). This process created a parallel dataset where descriptions were subtly elaborated upon without altering their fundamental meaning or factual content, allowing us to test BRIDGE's resilience to nuanced textual variations.

We conducted two evaluations to confirm the representations and key findings. First, we compared the numerical attribute representations derived by BRIDGE from the original and perturbed wine descriptions using the RV coefficient. Second, we reconstructed the variables for the anchoring analysis using the perturbed embeddings and re-estimated Models 1–4.

The RV coefficient analysis confirmed the robustness of BRIDGE's extracted representations to these perturbations. The RV coefficients comparing the original and perturbed representations were 0.971 for region representations and 0.984 for varietal representations, indicating a high degree of alignment. To assess whether these observed similarities were statistically significant, we conducted permutation tests comparing the observed RV coefficients to those expected under a null hypothesis of no systematic alignment. The resulting p-values were extremely low (all ps $< .001$ for both region and varietal representations), indicating that the alignment between original and perturbed representations was highly unlikely to be due to chance. These findings suggest that BRIDGE effectively treated the perturbations as nuisance variables, maintaining stable representations of the focal constructs of region and varietal.

The re-estimated statistical models using the perturbed embeddings also yielded results consistent with those obtained from the original data. As shown in Table 4, the similarity to the first task (i.e., the anchoring effect) remained significant across all models, with estimates ranging from 0.036 to 0.042 (ps range from .002 to .005). Other similarity measures, such as similarity to the previous or next tasks, remained nonsignificant, consistent with the original findings.

Benchmark Comparisons

To assess whether BRIDGE's neural architecture provides meaningful gains over simpler alternatives, we estimated benchmark models using similarity measures derived from three alternative representation methods: raw embedding similarity, nuisance-only similarity, and linear projection. All benchmarks used the same underlying embeddings (OpenAI text-embedding-3-large, 3,072 dimensions).

Table 4: Results: Perturbation Analyses of Wine Study

Effect	**Model 1**	**Model 2**	**Model 3**	**Model 4**
Intercept	0.082	0.083	0.083	0.084
	(0.011)	(0.012)	(0.011)	(0.012)
Similarity to First Task	0.036	0.041	0.036	0.042
	(0.013)	(0.014)	(0.013)	(0.013)
Similarity to Previous Task	0.012	0.017	0.012	0.018
	(0.013)	(0.014)	(0.013)	(0.014)
Similarity to Next Task	-0.004	-0.007	-0.005	-0.008
	(0.013)	(0.014)	(0.013)	(0.014)
Similarity to Final Task	-0.008	-0.008	-0.009	-0.008
	(0.013)	(0.014)	(0.013)	(0.014)
Similarity to Second Task	—	0.000	—	0.001
	—	(0.014)	—	(0.014)
Similarity to Two Tasks Before	—	-0.014	—	-0.013
	—	(0.014)	—	(0.014)
Similarity to Two Tasks After	—	-0.006	—	-0.006
	—	(0.014)	—	(0.014)
Similarity to Second-to-Final Task	—	0.018	—	0.018
	—	(0.014)	—	(0.014)
Nuisance Controls Included	No	No	Yes	Yes

Note. Fixed-effects estimates (posterior means). Standard errors reported in parentheses. Dashes (—) indicate the variable is not included. Nuisance control fixed-effect estimates suppressed for brevity.

Raw Embedding Similarity. Computing cosine similarity directly in the full 3,072-dimensional embedding space yields a significant anchoring coefficient when nuisance controls are included ($\hat{\beta} = 0.215$, 95% CI [0.030, 0.398]). However, without nuisance controls, the same full-embedding similarity measure fails to reach significance ($\hat{\beta} = 0.177$, 95% CI [-0.005, 0.354]). This contrast is meaningful: recall that adding nuisance controls to the BRIDGE-derived similarity measures barely shifts the anchoring estimate (0.039 vs. 0.038). As noted above, BRIDGE's representations isolate attribute signal by construction; unpartitioned embeddings conflate attributes with stylistic variation, so the nuisance controls become essential for recovering the effect. In both cases, the credible intervals remain substantially wider than BRIDGE's (0.368 and 0.359 vs. 0.059), reflecting the noise introduced by irrelevant dimensions.

Nuisance-Only Similarity. Computing similarity using only the nuisance dimensions—which capture description style, tone, and length orthogonal to the focal attributes—yields a null result ($\hat{\beta} = -0.007$, 95% CI [-0.030, 0.015]). This confirms that the anchoring effect is driven by attribute similarity, not by superficial textual resemblance between descriptions.

Linear Projection. As an alternative to BRIDGE's neural architecture, we estimated a linear projection baseline using Ridge regression to predict the embeddings from the attributes, followed by SVD on the residuals. This approach produced a marginal result ($\hat{\beta} = 0.135$, 95% CI [-0.000, 0.269], $p = .051$), with a credible interval approximately 5 times wider than BRIDGE's. While the linear projection recovers the correct sign, its substantially wider credible interval reflects greater noise in separating attribute signal from nuisance variation. This empirical pattern aligns with the Monte Carlo simulations: with 426 regions and 708 varietals, the high-cardinality attribute space and the rich, diverse language of wine descriptions create precisely the conditions under which BRIDGE's nonlinear architecture provides substantive gains over linear decomposition.

Replacing BRIDGE's nuisance controls with 10 SVD-reduced Empath features does not alter the anchoring estimate (0.040 vs. 0.039 without controls), confirming that the effect is robust to the choice of control specification; full benchmark results, including Empath psycholinguistic controls and attribute-only (Jaccard) similarity comparisons, are detailed in Web Appendix §F.5.

Validation: Replication with Controlled Coffee Stimuli and Conventional Analytical Methods

A study using controlled stimuli and conventional analytical methods replicated the observed product-anchoring effect in a different product category (coffee), providing converging evidence that the effect is not an artifact of BRIDGE. In this study, similarity was defined by attribute overlaps rather than AI-derived representations. Participants again exhibited preferences aligned with the attributes of options presented in their initial task. Full details of the experimental design, variable operationalization, model specifications, and results are provided in Web Appendix §G.

Study Discussion

We examine the dynamics of consumer preferences in sequential decision making for complex, verbally described products. Our findings support the notion that preferences are initially malleable but become defined by early (though arbitrary) product exposures, and remain largely invariant in later choices—thus exhibiting consistency. The results rule out revealed-preference, fatigue, and preference-updating accounts. Perturbation analyses verify that our findings are robust to irrelevant modifications of the stimuli. A follow-up coffee experiment replicates the effect using a conventional, controlled design, establishing the effect as robust and generalizable across product categories and analytical approaches.

General Discussion

We introduce BRIDGE, an experimental design that accommodates many diverse real-world verbal stimuli while exerting statistical control for stimulus variability. At its core is an interpretable AI model that draws on evidence that LLM semantic encodings closely mirror the human brain's processing of meaning (Goldstein et al. 2025) and that transforms unstructured verbal stimuli into structured numerical representations of (1) focal variables and (2) orthogonal statistical controls for non-focal variation. These representations facilitate the analysis of participant responses to these stimuli in various theory-testing paradigms, including within-subjects, between-subjects, and mixed designs, thereby addressing the stimulus-sampling problem and strengthening a study's internal and external validity.

We conduct Monte Carlo simulations to investigate BRIDGE's effectiveness in establishing statistical control. Even when nuisance variation directly shifts preferences (main effect confounding) and moderates the strength of focal attributes (interaction effect confounding)—the two channels by which nuisance variation can confound experimental measurement—BRIDGE accurately and precisely recovers structural parameters, including both treatment effects and interaction effects. It matches an infeasible Oracle estimator, which is efficient (and hence ideal) but impracticable as it relies on information that is typically unavailable to the researcher (e.g., full confound structure).

Yet BRIDGE remains feasible, as it develops both nuisance controls and learned attribute representations using only information that is observed and available to the researcher. In contrast, traditional approaches—baseline models with no controls, and controls-only approaches based on Empath lexical variables or linear projection—provide biased inference.

In coffee certification experiments (with human participants) where a stimulus confound is deliberately introduced, BRIDGE—operating without explicit knowledge of this confound—recovers treatment effect estimates comparable to a stimulus-controlled, "unconfounded" baseline, when a standard covariate adjustment is inadequate. These experiments show that BRIDGE can be applied to conventional experimental designs with a small number of carefully constructed stimuli, as it safeguards against remaining nuisance variation.

To demonstrate BRIDGE with many diverse real-world stimuli in a large-scale study, we investigate preference dynamics in wine, where 1,000 consumers each evaluate 32 pairs randomly drawn from a corpus of nearly 120,000 real-world tasting notes. Across participant choices among approximately 50,000 unique wines, the results reveal that preferences for verbally described products are initially malleable. Despite being entirely incidental, options presented at the outset become product anchors that shape later decisions. A controlled anchoring study replicates the effect. The results support the notion that preference dynamics are affected by the selective accessibility of knowledge about key attributes (e.g., region and varietal) as shaped by these anchors.

Contributions

We contribute to consumer behavior research in four ways. First, BRIDGE addresses the stimulus-sampling problem. To mitigate this problem, scholars recommend enlarging the stimulus sample (Westfall, Judd, and Kenny 2015) and treating it as a random factor, viewing each stimulus as one of many possible items sampled from a larger population (e.g., Baribault et al. 2018; Judd, Westfall, and Kenny 2012). BRIDGE helps implement those recommendations. While traditional experiments rely on orthogonal, factorial designs for causal inference, it combines large-scale randomization, continuous attribute representation, and statistical nuisance controls to minimize the likelihood that the observed effects are driven by extraneous features specific to a particular stimulus—a common concern in conventional experiments that rely on a small set of carefully pretested stimuli (Pham 2013; Simonsohn, Montealegre, and Evangelidis 2025)—improving internal validity. As stimuli are chosen randomly and presented in their original form, the observed effects are more likely to generalize beyond the sampled stimuli, improving external validity. Furthermore, existing methods of manual coding are often impractical at scale. In our wine experiment, for instance, manually coding the relationships between randomly sampled wines would require computing approximately 2.5 billion distances. BRIDGE efficiently computes these distances, accounting for nuisance variables. Finally, presenting each participant with a distinct, randomly sampled subset of stimuli leverages the between-item variance highlighted by Judd, Westfall, and Kenny (2012), thereby increasing statistical power without increasing the participant sample size.

Second, BRIDGE offers an alternative experimental design for studying consumer behavior in information-rich environments where products are commonly conveyed in prose in the real world, such as in hedonic consumption, complex financial products, and sustainability initiatives. In these contexts, as Alba and Williams (2013) observe of hedonic consumption, "consumer researchers have been inclined to frame the issue narrowly, in part because many integral characteristics … can be devilishly difficult to investigate via traditional experimental paradigms" (p. 3). BRIDGE makes such contexts experimentally tractable in both large-scale designs with unaltered real-world descriptions (as exemplified by our wine study) and conventional designs with controlled stimuli (as exemplified by our coffee certification experiments). Therefore, it can be used to resolve conflicting theoretical predictions arising from discrepancies in product presentations between simpler, stylized stimuli and detailed, real-world stimuli.

Third, BRIDGE enhances research objectivity and reproducibility. Traditional studies can involve numerous researcher decisions regarding (1) experimental design, such as stimulus simplification and selection, and (2) data analysis. These researcher degrees of freedom can inadvertently compromise reproducibility. Our framework allows for the inclusion of real-world stimuli without abbreviation or stylization, broad sampling to reflect ecological distributions and achieve market coverage, and a structured, end-to-end methodological pipeline that can be pre-specified and preregistered. These features can help mitigate the "garden of forking paths" problem (Gelman and Loken 2013), enhancing replicability and the overall credibility of the findings.

Fourth, our research builds on recent evidence in cognitive psychology and neuroscience demonstrating that semantic embeddings align closely with the human brain's hierarchical codes for speech and meaning (Goldstein et al. 2025), and can serve as a proxy for consumer conceptual knowledge in downstream tasks, such as food-health judgments (Gandhi et al. 2022). However, these embeddings are inherently unstructured and high-dimensional, comprising thousands of unlabeled dimensions, making it unclear which specific constructs or semantic features align with consumer knowledge. BRIDGE transforms these embeddings into structured, lower-dimensional, and interpretable representations—adaptable to specific contexts and dataset features through automatic tuning (see Web Appendix §F.2)—for use in theory testing. Our Monte Carlo simulations establish that BRIDGE's nonlinear architecture is not merely a design choice but a methodological necessity; simpler linear decompositions of the same embeddings provide an incomplete account of the confounding factors, while BRIDGE recovers the full effect structure.

Limitations, Extensions, and Directions for Future Research

Expanding research contexts

Future research can use BRIDGE to study other text-rich consumer contexts. For example, researchers could examine what makes online consumer reviews more useful by (a) collecting a corpus of reviews, (b) using BRIDGE to identify the underlying dimensions that affect review usefulness, and (c) validating the impact of these dimensions. To further verify these findings, researchers could develop controlled manipulations of the identified dimensions for use in a traditional laboratory study. This parallels our overall empirical strategy for assessing product anchoring with the BRIDGE-based wine experiment and its accompanying coffee validation study.

Consumer contexts such as insurance and financial products often involve rich descriptions spanning numerous attributes and remain challenging to study using conventional methods. Given that BRIDGE can handle large, diverse product spaces (with high-cardinality categorical variables) through continuous representation, it offers a unique opportunity to examine how consumers navigate multiple dimensions in complex decision-making environments.

Our investigations focused on textual stimuli because they are prevalent, and embedding-based measurement is currently more developed in text settings. However, as multimodal embedding models mature, BRIDGE can be extended to handle other forms of unstructured stimuli, such as images, audio, video, and even virtual and augmented reality (e.g., Chang, Mukherjee, and Chattopadhyay 2023).

Methodological extensions

A promising direction is to use BRIDGE to inductively uncover latent dimensions from unstructured texts. To test preference dynamics in our wine study, we chose a context in which we know which attributes are likely important to consumers; this allowed us to deductively transform the unstructured tasting notes into a theory-based attribute representation. Future research could compare how theory-driven versus data-driven dimensions shape preference dynamics.

Researchers can also expand upon the stimulus sampling strategies employed in our studies. While large-scale random sampling ensures sufficient variation across the attribute space and approximates balance across attribute levels, typical sampling methods for participants (e.g., simple random, stratified, or cluster sampling) can be used for stimulus selection. Some may be better suited to specific research questions—for instance, stratified sampling by wine provenance to ensure balanced groupings when studying Old World versus New World wines. The stimulus sampling strategy is among the design choices the researcher's guide recommends pre-specifying and preregistering (Web Appendix §B).

Generative AI can create descriptions when real-world stimuli are limited, such as in emerging or novel product categories[10], or when study objectives require highly controlled descriptions that

[10]We leveraged this capability in our simulations and in our coffee certification studies, and illustrate it in our researcher's guide (see Web Appendix §B).

mimic real products (e.g., the coffee descriptions in our certification and anchoring experiments). In these cases, researchers could adopt a human-in-the-loop framework, where humans guide the AI in generating stimuli or assist in selecting from AI-generated options (e.g., using platforms like MTurk). This approach enables researchers to craft realistic stimuli with ease and at a low cost, further broadening the applicability of BRIDGE-based experimental designs.

Managerial Implications

The use of real-world stimuli enhances the potential for counterfactual analysis, as inference is drawn relative to actual products rather than simplified or stylized abstractions. For example, our wine study estimates the preference structure of 1,000 consumers in the US, Australia, and New Zealand across a catalog of 119,955 wines. This detailed preference mapping could be used to predict sales and market share under different product assortments or pairings, providing a foundation for decision tools to optimize marketing strategies.

As AI evolves, we hope BRIDGE enables researchers to design experiments with naturalistic stimuli that parallel how products are presented in the real-world marketplace, advancing the study of consumer behavior in information-rich environments.

WEB APPENDIX

Web Appendix A: Introduction and Roadmap

This web appendix accompanies the manuscript, *Behavioral Research Through Interpretable, Dimensionality-reduced Generative AI Embeddings (BRIDGE): A Method to Incorporate Real-World Stimuli in Consumer Experiments*. It is organized into six sections: (§B) a practical guide to applying the methodology, (§C) perturbation analyses as a validation tool, (§D) simulation studies evaluating BRIDGE's robustness and applicability, (§E) supplementary materials for the coffee certification experiments, (§F) supplementary details and analyses for the wine choice experiment, and (§G) a validation study using controlled stimuli.

§B offers a practical guide for implementing the methodology using the accompanying `bridge` Python package. It describes a five-step workflow: (1) obtaining product descriptions, (2) collecting participant data, (3) converting descriptions to numerical representations using large language models (LLMs), (4) refining these high-dimensional encodings into low-dimensional interpretable representations and statistical controls using BRIDGE, and (5) analyzing participant data with the resulting representations. The section includes working code examples based on the wine study in the manuscript, demonstrates both API-based (OpenAI) and local (Gemma) embedding options, and shows how to adapt the methodology to new product domains. For applications with few base descriptions, it also illustrates LLM-based augmentation. The section concludes with a preregistration template, guidance for validating BRIDGE in a new application, and a description of the accompanying reproducibility bundle, which reproduces the Coffee Certification results (Table 2 in the manuscript) from pre-computed artifacts with a single command and supports step-by-step verification of the full pipeline.

§C introduces perturbation analyses, a validation tool for assessing whether BRIDGE robustly isolates and preserves focal constructs across diverse applications. These analyses systematically modify non-focal textual elements in product descriptions—such as tone, style, verbosity, or extraneous details—while keeping focal constructs unchanged. The modified descriptions are then processed through the same pipeline as the original data: re-encoded using the same embedding method and transformed by the trained BRIDGE model to produce new focal-construct representations (e.g., attribute representations) and nuisance controls. Representations derived from the perturbed text are compared to those from the original text to evaluate stability. The final statistical analysis is also replicated using the perturbed representations, and the resulting inferences (e.g., coefficient estimates) are compared to those from the original representations. If BRIDGE's representations and inferences remain stable despite these controlled modifications, researchers can be confident that observed effects reflect focal constructs rather than non-focal textual artifacts. To facilitate implementation, we provide a concise Python workflow showing how to create perturbed descriptions, obtain their representations, compare them, and replicate the final analysis.

§D presents a Monte Carlo simulation study that evaluates BRIDGE's inferential performance under a unified confounding regime in which unobserved description style both shifts baseline utility (main effect confounding) and moderates the effective strength of a focal attribute (interaction effect confounding). Using a hypothetical study of environmental spillover effects on preferences for organic versus non-organic packaged salads, we introduce unobserved stylistic variations (Factual, Engaging, Creative) that correlate with the treatment condition. BRIDGE's nuisance controls and attribute representations, interacted with the organic indicator, recover all structural parameters (including style-specific organic effects), matching the infeasible Oracle. We benchmark BRIDGE against four alternatives: (1) a model without controls, (2) an infeasible Oracle with style dummies, (3) Empath-derived lexical controls, and (4) a Linear Projection benchmark using Ridge regression plus SVD. This section provides the full programming framework and the complete results (Table W1, Figure W2); the manuscript presents summary estimates (Table 1).

§E describes the estimation approach for the coffee certification experiments. It includes the fair trade and organic coffee product descriptions used as stimuli, the experimental design, and the model specifications, with separate estimations for each certification type. Results are presented in the manuscript.

§F provides supplementary materials and analyses for the wine choice experiment. It offers verbatim examples of the real-world wine tasting notes used as stimuli, details the process of using Optuna to determine the BRIDGE neural network architecture, and reports classification metrics quantifying how well BRIDGE recovers focal attribute distinctions. The section then presents a sensitivity analysis examining how training data size affects representation quality, as well as quantitative model fit and sequence analyses (including LOOIC comparisons) to support the anchoring model specification. Finally, it decomposes the anchoring effect into chosen and unchosen components to clarify the underlying mechanism, and concludes with benchmark comparisons evaluating BRIDGE against alternative similarity measures (e.g., raw embeddings, Jaccard matching, linear projection, and Empath-based psycholinguistic controls), an embedding-backend sensitivity analysis (BERT vs. OpenAI inputs), and name-cue robustness analyses.

§G presents validation evidence for the anchoring effect from a controlled experiment using coffee descriptions as stimuli. Unlike the primary wine study, which used real-world stimuli, this experiment used carefully constructed product descriptions with minimal extraneous variation, enabling conventional analysis without BRIDGE's nuisance controls. We created 36 coffee profiles by crossing two attributes (aroma and taste) with six levels each, yielding standardized single-sentence descriptions with comparable length and consistent tone and structure. This design allowed direct measurement of attribute similarity by counting matching attributes between options. The results confirm that initial exposure significantly anchors subsequent preferences—the same effect observed in the main wine study. Controls for recency effects and placebo tests (calculating similarity to options in future tasks that participants had not yet seen) rule out alternative explanations such as preference stability. This validation demonstrates that the anchoring effects observed in the wine study (using real-world, richly varied product descriptions as stimuli) are not artifacts of the experimental design or the analytical approach (BRIDGE) but reflect a genuine behavioral phenomenon.

Accompanying code and data are openly available at **https://doi.org/10.17605/OSF.IO/5D6KX**. The repository carries the code and data for all studies reported in the paper. We also provide the `bridge` Python package, which implements the full BRIDGE pipeline, and a reproducibility bundle for the coffee certification experiments, both under an open-source license. The bundle packages the pipeline code, experiment data, and pre-computed outputs at every stage. Researchers can use it to reproduce all reported Coffee Certification results (Table 2 in the manuscript) with a single command. The `bridge` package is modular, making it straightforward to modify key components (such as product descriptions, experimental conditions, or statistical models) without extensive reconfiguration. It relies solely on standard Python packages, ensuring compatibility across a wide range of environments without specialized hardware or software.

Web Appendix B: A Researcher's Guide to Implementing BRIDGE

BRIDGE targets contexts in which real-world option descriptions are typically rich and complex. Common examples include hedonic experiences, complex financial products, and sustainability initiatives. In such contexts, traditional experimental designs often rely on a few, simplified stimuli that abstract product descriptions to key attributes (e.g., quality rating or brand) to isolate specific variables of interest and ensure stimulus comparability. While this approach enhances experimental control, it diminishes mundane realism, reducing relevance to actual marketplace settings and limiting both internal (through stimulus sampling) and external validity. BRIDGE addresses these concerns by incorporating diverse, real-world stimuli into experimental designs, mirroring the information environments consumers encounter on e-commerce platforms and in retail stores.

This guide demonstrates how to apply the methodology using the accompanying `bridge` Python package. The package automates the full pipeline—from embedding generation through neural network training to representation extraction—without requiring researchers to write neural network code. The code examples below draw on the wine study in the manuscript, which uses nearly 120,000 real-world tasting notes and models two focal attributes: region (426 unique wine-growing regions) and varietal (708 grape varieties). For smaller-scale applications, such as the coffee certification experiments, the package includes an augmentation module that creates additional description variants from a small number of base stimuli.

B.1 Research Methodology

We propose a five-step methodology: (1) obtain product descriptions; (2) conduct an experiment to collect participant responses; (3) convert descriptions to numerical representations using an LLM; (4) refine the high-dimensional encodings into interpretable representations using BRIDGE; and (5) analyze the data using these representations to draw substantive conclusions. Figure W1 depicts this workflow, though the sequence of steps is flexible. For example, one may obtain product descriptions (Step 1) and transform them using BRIDGE (Steps 3 and 4) before collecting participant data (Step 2) and conducting the participant data analysis (Step 5).

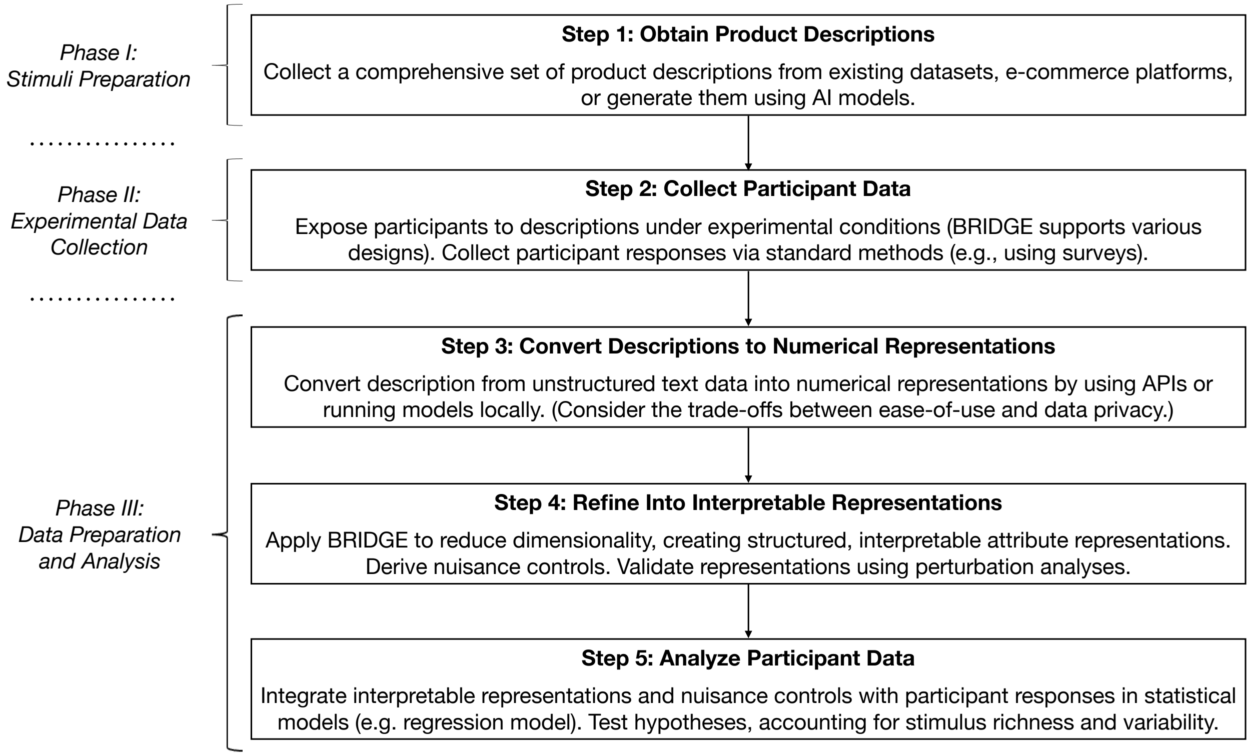


Figure W1: General Procedure of a Study Using the BRIDGE Model

B.1.1 Step 1: Obtain Product Descriptions

Researchers can source product descriptions from existing datasets in digital libraries such as the UCI Machine Learning Repository, Kaggle, or other academic databases. Alternatively, custom data can be collected from e-commerce platforms or manufacturers' websites, though this approach requires more time and expertise.[1]

For novel or emerging product categories, pre-existing descriptions may be unavailable, requiring researchers to create custom descriptions. One approach is to employ research assistants to develop product descriptions tailored to specific categories and research questions (e.g., descriptions emphasizing key attributes relevant to the study). However, this approach is labor-intensive

[1]Researchers must comply with platforms' terms of service, maintain rigorous privacy protocols, and uphold ethical standards. This includes adhering to legal requirements and safeguarding personally identifiable information in accordance with applicable regulations (e.g., GDPR in Europe). Researchers should secure necessary approvals (e.g., IRB), practice transparent data collection and responsible data management, and report data usage clearly.

and costly. Alternatively, generative AI models (e.g., OpenAI's GPT family or Meta's open-source models) can automatically generate product descriptions. This method is faster and more cost-effective, though description quality may vary. A hybrid human-in-the-loop approach can combine the precision of human input with the scalability of AI generation, balancing quality and efficiency.

Crucially, to train the BRIDGE model (Step 4), the dataset must include not only the unstructured descriptions but also the known values (labels) for each focal attribute (e.g., region and varietal in the wine study, or treatment condition and flavor profile in the coffee experiments). The quality and quantity of this labeled data directly affect the resulting representations and nuisance controls.

The exact amount of data required depends on several factors—the number of attributes and levels, the subtlety of linguistic differences between levels, and the desired precision—but some general guidelines apply. As a rule of thumb, researchers should collect at least 10–20 diverse examples for *each level* of *each* focal attribute, giving the model enough data to learn reliable patterns. For instance, if studying the effect of TV screen size, fewer examples might suffice if the size (e.g., 32, 55, 75 inches) is explicitly stated numerically in descriptions. In contrast, if screen size is described categorically (e.g., Small/Medium/Large) using varied language, more examples per level are needed to capture these linguistic nuances. The dataset should also reflect the diversity of language used to describe each attribute level in practice.

Ultimately, dataset adequacy is best assessed during model training (Step 4) by monitoring performance on a held-out validation set; poor validation performance may indicate insufficient labeled data. For empirical guidance, Web Appendix §F.3 presents a sensitivity analysis of training data size and representation quality in our wine study. For context, our wine study used descriptions from nearly 120,000 distinct wines, averaging roughly 280 examples per region and 170 per varietal across 426 regions and 708 varietals.

B.1.2 Step 2: Collect Participant Data

Our methodology accommodates within-subjects, between-subjects, and mixed designs, and it integrates into standard data collection procedures in consumer research. Standard procedures—participant recruitment, random assignment, questionnaire design, and incentive structures—are well documented in the literature and not elaborated here. We recommend following established best practices while making context-specific adjustments.

B.1.3 Step 3: Convert Descriptions to Numerical Representations

Data from conventional experimental designs that use a few simplified stimuli can be analyzed using traditional methods such as ANOVA. In contrast, when participants are exposed to many distinct, real-world descriptions, data analysis requires interpretable attribute representations and nuisance controls. To generate these representations and controls, BRIDGE leverages LLMs to convert the unstructured product descriptions into high-dimensional numerical encodings.

Two primary methods are available: using application programming interfaces (APIs) or running models locally. APIs, such as those provided by OpenAI (e.g., `text-embedding-3-large`), Google, or Microsoft, allow researchers to send product descriptions to external services, which process the input and return numerical encodings. Although convenient—the provider handles hosting, scaling, and updates—APIs raise data privacy concerns. In addition, APIs charge usage fees, typically based on token volume. Current pricing for leading models like `text-embedding-3-large` is approximately $0.0001 per 1,000 tokens, making it affordable for typical experimental datasets—for instance, embedding the ~120,000 descriptions in our wine study cost approximately $2.

By contrast, running models locally (e.g., Gemma or `NV-Embed-v2`) requires installing and operating the model on the researcher's own computing infrastructure. This approach grants full control over data, enhances privacy, ensures model version stability, and eliminates recurring API costs. However, it demands greater technical expertise, potentially high-performance hardware (e.g., GPUs), and more setup time.

B.1.4 Step 4: Refine Into Interpretable Representations

The high-dimensional encodings generated in Step 3 capture rich semantic information but lack direct interpretability for hypothesis testing. To make these encodings interpretable and suitable for statistical analysis, we transform them using BRIDGE. The `bridge` package automates this process; the underlying steps are as follows. The BRIDGE network first maps the high-dimensional encoding onto a lower-dimensional intermediate representation. This intermediate representation then feeds into a partitioned sub-network, where separate branches generate interpretable, low-dimensional representations for each focal attribute (e.g., region and varietal in the wine study).

Generating statistical controls for nuisance variation—that is, variation unrelated to the focal attributes—involves projecting the original LLM embedding onto the orthogonal complement of the subspace spanned by the attribute representations. This ensures the resulting nuisance controls are orthogonal to—and thus uncorrelated with—the attribute representations.

Although libraries like PyTorch can leverage GPUs to accelerate training, the feedforward networks in BRIDGE run reasonably efficiently on standard CPUs. For context, the entire wine study analysis—data processing, embedding generation, BRIDGE training with hyperparameter optimization, and Bayesian modeling—ran on a Mac Studio with an M2 Ultra chip in approximately 3 hours. The BRIDGE training itself took less than 30 minutes, with the Bayesian analysis accounting for the majority of the runtime. This suggests that the core BRIDGE training step is computationally feasible for many researchers, even without dedicated high-performance computing resources.

Validating these representations and controls is essential. This may involve statistical comparisons with coded attributes, or qualitative checks that the representations capture the intended constructs and the nuisance controls isolate irrelevant variation. Additionally, perturbation analyses (§C) provide a robust method for evaluating the stability and fidelity of the representations. We discuss how to interpret these checks — and what they do and do not establish for a new application — under *Validating BRIDGE in a New Application* below.

B.1.5 Step 5: Analyze Participant Data

Researchers can analyze the collected participant data using the interpretable numerical representations and statistical controls generated in the prior steps. For instance, they can use regression models with interaction terms to examine how product attributes interact with experimental conditions, and hierarchical models to account for participant-level variability.

B.2 Python Code

The following code demonstrates the complete BRIDGE workflow using the `bridge` Python package. The code uses the wine tasting notes from the manuscript, modeling two focal attributes: region and varietal. However, for simplicity, the example considers a hypothetical between-subjects question similar to the coffee certification studies—whether mentioning "organic" in a wine description affects consumer preference—rather than the full sequential-choice design reported in the paper.

B.2.1 Installation and Setup

```

################################################################################
# BRIDGE Researcher's Guide - Setup
################################################################################
# Install the bridge package from the project source directory:
#   pip install -e "/path/to/bridge/source"
#
# The package requires: torch, numpy, pandas, scikit-learn
# (plus optuna for hyperparameter tuning)
# For OpenAI embeddings: openai  |  For local embeddings: sentence-transformers

import pandas as pd
import numpy as np
from bridge import BRIDGEPipeline, BRIDGEConfig

# Load the product description dataset
# Must contain: (1) unstructured descriptions, (2) structured attribute labels
df = pd.read_csv("wine_descriptions.csv")

print(f"Loaded {len(df)} wine descriptions")
```

```python
print(f"Unique regions: {df['region'].nunique()}")           # 426 regions
print(f"Unique varietals: {df['variety'].nunique()}")          # 708 varietals

```

B.2.2 Steps 3–4: Embedding Generation and BRIDGE Training

BRIDGE's pipeline automates the conversion of descriptions to numerical representations (Step 3) and the refinement into interpretable representations with nuisance controls (Step 4). The `fit()` method runs the complete pipeline: it generates semantic embeddings, trains the BRIDGE neural network with Bayesian hyperparameter optimization via Optuna (Akiba et al. 2019), extracts attribute representations, and computes orthogonal nuisance controls.

```python

################################################################################
# BRIDGE Researcher's Guide - Steps 3 and 4: Embed, Train, Extract
################################################################################

# Initialize the BRIDGE pipeline
# - attributes: names of the focal variables to model
# - config: hyperparameters (defaults use OpenAI embeddings)
# - output_dir: where to save trained model, representations, and metadata
pipeline = BRIDGEPipeline(
    attributes=["region", "varietal"],
    config=BRIDGEConfig(),                # OpenAI text-embedding-3-large (3072-dim)
    output_dir="./bridge_output"
)

# Alternative: use local Gemma embeddings (768-dim, free, privacy-preserving)
# pipeline = BRIDGEPipeline(
#     attributes=["region", "varietal"],
#     config=BRIDGEConfig.for_gemma(),    # Local embeddings, no API key needed
#     output_dir="./bridge_output"
# )

# Fit the pipeline: embed descriptions, train neural network, extract
# - descriptions: the unstructured text to process
# - labels: dict mapping each attribute name to its structured label array
# - tune: if True, runs Bayesian hyperparameter search (Optuna TPE)
# - n_trials: number of optimization trials (75 recommended for publication)
pipeline.fit(
    descriptions=df["description"],
```

```
    labels={
        "region": df["region"],                # e.g., "Napa Valley"
        "varietal": df["variety"],              # e.g., "Pinot Noir"
    },
    tune=True,
    n_trials=75,
)

# Extract representations (returns a dict of NumPy arrays)
representations = pipeline.transform()
# representations["region"]:    shape (N, d_r) -- interpretable region representations
# representations["varietal"]:  shape (N, d_v) -- interpretable varietal representations
# representations["nuisance"]:  shape (N, d_n) -- orthogonal nuisance controls
# Dimensions d_r, d_v are determined by hyperparameter optimization; d_n
# defaults to 5 (an SVD elbow plot is printed to guide adjustment via
# num_nuisance_dims)

# Export .npy files for downstream analysis (Python or R)
pipeline.export()
# Creates: bridge_output/representations/region_embedding.npy
#          bridge_output/representations/varietal_embedding.npy
#          bridge_output/representations/nuisance_embedding.npy
#          bridge_output/metadata/classification_metrics.json
#          (+ encoder/hyperparameter/config/manifest JSONs; the trained model
#          bridge_output/model/bridge_model.pt is saved earlier, by fit())

```

The `BRIDGEPipeline` class provides several configuration options:

- **Embedding backend**: BRIDGE defaults to OpenAI's `text-embedding-3-large` (3,072 dimensions). For privacy-sensitive applications or to avoid API costs, use `BRIDGEConfig.for_gemma()` for local Gemma embeddings (768 dimensions).
- **Pre-computed embeddings**: If embeddings are already available, pass them via the `embeddings` parameter in `fit()`, skipping the embedding generation step entirely.
- **Cached models**: After initial training, the model is saved to disk, and subsequent calls to `fit()` automatically load the cached model for reproducible analysis without retraining. Set `use_cached_model=False` to force retraining.

- **Nuisance dimensions**: BRIDGE extracts nuisance dimensions using Singular Value Decomposition (SVD) on the orthogonal complement of the attribute subspace, with an elbow plot to guide dimension selection. The number of dimensions can be customized via `BRIDGEConfig(num_nuisance_dims=k)`.
- **Classification metrics**: After training, `fit()` automatically computes and prints classification accuracy for each attribute, providing a quality check on the learned representations.

B.2.3 Understanding the Outputs

The `transform()` method returns a dictionary with one entry per attribute plus a `"nuisance"` key:

- **Attribute representations** (e.g., `representations["region"]`): Dense, low-dimensional vectors that encode the focal construct. Products with similar attributes (e.g., wines from neighboring regions) receive similar representations, which can serve as predictors, as inputs for similarity measures, or for other downstream analyses.
- **Nuisance controls** (`representations["nuisance"]`): Components orthogonal to all attribute representations that capture textual variation unrelated to the focal constructs (e.g., writing style, description length, vocabulary). Including these as covariates in downstream models controls for nuisance variation without distorting the focal attribute signals.

Orthogonality between attribute representations and nuisance controls is guaranteed by construction, because nuisance controls are computed as the projection of the original LLM embedding onto the orthogonal complement of the attribute subspace.

B.2.4 Step 5: Analyzing Participant Data

Once representations are exported, they integrate directly into standard statistical workflows. To illustrate, suppose a researcher wants to study whether mentioning “organic’ ’ in a wine description affects consumer preference, using real tasting notes as stimuli. The BRIDGE nuisance controls absorb stylistic variation across descriptions, helping isolate the treatment effect. Stylistic variation that covaries with the treatment (e.g., if organic wines tend to be described more elaborately) would otherwise confound the estimate; stylistic variation unrelated to the treatment inflates error variance. The nuisance controls address both: they adjust for the text-based confounding component while absorbing idiosyncratic variation. The code below shows how these controls are loaded and merged with experimental data, just as in any standard regression analysis.

```python
################################################################################
# BRIDGE Researcher's Guide - Step 5: Analyze Participant Data (Python)
################################################################################

import numpy as np
import pandas as pd
import statsmodels.formula.api as smf

# Load BRIDGE representations (saved by pipeline.export())
nuisance = np.load("bridge_output/representations/nuisance_embedding.npy")

# Load experimental data (one row per participant-wine observation)
exp_df = pd.read_csv("experimental_data.csv")
# Columns: participant_id, wine_index (0-based index into description dataset),
#          organic (0/1), preference (1-7)

# Add nuisance controls as covariates
for dim in range(nuisance.shape[1]):
    exp_df[f"INTN{dim+1}"] = nuisance[exp_df["wine_index"], dim]

# Estimate treatment effect with nuisance controls
# Formula assumes 5 nuisance dims; adjust INTN terms to match num_nuisance_dims
model = smf.ols(
    "preference ~ organic + INTN1 + INTN2 + INTN3 + INTN4 + INTN5",
    data=exp_df
).fit(cov_type="cluster", cov_kwds={"groups": exp_df["participant_id"]})
print(model.summary())
```

For R users, the `.npy` files can be loaded via the `reticulate` package. Below, we fit an analogous Bayesian hierarchical model with `brms`, adding participant-level random intercepts and slopes:

```r
################################################################################
# BRIDGE Researcher's Guide - Step 5: Analyze Participant Data (R)
################################################################################

library(reticulate)
library(brms)
np <- import("numpy")

# Load nuisance controls (auto-converted to R matrix)
nuisance <- np$load("bridge_output/representations/nuisance_embedding.npy")

# Merge with experimental data
exp_df <- read.csv("experimental_data.csv")
# wine_index is 0-based (see the Python example); add 1 for R's 1-based indexing
for (d in 1:ncol(nuisance)) {
  exp_df[[paste0("INTN", d)]] <- nuisance[exp_df$wine_index + 1, d]
}

# Bayesian hierarchical model with nuisance controls
# Formula assumes 5 nuisance dims; adjust INTN terms to match num_nuisance_dims
model <- brm(
  preference ~ organic + INTN1 + INTN2 + INTN3 + INTN4 + INTN5 +
               (1 + organic | participant_id),
  data = exp_df,
  backend = "cmdstanr"
)
summary(model)

```

B.2.5 Adapting BRIDGE to New Domains

The `bridge` package is domain-agnostic. To apply BRIDGE to a new category—financial products, restaurant reviews, or real estate listings, for example—researchers need only specify the relevant attribute names and labels:

```python

# Example: Applying BRIDGE to real estate listings
pipeline = BRIDGEPipeline(
    attributes=["neighborhood", "property_type"],
    config=BRIDGEConfig(),
    output_dir="./realestate_output"
```

```
)
pipeline.fit(
    descriptions=listings_df["description"],
    labels={
        "neighborhood": listings_df["neighborhood"],
        "property_type": listings_df["type"],
    },
    tune=True, n_trials=75,
)

```

For applications with few base descriptions (e.g., 10–20 controlled stimuli), the `bridge.augmentation` module generates additional variants using large language models, varying writing style while preserving core attribute content. This gives BRIDGE sufficient training data to learn the mapping from text to attributes:

```

from bridge.augmentation import augment_descriptions_sync

# Generate 50 variants per base description in each of the 2 default strategies
result = augment_descriptions_sync(
    base_descriptions=base_df["description"].tolist(),
    num_variations=50,
    model="qwen2.5:32b-instruct-q8_0",          # local LLM via Ollama (free, for
    ↪  text generation)
)
augmented_df = result.to_dataframe()
# 20 base x 50 x 2 default strategies = 2,000 augmented variants
# (append the 20 originals before training)

```

B.3 Enhancing Objectivity and Preregistration in Experimental Research

A key advantage of our proposed research design is its ability to enhance objectivity and reproducibility in consumer research. Traditional experimental designs involving unstructured stimuli require numerous subjective decisions about stimulus selection, simplification, and analysis—decisions that can introduce researcher degrees of freedom and compromise reproducibility.

Our framework transforms this process into a structured, *end-to-end algorithmic pipeline* whose key components can be pre-specified before data collection. Researchers can pre-commit to:

- The **embedding method** (e.g., a specific versioned API or open-source model like `NV-Embed-v2`).
- The **BRIDGE architecture** or the systematic procedure for determining it (e.g., using automated hyperparameter search via Optuna within defined parameters, as demonstrated in Web Appendix §F.2).
- The **stimulus sampling strategy** (e.g., random sampling from a real-world corpus), reducing selection bias.
- A comprehensive **statistical analysis plan**, including dependent variables, derived representations, controls, and models for hypothesis testing.

With real-world stimuli and automated generation of representations and controls, the entire pipeline becomes algorithmic. In addition to mitigating the stimulus-sampling problem (Wells and Windschitl 1999), this approach addresses the "garden of forking paths' ' problem (Gelman and Loken 2013), in which seemingly innocuous analytical decisions can alter research conclusions. By reducing researcher degrees of freedom in stimulus processing and analysis, the approach supports open science practices such as preregistration, enhances replicability, and bolsters the credibility of findings by limiting post-hoc analytical decisions. The preceding section, including the accompanying code, is designed to facilitate such transparent and reproducible workflows.

B.4 Preregistration Template

The following template supplements a standard preregistration (e.g., AsPredicted, OSF) with the additional commitments needed when the analysis pipeline includes BRIDGE. Researchers should complete a standard preregistration covering hypotheses, design, sample size, and exclusion rules, then append this BRIDGE-specific supplement.

B.4.1 A. Embedding Model

1. **What are the stimuli and how are they sampled?** Describe the corpus, the stimulus sampling strategy (e.g., random sampling), and the text to be embedded.
2. **Which embedding model will be used?** (e.g., OpenAI `text-embedding-3-large`, a local model such as `all-MiniLM-L6-v2`)
3. **Will pre-computed embeddings be made available for replication?** (Yes / No / Conditional on data-sharing agreement)

B.4.2 B. Focal Attribute Specification

4. **List all focal attributes and their operationalization.** For each attribute: name, number of levels, and how ground-truth labels are assigned (e.g., expert coding, experimental manipulation, metadata).
5. **Held-out validation plan.** When the stimulus corpus is large enough to support a train/test split, specify the held-out fraction or cross-validation scheme and the performance metrics to report for each focal attribute (at minimum: accuracy and weighted F1).
6. **What is the hyperparameter tuning strategy?** (e.g., Optuna with N trials, grid search, defaults from the `bridge` package)

B.4.3 C. Nuisance Component Selection

7. **How will nuisance dimensions be identified?** (e.g., SVD on the residual embedding space after removing focal attribute representations, inspection of explained variance, semantic interpretation of top-loading texts)
8. **How many nuisance dimensions will be retained?** State the decision rule (e.g., scree plot elbow, fixed number, cumulative variance threshold).

B.4.4 D. BRIDGE Regression Specification

9. **What is the focal regression equation?** Write out the model, identifying treatment indicators, BRIDGE controls, and any additional covariates.
10. **Estimation method.** (e.g., Bayesian linear regression with default priors, OLS with robust SEs, mixed-effects model)
11. **Model comparison / fit metric.** (e.g., LOOIC, AIC/BIC, adjusted R^2)

B.4.5 E. Baseline Comparisons

12. **Which baseline specifications will be reported alongside the BRIDGE model?** At minimum one baseline is required. Options include:
    - Empath / lexical controls (e.g., SVD-reduced psycholinguistic features)
    - Linear projection (Ridge regression of embeddings on attribute labels, then SVD on residuals, as in the manuscript's benchmarks)
    - Full embedding (all embedding dimensions, with and without covariates)
13. **What comparisons will be drawn?** (e.g., coefficient magnitude and CI width across specifications; LOOIC differences; sign/significance stability)

B.4.6 F. Sensitivity and Robustness (Optional but Recommended)

14. **Controls-sensitivity analysis.** Will you report how focal estimates change when nuisance dimensions are added or removed?
15. **Alternative embedding model.** Will you replicate results with a second embedding model as a robustness check?

B.4.7 Checklist Summary

Before submitting, confirm: (1) standard preregistration filed covering hypotheses, design, sample size, and exclusion rules; (2) embedding model, input text, and sampling strategy specified (A); (3) all focal attributes listed with labeling procedure (B); (4) held-out validation plan stated with metrics

(B); (5) hyperparameter tuning strategy stated (B); (6) nuisance selection procedure described (C); (7) regression specification written out (D); (8) at least one baseline comparison pre-specified (E); (9) optional sensitivity and robustness checks considered (F).

B.5 Validating BRIDGE in a New Application

The validation built into the steps above — held-out classification metrics (Step 4; preregistration item 5), perturbation analyses (§C), controls-sensitivity checks (item 14), and baseline comparisons (preregistration Section E) — shows that a trained model recovers the focal attributes and that its conclusions are stable. These checks are necessary, but on their own they do not establish that BRIDGE is *appropriate* for a new research question. Because the focal attributes, the schema used to label them, and how informatively the stimuli express them are all specific to a study, validation is necessarily local: it must be re-established in each application rather than inherited from ours.

Concretely, high held-out accuracy and weighted F1 show that the focal attribute is *recoverable from the available text*, but they do not by themselves show three things a researcher needs to assume:

- that the learned representation captures the construct **as participants perceive it**, rather than a correlated but distinct property;
- that the model relies on the construct rather than on **superficial cues** (e.g., an explicit label or a give-away token in the description); and
- that the **nuisance controls absorb the particular textual confounds** that threaten the inference, rather than incidental variation.

We therefore recommend reporting, alongside the held-out metrics, positive evidence that the focal attribute is genuinely recoverable from the text rather than read off an explicit label, and that plausible textual confounds have been addressed. In our wine study, classification performance is high but is *not* higher for wines whose displayed names show the attribute (§F.7), and similarity computed from the categorical attributes alone is too sparse to detect the behavioral effect — so the continuous representation performs genuine measurement rather than re-reading visible labels (§F.5).

Harder applications. Some settings widen the gap between "predicts the label" and "measures the construct" — for example, when the focal attribute is weakly signaled, cue-dependent, domain-specific, inconsistently labeled, or plausibly confounded with writing style. There, held-out accuracy is especially likely to overstate appropriateness, and additional evidence is warranted. Useful options include:

- **Partial-stimulus comparison.** Evaluate on a reduced version of the stimulus (e.g., with explicit labels removed) to confirm the attribute is recovered from the substantive description, not a cue.
- **Restricted-sample analysis.** Re-estimate the focal model on the subset of trials in which explicit cues are absent, reporting any loss of sample or class coverage (in the wine study the cue-free subset proved too small to be informative, so we used a full-sample interaction test instead — §F.7).
- **Alternative embedding** (preregistration item 15). Replicate with a second embedding to confirm conclusions do not hinge on one representation.
- **Human coding.** Hand-code a subset of stimuli for the focal attribute and benchmark the learned representation against those independent judgments.

The most informative choice depends on the suspected failure mode; in each case the goal is application-level evidence that the focal attributes are represented in the available text and that plausible confounds have been handled.

A boundary condition. These checks matter because BRIDGE's confound control, like any data-reduction-based control, cannot be guaranteed to be exhaustive: it absorbs the non-focal variation that the underlying embedding represents, and a researcher can rarely know that this captures every stylistic feature correlated with the treatment. For the same reason, although BRIDGE is embedding-agnostic in construction, its ability to control confounds depends on what the embedding encodes — a legacy BERT embedding supports weaker, less precise inference (§F.6), and current embeddings may have blind spots that future models will close. High predictive accuracy should accordingly be read as a precondition for *considering* BRIDGE in a new setting, not as proof that its estimates are unconfounded there.

Illustration. Suppose the focal construct is the humor of an advertisement, with stimuli drawn from real campaigns. High accuracy in predicting human humor ratings would establish that humor is recoverable from the ad text — a precondition — but not that BRIDGE has measured humor as viewers experience it, nor that it has separated humor from correlated features such as length, brand familiarity, or production polish. The researcher would revalidate locally: confirm that accuracy does not collapse when give-away cues are removed (partial-stimulus or restricted-sample), benchmark against human coding of a subset, reproduce the result under a second embedding, and use perturbation analysis to verify that the humor representation is stable when incidental wording changes while the nuisance controls absorb it. Only with that application-level evidence would the humor estimates carry the intended interpretation.

B.6 Reproducibility Bundle

To facilitate adoption and enable end-to-end verification, we provide a self-contained reproducibility bundle alongside the `bridge` package, both released under an open-source license (CC BY-SA 4.0). The bundle contains the complete pipeline code, experiment data, and pre-computed outputs at every stage, and reproduces the Coffee Certification Experiments (Table 2 in the manuscript): a complete application of the BRIDGE pipeline from description augmentation through nuisance control extraction to Bayesian estimation. We chose this study because (a) the data are shareable and compact, (b) the pipeline demonstrates LLM-based augmentation, a key BRIDGE feature for applications with few base stimuli, and (c) it illustrates how BRIDGE handles treatments embedded in text with induced confounding. The data and code for all studies reported in the paper are available in the OSF repository at **https://doi.org/10.17605/OSF.IO/5D6KX**. The bundle is identical to the deposited code and data for the coffee certification experiments, so researchers can work from either.

B.6.1 Quick Reproduction

Researchers can reproduce all reported Coffee Certification results with a single command, without installing Python or the `bridge` package:

```
cd 02_Coffee_Certification
Rscript code/05_coffee_certification_estimate.R
```

This loads pre-computed analysis data (with BRIDGE-derived nuisance controls already merged) and cached Bayesian model fits, then prints the full results table. The output reproduces Table 2: Naïve (matched, shorter, longer, pooled), Word Count, and BRIDGE estimates with 95% credible intervals for both the Fair Trade ($N = 353$) and Organic ($N = 352$) experiments.

B.6.2 Bundle Structure

The bundle is organized as a five-step pipeline. Each step can be run independently using the pre-computed outputs from the previous step, or skipped entirely for the quick reproduction above.

- **Step 1: Define Base Descriptions** (`01_coffee_certification_prepare_descriptions.py`). Creates 16 base coffee descriptions (8 Fair Trade, 8 Organic) varying by certification framing, flavor profile, and description length—the stimuli shown to participants.

- **Step 2: Augment Descriptions** (`02_coffee_certification_augment_descriptions.py`). Uses a local language model (Qwen 2.5 32B via Ollama) to generate 150 variations per base description (50 per strategy: summarize, paraphrase, elaborate), yielding 2,416 total descriptions for BRIDGE training (the 2,400 generated variations plus the 16 originals).

- **Step 3: Train BRIDGE** (`03_coffee_certification_train_bridge.py`). Trains the BRIDGE neural network on the augmented descriptions, learning attribute representations (profile, condition, experiment) and extracting orthogonal nuisance controls via SVD. The trained model, embeddings, and metadata are saved.

- **Step 4: Map Controls to Data** (`04_coffee_certification_merge_controls.py`). Matches each participant's treatment and control descriptions to the 16 base descriptions and computes nuisance control differences and word-count differences.
- **Step 5: Bayesian Estimation** (`05_coffee_certification_estimate.R`). Fits four Bayesian linear regressions per experiment: Naïve (condition indicators), Word Count (controls for word-count difference), and BRIDGE with one and with two nuisance controls.

All intermediate outputs—base descriptions, augmented descriptions, trained model weights, nuisance controls, and cached Bayesian model fits—are included as pre-computed files, enabling researchers to inspect any stage of the pipeline or re-run individual steps. The bundle requires only R (4.3+) with `brms` (plus `cmdstanr` and `dplyr`) for the quick reproduction; re-running the full pipeline additionally requires Python (3.14+) with the `bridge` package and Ollama for local text generation.

B.6.3 Adapting the Bundle to New Applications

To apply the methodology to a new product category (e.g., financial products, restaurant reviews, real estate listings), researchers adapt each step: (1) collect or create base descriptions with labeled focal attributes, (2) augment if the corpus is small, (3) train BRIDGE on the augmented corpus, (4) merge nuisance controls with experiment data, and (5) estimate models of interest. The code examples in §B.2 (Python and R) and the reproducibility bundle provide working templates for each step.

Web Appendix C: Perturbation Analyses

We introduce perturbation analyses as a validation tool to assess the robustness of BRIDGE. In experiments using rich, unstructured stimuli, key features—such as tone, writing style, or extraneous content—if not properly controlled, can bias results and undermine theoretical conclusions. Perturbation analyses address this concern by (1) systematically modifying these aspects of the stimuli, (2) assessing the stability of attribute representations derived from the original versus modified stimuli, and (3) examining whether these changes alter the conclusions of the study. This directly tests the reliability of the findings, ensuring that observed effects reflect theoretical constructs rather than artifacts of the input data.

This section demonstrates the perturbation analysis framework using the wine study from the researcher's guide (§B), providing a complete, replicable example. The manuscript reports the results from applying this technique to the full wine study data. The same framework extends to other domains. For instance, in studies of consumer responses to financial product descriptions, researchers could use perturbation analyses to test whether BRIDGE captures focal constructs like risk tolerance while controlling for irrelevant factors such as tone or verbosity.

C.1 Python Code

The following Python code implements the perturbation analyses. It adapts the workflow from the researcher's guide (§B) into six steps. We first summarize these steps, then describe each in detail.

1. **Environment Setup and Data Preparation**: As in Step 1 of the researcher's guide, we import the necessary libraries and load the original dataset containing wine descriptions and their corresponding attributes (e.g., region, varietal).

2. **Generate Perturbed Descriptions**: We create perturbed copies of the original product descriptions by modifying tone, style, or extraneous content while leaving the focal attributes unchanged. For example, we append an irrelevant sentence like *"Limited allocation - order now!"* to an original description. We provide two approaches: (1) a deterministic function

that appends a sentence randomly sampled from a predefined set of extraneous sentences, and (2) an LLM-based function that dynamically modifies the text by adding extraneous content, rephrasing, or shifting tone, while preserving the factual attributes.

3. **Fit the BRIDGE Pipeline**: Mirroring Step 3 of the researcher's guide, we convert both sets of descriptions (original and perturbed) into high-dimensional embeddings using the same embedding method (e.g., OpenAI's `text-embedding-3-large`), ensuring consistency and comparability (the originals are embedded when the pipeline is fitted; the perturbed set is embedded by the same method inside `transform()` in Step 4).

4. **Obtain Attribute Representations and Nuisance Controls**: Corresponding to Step 4 of the researcher's guide, a trained BRIDGE model processes the embeddings to extract (a) attribute-specific representations (e.g., region, varietal), and (b) nuisance controls. Attribute-specific representations focus on the constructs of interest; nuisance controls (derived via orthogonalization) isolate irrelevant variation.

5. **Analyze Stability of Attribute Representations and Effectiveness of Nuisance Controls**: We compute the differences between the attribute-specific representations derived from the original versus perturbed descriptions. Smaller differences signal greater robustness to extraneous changes. We also evaluate differences in the nuisance controls to confirm that BRIDGE captures the introduced perturbations rather than allowing them to distort the focal attribute representations.

6. **Compare Findings From Original vs. Perturbed Representations**: We evaluate the robustness of the study's findings by comparing results derived from original-description representations with those from perturbed-description representations. Specifically, we replicate the study's key analyses (e.g., regression models, hypothesis tests) using attribute-specific representations from perturbed descriptions and compare results across the two sets of representations. We then assess whether the observed differences are minor and

attributable to random noise or whether they indicate sensitivity of the model findings to the irrelevant variation.

C.1.1 Step 1: Environment Setup and Data Preparation

We set up the Python environment, import the necessary libraries, and load the dataset containing the original wine descriptions along with their structured attributes (e.g., region, varietal). This prepares the data for subsequent processing.

```python
################################################################################
# Perturbation Analysis - Step 1: Environment Setup and Data Preparation
################################################################################
# Import the bridge package and standard libraries for perturbation analysis.

# Standard library imports
import os
import random
import asyncio

# Third-party imports
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.formula.api as smf

# BRIDGE package
from bridge import BRIDGEPipeline, BRIDGEConfig

# OpenAI client for LLM-based perturbation generation
from openai import AsyncOpenAI

# --- Load Original Wine Data ---
ORIGINAL_DATA_FILE = "wine_descriptions.csv"
print(f"Loading original wine data from: {ORIGINAL_DATA_FILE}...")
df = pd.read_csv(ORIGINAL_DATA_FILE)
print(f"Successfully loaded. Shape: {df.shape}")
print(df.head())

print("\nStep 1 complete: Environment setup and original data loaded.")
```

C.1.2 Step 2: Generate Perturbed Descriptions

We create perturbed versions of the original wine descriptions by modifying irrelevant aspects such as adding extraneous sentences, changing the tone, or introducing stylistic variations. These perturbations simulate irrelevant variation in the descriptions, allowing us to assess BRIDGE's robustness in differentiating relevant from irrelevant information when extracting attribute representations.

```
##############################################################################
# Perturbation Analysis - Step 2: Generate Perturbed Descriptions
##############################################################################
# Create modified versions of the original wine descriptions by altering
# irrelevant aspects (tone, style, extraneous content) while keeping the core
# wine attributes (e.g., region, varietal) unchanged.
# Two methods are provided: deterministic and LLM-based.

# --- Method 1: Deterministic Perturbation ---
def perturb_description_deterministic(description):
    'Append a randomly chosen irrelevant marketing sentence.'
    irrelevant_sentences = [
        "A sommelier's top pick this season!",
        "Limited allocation - order now!",
        "Pairs beautifully with any occasion.",
        "A must-have for your cellar.",
        "Experience the terroir difference.",
    ]
    return f"{description.strip()} {random.choice(irrelevant_sentences)}"

# --- Method 2: LLM-Based Perturbation ---
aclient = AsyncOpenAI(api_key=os.getenv("OPENAI_API_KEY"))

async def perturb_description_llm(
    description,
    product_category="wine",
    perturbation_type="add",  # "add", "rephrase", "tone"
    model="gpt-5.2",
):
    'Perturb a description using an LLM while preserving core attributes.'
    prompts = {
        "add": (
            f"Slightly modify this {product_category} description by adding "
            f"a short, irrelevant marketing phrase. Do NOT change any existing "
```

```
            f"facts. Original: \"{description}\""
        ),
        "rephrase": (
            f"Rephrase this {product_category} description slightly, using "
            f"synonyms or changing sentence structure, but keep all factual "
            f"attributes identical. Original: \"{description}\""
        ),
        "tone": (
            f"Rewrite this {product_category} description in a slightly more "
            f"enthusiastic tone. Do NOT change core factual attributes. "
            f"Original: \"{description}\""
        ),
    }
    response = await aclient.chat.completions.create(
        model=model,
        messages=[
            {"role": "system", "content": (
                f"You are modifying {product_category} descriptions. "
                f"Alter the text WITHOUT changing core factual attributes."
            )},
            {"role": "user", "content": prompts[perturbation_type]},
        ],
        temperature=0.5,
    )
    return response.choices[0].message.content.strip()

async def perturb_all_descriptions(descriptions, **kwargs):
    'Perturb all descriptions concurrently.'
    tasks = [perturb_description_llm(d, **kwargs) for d in descriptions]
    return await asyncio.gather(*tasks)

# --- Apply Perturbation ---
# !! Researcher Action: Choose "deterministic" or "llm" !!
perturbation_method = "deterministic"

description_col = "description"
if perturbation_method == "deterministic":
    df["Perturbed_Descriptions"] = df[description_col].apply(
        perturb_description_deterministic
    )
else:
    df["Perturbed_Descriptions"] = asyncio.run(
        perturb_all_descriptions(df[description_col].tolist())
    )

```

```python
print("\nSample of Original vs. Perturbed Descriptions:")
print(df[[description_col, "Perturbed_Descriptions"]].head())
print("\nStep 2 complete: Perturbed descriptions generated.")
```

C.1.3 Step 3: Fit the BRIDGE Pipeline

We convert both original and perturbed wine descriptions into high-dimensional embeddings using an LLM. Using the same embedding method for both sets ensures consistency and enables direct comparison. The original descriptions are embedded when the pipeline is fitted; the perturbed descriptions are embedded by the same method inside `transform()` in Step 4. This encoding enables us to assess whether perturbations affect focal attribute representations (region, varietal) and whether nuisance controls effectively capture irrelevant variation.

```python
##############################################################################
# Perturbation Analysis - Step 3: Fit the BRIDGE Pipeline
##############################################################################
# Load a pre-trained BRIDGE model and use it to generate embeddings and
# extract representations for both original and perturbed descriptions.
# The bridge package handles embedding generation, model loading, and
# representation extraction through a unified pipeline API.

# --- Initialize BRIDGE Pipeline ---
# Configure with your attributes. The pipeline will load a cached model
# if one exists at the specified output directory.
pipeline = BRIDGEPipeline(
    attributes=["region", "varietal"],
    config=BRIDGEConfig(),
    output_dir="./bridge_output"
)

# --- Fit the Pipeline (loads cached model if available) ---
# !! Researcher Action: Ensure the model was previously trained and saved !!
pipeline.fit(
    descriptions=df["description"],
    labels={
        "region": df["region"],              # e.g., "Napa Valley"
        "varietal": df["variety"],             # e.g., "Pinot Noir"
    },
    use_cached_model=True,  # Load pre-trained model
)
```

```
print("\nStep 3 complete: BRIDGE pipeline loaded and ready.")
```

C.1.4 Step 4: Obtain Attribute Representations and Nuisance Controls

We use the trained BRIDGE model to extract attribute representations (region, varietal) and compute nuisance controls for both original and perturbed embeddings.

Using the BRIDGE pipeline fitted in Step 3, we call its `transform()` method to extract attribute representations for both original and perturbed descriptions. This allows us to analyze how perturbations affect attribute representations and whether nuisance controls effectively capture the introduced noise. Following the procedure in the researcher's guide (Step 4), we then derive nuisance controls by orthogonalizing each set of LLM embeddings (original and perturbed) and extracting the components unrelated to the focal attributes. These controls are then ready for inclusion in subsequent econometric models to account for extraneous factors.

```
################################################################################
# Perturbation Analysis - Step 4: Extract Representations for Both Sets
################################################################################
# Use the fitted BRIDGE pipeline to extract attribute representations and
# nuisance controls for both original and perturbed descriptions.
# The transform() method handles embedding generation, representation
# extraction, and nuisance control computation automatically.

# --- Extract Representations for Original Descriptions ---
print("Extracting representations for ORIGINAL descriptions...")
reps_original = pipeline.transform()  # Uses training descriptions
print("Original representations extracted.")
for key, arr in reps_original.items():
    print(f"  {key}: shape {arr.shape}")

# --- Extract Representations for Perturbed Descriptions ---
print("\nExtracting representations for PERTURBED descriptions...")
reps_perturbed = pipeline.transform(
    descriptions=df["Perturbed_Descriptions"]
)
print("Perturbed representations extracted.")
for key, arr in reps_perturbed.items():
    print(f"  {key}: shape {arr.shape}")
```

```python

print("\nStep 4 complete: Representations and nuisance controls "
      "obtained for both original and perturbed data.")
```

C.1.5 Step 5: Analyze Stability of Attribute Representations and Effectiveness of Nuisance Controls

We assess the stability of attribute representations (region, varietal) and the effectiveness of nuisance controls in capturing irrelevant variation introduced by the perturbations. This validates BRIDGE's robustness in isolating focal constructs from noise. We use both visual comparisons and the RV coefficient to assess alignment between original and perturbed representations.

We first compute differences between attribute representations extracted from original and perturbed embeddings. Small differences indicate that the representations are stable and unaffected by perturbations, suggesting that BRIDGE effectively captures the focal attributes despite the noise. Next, we analyze how the nuisance controls differ between the original and perturbed embeddings. Substantial differences suggest that the controls effectively capture irrelevant variation introduced by the perturbations. Visualizing the attribute-representation differences via histograms makes it easy to identify patterns and anomalies, revealing their distribution and magnitude.

Next, we calculate the RV coefficient between the original and perturbed representations for each attribute (region, varietal). The RV coefficient is a multivariate generalization of the squared Pearson correlation coefficient that measures similarity between two sets of variables. Values close to 1 indicate strong alignment; values close to 0 indicate poor alignment. High RV coefficients for attribute representations confirm that they are stable and largely unaffected by perturbations. Similarly, we compute the RV coefficient for the original versus perturbed nuisance controls. Here, a lower RV coefficient is expected, indicating that the controls capture different variation in each set of descriptions. A permutation test then assesses whether each observed RV coefficient exceeds the alignment expected by chance.

```python
###############################################################################
# Perturbation Analysis - Step 5: Analyze Stability of Representations
###############################################################################
# Evaluate whether attribute representations are stable (high RV coefficient)
```

```python
# and whether nuisance controls capture the perturbation noise (lower RV).

# --- RV Coefficient Function ---
def rv_coefficient(X, Y):
    'Compute the RV coefficient (multivariate squared correlation).'
    X_c = X - X.mean(axis=0)
    Y_c = Y - Y.mean(axis=0)
    S_XY = X_c.T @ Y_c
    numerator = np.trace(S_XY @ S_XY.T)
    denominator = np.sqrt(
        np.trace((X_c.T @ X_c) @ (X_c.T @ X_c)) *
        np.trace((Y_c.T @ Y_c) @ (Y_c.T @ Y_c))
    )
    return numerator / denominator if denominator > 0 else 0.0

# --- Compute RV Coefficients for Each Representation ---
attribute_names = [k for k in reps_original if k != "nuisance"]
print("RV Coefficients (Original vs. Perturbed):")
print("-" * 50)

rv_results = {}
for attr in attribute_names:
    rv = rv_coefficient(reps_original[attr], reps_perturbed[attr])
    rv_results[attr] = rv
    print(f"  {attr:>12s}: {rv:.4f}")

# RV for nuisance controls (expect lower values)
rv_nuisance = rv_coefficient(
    reps_original["nuisance"], reps_perturbed["nuisance"]
)
rv_results["nuisance"] = rv_nuisance
print(f"  {'nuisance':>12s}: {rv_nuisance:.4f}")
print("-" * 50)
# High RV for attributes (~0.95+) indicates stability.
# Lower RV for nuisance indicates it captured perturbation noise.

# --- Permutation Test for Statistical Significance ---
def permutation_test_rv(X, Y, n_permutations=1000):
    'Test significance of RV coefficient via permutation.'
    observed_rv = rv_coefficient(X, Y)
    count = 0
    for _ in range(n_permutations):
        Y_perm = Y[np.random.permutation(len(Y))]
        if rv_coefficient(X, Y_perm) >= observed_rv:
            count += 1
```

```python
    return observed_rv, count / n_permutations

for attr in attribute_names:
    rv, p = permutation_test_rv(reps_original[attr], reps_perturbed[attr])
    print(f"  {attr}: RV = {rv:.4f}, p = {p:.4f}")

# --- Visualize Differences ---
fig, axes = plt.subplots(1, len(attribute_names),
                         figsize=(5 * len(attribute_names), 4))
if len(attribute_names) == 1:
    axes = [axes]
for ax, attr in zip(axes, attribute_names):
    diffs = np.linalg.norm(
        reps_original[attr] - reps_perturbed[attr], axis=1
    )
    ax.hist(diffs, bins=30)
    ax.set_title(f"{attr.title()} Difference")
    ax.set_xlabel("Euclidean Distance")
    ax.set_ylabel("Frequency")
plt.suptitle("Attribute Representation Stability", y=1.02)
plt.tight_layout()
plt.show()

print("\nStep 5 complete: Stability and effectiveness analyzed.")
```

C.1.6 Step 6: Compare Findings From Original vs. Perturbed Representations

We evaluate the robustness of findings by comparing regression results obtained using representations from original product descriptions with those from perturbed descriptions. This comparison reveals whether key findings remain consistent when stimuli are altered in ways that should not affect the underlying theoretical constructs. We estimate the study's focal regression (cf. §B, Step 5), first using representations and nuisance controls from original embeddings, and then using those from perturbed embeddings. By examining the coefficients, their significance, and overall model fit, we assess whether observed differences are minor and attributable to noise or whether they indicate sensitivity to irrelevant variation introduced by perturbation.

```python
###############################################################################
# Perturbation Analysis - Step 6: Compare Findings (Original vs. Perturbed)
###############################################################################
```

```python
# Replicate the primary statistical analysis using representations from
# both original and perturbed descriptions. Consistent results indicate
# that findings are robust to irrelevant textual variations.

# --- Prepare Analysis DataFrames ---
# !! Researcher Action: "Choice" and "Condition" are the outcome and treatment
# columns of your experimental data, merged with the representations as in the
# researcher's guide (Step 5); they are not created by the steps above. !!
dependent_var = "Choice"
condition_col = "Condition"

def build_analysis_df(reps, df_base, suffix=""):
    'Build a regression-ready DataFrame from BRIDGE representations.'
    df_out = df_base[[dependent_var, condition_col]].copy()
    for attr in [k for k in reps if k != "nuisance"]:
        for i in range(reps[attr].shape[1]):
            col = f"{attr}{suffix}_{i}"
            df_out[col] = reps[attr][:, i]
    for i in range(reps["nuisance"].shape[1]):
        col = f"nuisance{suffix}_{i}"
        df_out[col] = reps["nuisance"][:, i]
    return df_out

df_orig = build_analysis_df(reps_original, df, "_orig")
df_pert = build_analysis_df(reps_perturbed, df, "_pert")

# --- Build and Fit OLS Models ---
def fit_ols(df_analysis, suffix, attr_names):
    'Fit OLS with attribute representations and nuisance controls.'
    attr_cols = [c for c in df_analysis.columns
                 if any(c.startswith(a + suffix) for a in attr_names)]
    nuisance_cols = [c for c in df_analysis.columns
                     if c.startswith(f"nuisance{suffix}")]
    predictors = [condition_col] + attr_cols + nuisance_cols
    formula = f"{dependent_var} ~ " + " + ".join(predictors)
    return smf.ols(formula, data=df_analysis).fit()

attribute_names = [k for k in reps_original if k != "nuisance"]
result_orig = fit_ols(df_orig, "_orig", attribute_names)
result_pert = fit_ols(df_pert, "_pert", attribute_names)

# --- Compare Key Results ---
print("\n--- Comparison of Model Fit ---")
print(f"  Original R-squared:  {result_orig.rsquared:.4f}")
print(f"  Perturbed R-squared: {result_pert.rsquared:.4f}")
```

```python

print(f"\n--- Condition Coefficient ---")
cond_orig = result_orig.params[condition_col]
cond_pert = result_pert.params[condition_col]
p_orig = result_orig.pvalues[condition_col]
p_pert = result_pert.pvalues[condition_col]
print(f"  Original:  {cond_orig:.4f} (p={p_orig:.4f})")
print(f"  Perturbed: {cond_pert:.4f} (p={p_pert:.4f})")

# Consistent results across original and perturbed analyses indicate
# that the findings are robust to irrelevant textual modifications.
print("\nStep 6 complete: Findings compared.")
```

Web Appendix D: Monte Carlo Studies Evaluating BRIDGE's Performance

This section provides the programming framework for the Monte Carlo simulation presented in the manuscript. The manuscript reports summary estimates (Table 1); the complete results are presented below (Table W1 and Figure W2).

The simulations are grounded in a marketing research scenario: investigating whether exposure to information about an environmental concern (factory-farmed meat) shifts consumer preferences for organic packaged salads. To create diverse, realistic stimuli, we used an LLM to generate product descriptions in three styles (Factual, Engaging, Creative) for each unique combination of product attributes. The stimuli comprise 640 attribute combinations (2 organic levels $\times$ 4 sizes $\times$ 5 types $\times$ 16 weights), with 100 variations per combination in each style, yielding 192,000 product descriptions.

Simulated participant responses follow a linear utility framework in which participant i's preference for product j is:

$$\begin{aligned}\text{preference}_{ij} = \alpha + \beta_1 \text{Treatment}_j + \beta_2\Big(1 + \lambda \cdot \text{Intensity}_j\Big) \cdot \text{Organic}_j + \beta_3(\text{Treatment}_j \times \text{Organic}_j) \\ + \beta_4 \text{Size}_j + \beta_5 \text{Type}_j + \beta_6 \text{Weight}_j + \theta \cdot (\text{Intensity}_j - 1) + \varepsilon_{ij},\end{aligned} \tag{W1}$$

where β_1, β_2, and β_3 capture the treatment effect, organic premium, and their interaction (spillover), respectively; β_4, β_5, and β_6 represent size, type, and weight effects; and $\varepsilon_{ij} \sim \mathcal{N}(0, 1)$. $\text{Intensity}_j \in \{0, 1, 2\}$ maps description styles to increasing elaboration (Factual = 0, Engaging = 1, Creative = 2). The design parameter θ controls the *additive* effect of style on preferences, while λ controls the *moderating* effect of description intensity on the organic premium. Both channels operate simultaneously: $\theta = 0.25$ shifts utility by -0.25, 0, and $+0.25$ for Factual, Engaging, and Creative descriptions, respectively, while $\lambda = 0.25$ makes the organic effect style-specific: $\beta_2 = 0.50$ for Factual, $\beta_2(1 + \lambda) = 0.625$ for Engaging, and $\beta_2(1 + 2\lambda) = 0.75$ for Creative.

The ground-truth parameters are: $\beta_1 = 0.25$ (treatment effect), $\beta_2 = 0.5$ (organic premium),

$\beta_3 = 0.5$ (interaction), $\beta_4 = (0, 0.3, 0.5, 1)$ for small, medium, large, and extra-large, respectively, $\beta_5 = (0, -0.75, 0.4, 0.6, -0.5)$ for arugula, iceberg, radicchio, spinach, and spring mix, respectively, and $\beta_6 = 0.1$ (weight). Responses are generated on a 1-to-13 Likert scale for each of 24 evaluation tasks. We evaluate the five estimators—one BRIDGE specification and four benchmarks—defined in the manuscript.

D.1 Programming Framework

We implement these simulations in a modular framework comprising eight steps:

1. **Setup, Ground Truth, and Base Data:** We set up the computational environment in Python, using libraries such as `NumPy` and `Pandas` for data manipulation, and set a random seed to ensure reproducibility. We define the ground-truth parameters specifying the true effects of product attributes and experimental conditions on consumer preferences. A balanced factorial design crosses all attribute levels: organic (yes/no), size (small, medium, large, extra-large), type (spring mix, iceberg, radicchio, spinach, arugula), and weight (5–20 oz, integer-valued). From these combinations, we generate base product descriptions.

2. **LLM-Based Product Description Generation:** We use OpenAI's GPT model to generate multiple variations of each base description. For each unique combination of product attributes, the LLM generates 100 variations in each of the three styles (Factual, Engaging, Creative). This yields varied, naturalistic descriptions containing both focal attributes and stylistic variation.

3. **Embed the Product Descriptions:** We embed the product descriptions using OpenAI's `text-embedding-3-large` model. These embeddings capture both focal attributes and stylistic variation. We perform quality checks (e.g., detecting missing values or outliers) and document all steps for reproducibility.

4. **Apply Empath and Generate SVD Controls:** We score each description across the Empath library's approximately 200 psycholinguistic categories. We then apply Singular Value

Decomposition (SVD) to extract 10 low-dimensional components. These SVD components serve as general-purpose nuisance controls for the Empath benchmark.

5. **Train the Interpretable AI Model (BRIDGE):** We train BRIDGE to map the LLM embeddings (Step 3) to lower-dimensional, interpretable attribute representations and nuisance controls. The model extracts product attributes from the embeddings while isolating unrelated variation via orthogonalization. The architecture includes shared layers for dimensionality reduction and attribute-specific output layers for predicting each attribute. We use categorical cross-entropy for categorical attributes and mean squared error for continuous attributes, with the `Adam` optimizer, and use early stopping and model checkpointing to prevent overfitting.

6. **Simulate Participant Responses:** We generate participant responses from the utility function defined above, incorporating product attributes, experimental condition, description style, and random error. We simulate responses for 1,000 participants across 24 evaluation tasks each. Participants are randomly assigned to a treatment or control condition, which determines the distribution of styles they encounter and introduces the confound. Treatment participants see 8 Engaging, 12 Creative, and 4 Factual descriptions, while control participants see 12 Factual, 8 Engaging, and 4 Creative descriptions. Responses are recorded on a 1-to-13 Likert scale.

7. **Apply BRIDGE and Evaluate Performance:** Using BRIDGE's pre-computed nuisance controls and attribute representations, we estimate β_1 (treatment effect) and β_3 (interaction effect) via OLS regression and compare the estimates against the known ground truth.

8. **Estimate Benchmark Models and Compare Performance:** Finally, we estimate the benchmark models (No Controls, Empath Controls, Oracle, and Linear Projection) using the simulated responses (Step 6) and each model's designated predictors; the Linear Projection controls are constructed at this stage by Ridge regression of the embeddings on the attribute

labels, followed by SVD of the residuals (2 components retained). We compare each model's recovery of β_1 and β_3 against the known ground truth.

D.2 Coefficient Estimates

Table W1 reports the complete OLS coefficient vector for the BRIDGE specification. The intercept and 25 regressors are organized into five groups: focal variables, product attribute controls, nuisance controls interacted with the non-organic indicator $(1 - \text{Organic})$, nuisance controls interacted with the organic indicator, and attribute representations interacted with the organic indicator. The nuisance controls are centered at their respective population means (computed from the full 192,000-description corpus) and split into two organic-specific channels so that the effective nuisance adjustment can differ for organic versus conventional descriptions. The attribute representations (Size, Type, Weight) enter only the organic channel.

D.3 Style-Specific Organic Effects Across Confounding Levels

Figure W2 presents the estimation bias for the three style-specific organic effects across the same four confounding scenarios examined in the manuscript. Only BRIDGE and the infeasible Oracle can decompose the organic premium by description style; simpler estimators yield a single pooled coefficient. Across all four scenarios and all three style levels, BRIDGE and Oracle are virtually indistinguishable, with both centered on zero bias.

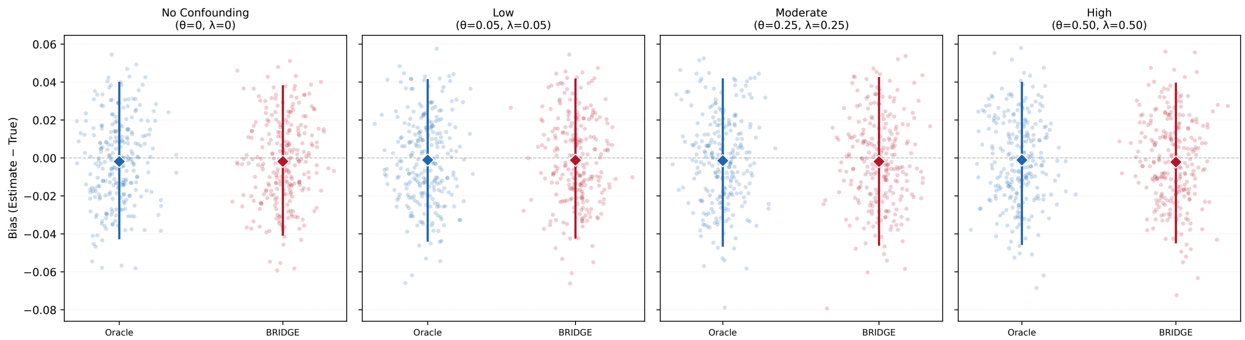
Table W1: BRIDGE Coefficient Estimates — Monte Carlo Simulation ($\theta = 0.25$, $\lambda = 0.25$)

Variable	**Estimate**	**SE**	p**-value**
Panel A: Focal Variables			
Intercept	4.693	0.067	$<$.001
Treatment (β_1)	0.248	0.021	$<$.001
Organic	0.605	0.020	$<$.001
Treatment $\times$ Organic (β_3)	0.512	0.029	$<$.001
Panel B: Product Attribute Controls			
Weight (oz)	0.100	0.002	$<$.001
Size: Medium	0.299	0.024	$<$.001
Size: Large	0.576	0.024	$<$.001
Size: Extra-Large	1.076	0.024	$<$.001
Type: Iceberg	-0.559	0.112	$<$.001
Type: Radicchio	0.381	0.045	$<$.001
Type: Spinach	0.670	0.095	$<$.001
Type: Spring Mix	-0.462	0.058	$<$.001
Panel C: Nuisance Controls $\times$ (1 $-$ Organic)			
$\text{Nuisance}_1 \times (1 - \text{Organic})$	0.837	0.040	$<$.001
$\text{Nuisance}_2 \times (1 - \text{Organic})$	-0.156	0.184	.398
$\text{Nuisance}_3 \times (1 - \text{Organic})$	0.144	0.174	.406
$\text{Nuisance}_4 \times (1 - \text{Organic})$	-0.481	0.065	$<$.001
Panel D: Nuisance Controls $\times$ Organic			
$\text{Nuisance}_1 \times \text{Organic}$	1.240	0.049	$<$.001
$\text{Nuisance}_2 \times \text{Organic}$	-0.167	0.187	.373
$\text{Nuisance}_3 \times \text{Organic}$	0.074	0.178	.678
$\text{Nuisance}_4 \times \text{Organic}$	-0.574	0.071	$<$.001
Panel E: Attribute Representations $\times$ Organic			
Size $\text{Rep}_1 \times$ Organic	-0.010	0.011	.360
Size $\text{Rep}_2 \times$ Organic	-0.002	0.004	.583
Type $\text{Rep}_1 \times$ Organic	0.001	0.032	.964
Type $\text{Rep}_2 \times$ Organic	-0.000	0.018	.983
Weight $\text{Rep}_1 \times$ Organic	-0.058	0.014	$<$.001
Weight $\text{Rep}_2 \times$ Organic	-0.008	0.027	.773

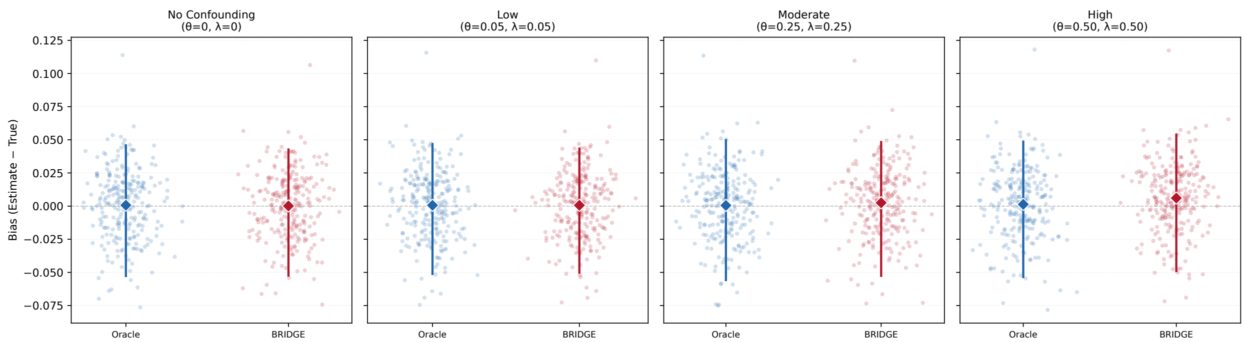
$N = 24{,}000$ observations (1,000 participants $\times$ 24 tasks). $R^2 = 0.481$.

Reference categories: Size = Small, Type = Arugula.

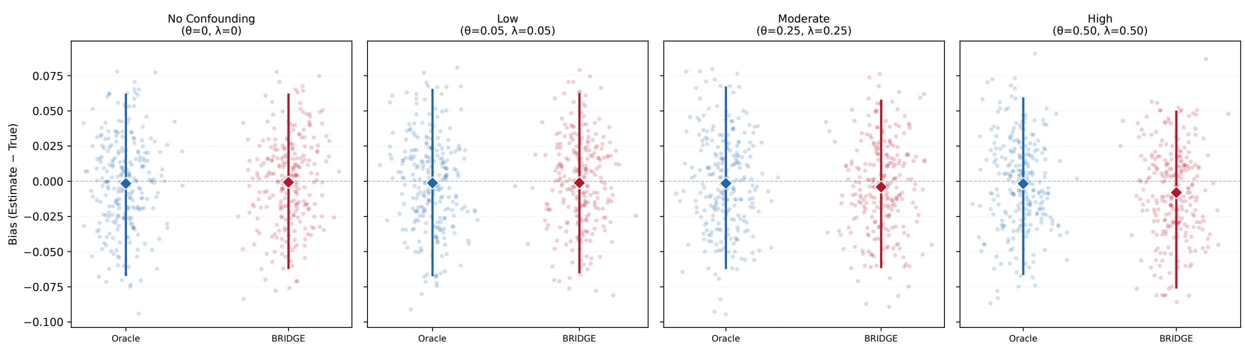
Nuisance controls and attribute representations centered at population means from the full 192,000-description corpus, computed separately for organic and conventional descriptions.



(a) Organic, Factual ($\beta_2 = 0.50$)



(b) Organic, Engaging ($\beta_2(1 + \lambda)$)



(c) Organic, Creative ($\beta_2(1 + 2\lambda)$)

Figure W2: Style-specific organic effects: Oracle vs. BRIDGE estimation bias across four confounding levels (no confounding: $\theta = 0$, $\lambda = 0$; low: $\theta = 0.05$, $\lambda = 0.05$; moderate: $\theta = 0.25$, $\lambda = 0.25$; high: $\theta = 0.50$, $\lambda = 0.50$). Each dot represents one of 250 random seeds; diamonds indicate the mean bias; vertical bars span the 2.5th–97.5th percentile interval. The dashed line at zero indicates no bias.

Web Appendix E: Coffee Certification Experiments

Having established BRIDGE's performance in controlled Monte Carlo simulations, we now turn to experiments with real participants. Product descriptions that disclose certification information may differ not only in whether a certification is included but also in how much text is used to describe it. For example, a fair trade or organic disclosure may be conveyed succinctly (a brief label) or elaborately (with detailed sourcing and standards information). When description length covaries with the certification framing and also affects consumer responses, it becomes a text-length confound that can obscure the focal treatment effect. If researchers anticipate this possible confound, a typical approach is to design matched stimuli (pairing treatment text with a length-matched control text) to control for the confound explicitly. However, researchers may not always know in advance the relevant confounds; even if they do, designing perfectly controlled stimuli may be difficult or not always feasible. A method like BRIDGE can offer a means to control for nuisance variables — non-focal variation in the stimulus text that can affect the outcome and that, when correlated with the focal treatment, confounds the estimate — with two benefits: (1) it does not require privileged information about specifics of the confounds in advance and (2) it operates based on observed information, deriving nuisance controls from the product descriptions themselves.

To establish the efficacy of the proposed method (BRIDGE) with regard to experimental control, we examine whether a certified framing (treatment) in coffee descriptions affects consumer preferences, while deliberately varying treatment-text length (confound). Each participant was shown a pair of coffee descriptions: one with certification information (treatment) and one without (control). We consider three experimental conditions in which the treatment text is shorter than, matched to, or longer than the control text. This design allows us to know the confound structure, to examine the performance of conventional estimators in the different conditions, and to investigate whether the use of BRIDGE enables us to infer the same findings in the confounded conditions (shorter/longer text) as are inferred using the conventional estimators in the "unconfounded" condition (matched text length).

This section provides supplementary materials for the coffee certification experiments reported in the manuscript, including detailed methods, experimental stimuli, and model specifications for each experiment.

E.1 Experiment 1: Fair Trade Coffee Descriptions

E.1.1 Method

E.1.1.1 Participants and design. A total of 353 U.S.-based Prolific participants (43.9% women; 55.5% men; 0.6% non-binary; M_{age} = 44.9) completed the study for monetary compensation. They were randomly assigned to one of three between-subjects conditions (fair-trade framing length: shorter-than-control, length-matched, longer-than-control; preregistered: https://aspredicted.org/7dt3ev.pdf).

E.1.1.2 Procedure and measures. Participants evaluated a pair of coffee descriptions—a "Base Option" (control, without certification framing) and a "Comparison Option" (treatment, with certified fair-trade framing)—and indicated their relative preference on a 7-point scale (1 = strongly prefer Option A, 7 = strongly prefer Option B). All participants were presented with the same two underlying coffee profiles (nutty/balanced; chocolatey/full-bodied). For each participant, fair-trade framing was randomly assigned to one of the two profiles, and the left–right placement of options was randomized. To introduce a text-length confound, we varied the length of the Comparison Option (10, 21, or 52 words) while holding the Base Option constant at a medium length of 21 words. This yielded the three preregistered between-subjects conditions in which the fair-trade description was shorter than ("FairTrade+Short"), equal to ("FairTrade+Medium"), or longer than ("FairTrade+Long") the Base Option.

Relative preference was recoded to range from -3 (strongly prefer Base Option) to $+3$ (strongly prefer Comparison Option). Task involvement was measured using three items (e.g., "I read through the coffee descriptions carefully") on a 7-point scale (1 = "strongly disagree," 7 = "strongly agree"; $\alpha = .769$). Mood was measured with four 7-point items (e.g., "unpleasant/pleasant," "unhappy/happy"; $\alpha = .905$). As expected, neither task involvement ($F(2, 350) = 1.52$, $p = .220$) nor mood ($F(2, 350) = 0.32$, $p = .727$) differed across conditions. Participants also provided background information (e.g., gender, age).

E.2 Experiment 2: Organic Coffee Descriptions

E.2.1 Method

E.2.1.1 Participants and design. Experiment 2 employed a design parallel to Experiment 1, substituting certified USDA Organic framing for fair trade framing as the treatment. A total of 352 U.S.-based Prolific participants (49.7% women; 48.9% men; 1.1% non-binary; 0.3% preferred not to say; M_{age} = 44.7) completed the study for monetary compensation. They were randomly assigned to one of three between-subjects conditions (organic framing length: shorter-than-control, length-matched, longer-than-control; preregistered: https://aspredicted.org/fz6y8h.pdf).

E.2.1.2 Procedure and measures. The procedure and measures were identical to those in Experiment 1, except for the certification framing. Participants evaluated a pair of coffee descriptions—a Base Option without certification framing and a Comparison Option with certified organic framing—using the same coffee profiles (nutty/balanced; chocolatey/full-bodied). Participants indicated their relative preference on the same 7-point scale. Organic framing was randomly assigned to one of the two profiles, and option position (left–right) was randomized.

To introduce the text-length confound, we again varied the length of the Comparison Option. The Base Option was always medium length (24 words), whereas the Comparison Option was shorter (10 words; "Organic+Short"), matched (24 words; "Organic+Medium"), or longer (55 words; "Organic+Long"), depending on the condition. Relative preference was recoded to range from -3 (strongly prefer Base Option) to $+3$ (strongly prefer Comparison Option). As in Experiment 1, task involvement was measured with three items ($\alpha = .684$) and mood with four items ($\alpha = .903$). As expected, neither task involvement ($F(2, 349) = 0.55$, $p = .579$) nor mood ($F(2, 349) = 0.97$,

$p = .380$) differed across conditions. Participants also provided background information (e.g., gender, age).

E.3 Experimental Stimuli

Tables W2 and W3 present the complete stimuli used in the fair-trade and organic coffee certification experiments, respectively. Each experiment used two underlying coffee profiles (nutty/balanced and chocolatey/full-bodied). The Base Option was always medium length, whereas the Comparison Option varied across three length conditions (short, medium, long) to create the text-length confound. Together, these three conditions define the confound structure.

E.4 Model Specifications

Across both experiments, the key dependent variable is each participant's 7-point relative preference, recoded to range from -3 (strongly prefer the Base Option) to $+3$ (strongly prefer the Comparison Option). To evaluate how well different analytic models recover the treatment effect when description length varies, we compare three estimators that parallel those examined in the Monte Carlo simulation study:

1. **Naïve (matched/shorter/longer/pooled).** A single Bayesian linear regression with indicators for each description-length condition (medium-length as reference), nesting all condition-specific naïve estimates reported in the manuscript. Naïve (matched) is the intercept ($\hat{\beta}_0$), representing the treatment effect when description length is held constant across the Comparison Option and Base Option texts. Naïve (shorter) $= \hat{\beta}_0 + \hat{\beta}_1$ and Naïve (longer) $= \hat{\beta}_0 + \hat{\beta}_2$ give the treatment effects when the Comparison Option text is shorter or longer than the Base Option text, respectively. Naïve (pooled) is the weighted average of these three condition-specific effects, with weights proportional to the number of observations in each condition. We include this pooled estimate to illustrate the consequences of ignoring the length confound, as it is expected to yield a biased estimate if description length affects the certification effect.

Table W2: Experiment 1 (Fair Trade): Coffee Description Stimuli by Condition and Profile

Option	Coffee Profile 1: Nutty/balanced	Coffee Profile 2: Chocolatey/full-bodied
Base Option (21 words)	This coffee's nutty aroma and balanced taste form a smooth profile. The beans are sourced through conventional trade practices used commonly.	This coffee's chocolatey aroma and full-bodied taste deliver a deep finish. The beans are sourced through conventional trade practices used commonly.
Fair Trade Option (21 words)	This fair-trade coffee's nutty aroma and balanced taste form a smooth profile. The beans are sourced under Fair Trade Certified™ standards.	This fair-trade coffee's chocolatey aroma and full-bodied taste deliver a deep finish. The beans are sourced under Fair Trade Certified™ standards.
Fair Trade Option (10 words)	This fair-trade coffee is nutty and balanced, Fair Trade Certified™.	This fair-trade coffee is chocolatey and full-bodied, Fair Trade Certified™.
Fair Trade Option (52 words)	This fair-trade coffee's nutty aroma and balanced taste form a smooth profile with subtle hazelnut notes and a velvety finish. The beans are sourced under Fair Trade Certified™ standards that make trade more equitable, ensuring farmers, workers, and producers are paid fairly, work in safe conditions, and can invest in their communities.	This fair-trade coffee's chocolatey aroma and full-bodied taste deliver a deep finish with subtle cocoa notes and a rich character. The beans are sourced under Fair Trade Certified™ standards that make trade more equitable, ensuring farmers, workers, and producers are paid fairly, work in safe conditions, and can invest in their communities.

Note. The Base Option (non-fair-trade) was always presented at medium length (21 words). The Comparison Option varied across three length conditions: short (10 words), medium (21 words), and long (52 words). Fair-trade framing was randomly assigned to one of the two coffee profiles for each participant.

Table W3: Experiment 2 (Organic): Coffee Description Stimuli by Condition and Profile

Option	Coffee Profile 1: Nutty/balanced	Coffee Profile 2: Chocolatey/full-bodied
Base Option (24 words)	This coffee's nutty aroma and balanced taste form a smooth profile. The beans are grown following conventional agriculture practices, used in regular coffee production.	This coffee's chocolatey aroma and full-bodied taste deliver a deep finish. The beans are grown following conventional agriculture practices, used in regular coffee production.
Organic Option (24 words)	This organic coffee's nutty aroma and balanced taste form a smooth profile. The beans are grown following organic agriculture standards, and USDA Organic certified.	This organic coffee's chocolatey aroma and full-bodied taste deliver a deep finish. The beans are grown following organic agriculture standards, and USDA Organic certified.
Organic Option (10 words)	This organic coffee is nutty and balanced, USDA Organic certified.	This organic coffee is chocolatey and full-bodied, USDA Organic certified.
Organic Option (55 words)	This organic coffee's nutty aroma and balanced taste create a smooth profile with a velvety, rounded finish and subtle hazelnut notes. The beans are grown according to organic agriculture standards and are USDA Organic certified, meaning they are cultivated without synthetic pesticides or chemical fertilizers, using natural methods that promote biodiversity and prioritize soil health.	This organic coffee's chocolatey aroma and full-bodied taste deliver a deep finish with a robust, rich character and subtle cocoa notes. The beans are grown according to organic agriculture standards and are USDA Organic certified, meaning they are cultivated without synthetic pesticides or chemical fertilizers, using natural methods that promote biodiversity and prioritize soil health.

Note. The Base Option (non-organic) was always presented at medium length (24 words). The Comparison Option varied across three length conditions: short (10 words), medium (24 words), and long (55 words). Organic framing was randomly assigned to one of the two coffee profiles for each participant.

2. **Word Count.** A Bayesian linear regression that controls for the word-count difference between the Comparison Option and Base Option descriptions (Δ_{wc}). This model assumes that the effect of text length on preference is proportional to the word-count difference.

3. **BRIDGE.** A Bayesian linear regression that adds a vector of nuisance controls generated by BRIDGE. These controls are derived from orthogonalized residuals of the AI text embeddings after projecting out the learned focal attribute representations. BRIDGE requires neither knowledge of the confound structure (i.e., the experimental conditions) nor specification of the confounding dimension—it learns the nuisance structure directly from the text. This estimator is both feasible and flexible.

To train BRIDGE for these experiments, we augmented the 16 base coffee descriptions (8 per experiment) using the `bridge.augmentation` module, generating 50 variations per description in each of three strategies (summarize, paraphrase, elaborate) via Qwen 2.5 32B, yielding 2,416 training descriptions (the 2,400 generated variations plus the 16 originals). BRIDGE was trained on three focal attributes—coffee profile (2 levels), certification condition (2 levels), and experiment (2 levels)—using local Gemma embeddings (768 dimensions) with Optuna hyperparameter tuning (50 trials). The trained model achieved 100% classification accuracy for all three attributes. Five nuisance dimensions were selected via the SVD elbow criterion. The estimation models reported in the manuscript enter one nuisance control; results with two nuisance controls are similar (see the note to Table 2 in the manuscript). Crucially, description length was *not* specified as an attribute—the nuisance controls must discover this variation from the embeddings.

Results for both experiments are presented in the manuscript.

Web Appendix F: Wine Choice Experiment

This section provides supplementary materials for the wine choice experiment reported in the manuscript. Below, we present the experimental stimuli, describe the process for determining the BRIDGE neural-network architecture, summarize the resulting classification performance, evaluate sensitivity to training-data size, and report model fit comparisons, decomposition analyses, and benchmark evaluations, together with embedding-backend sensitivity and name-cue robustness analyses.

F.1 Experimental Stimuli

Table W4 lists eight representative wines drawn from our corpus of nearly 120,000 wines. Each entry shows a wine's name and descriptive tasting note (written by professional tasters during a blind taste test), reproduced verbatim from the corpus; participants were only presented with the wine's name and tasting note. They were not separately shown the wine's region and varietal.

These examples are selected for illustration: they span eight regions and eight widely consumed varietals. The actual stimulus set was randomly selected from the complete real-world corpus and therefore represents the diversity of wines that motivated our use of BRIDGE for textual feature extraction.

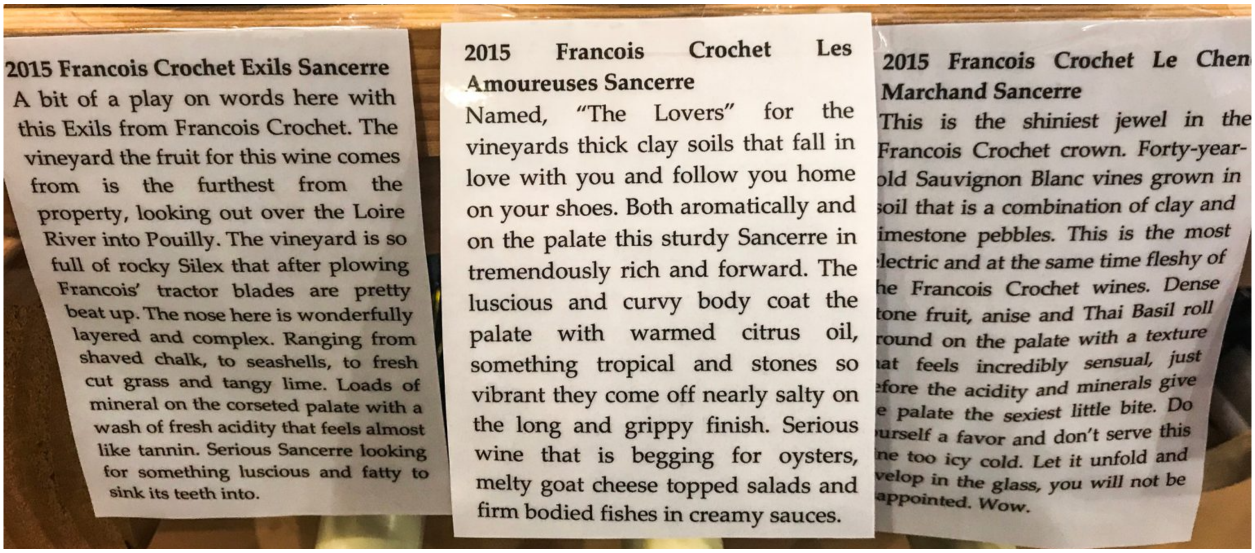
Tasting notes are commonly used in real-world retail settings. For instance, "shelf talkers"—small descriptive cards attached to store shelves—help consumers navigate their choices by providing descriptions of the wines' characteristics. Panel (a) in Figure W3 provides a close-up view of shelf talkers, while Panel (b) shows them in a retail context.

Figure W4, a word cloud of the top 1,000 taste descriptors, illustrates the range of vocabulary in the tasting notes. Terms range from common descriptors like "dry," "full-bodied," or "oaky" to more unusual ones such as "flinty," "barnyard," or "bruised," and occasionally to idiosyncratic notes like "green bell pepper," "petrol," "wet stone," or "bubblegum." This vocabulary poses a challenge for traditional experimental methods that rely on simpler stimuli, underscoring the need for BRIDGE.

Table W4: Examples of Wine Stimuli

Wine	Tasting Note
Avalon 2011 Cabernet Sauvignon (California)	This has rich flavors of black currants, licorice and sandalwood, and it finishes dry and spicy. This should be relatively easy to find thanks to its 50,000-case production.
Stevens 2013 StevensMerlot Merlot (Yakima Valley)	A blend of fruit from DuBrul and Meek vineyards, this wine brings light aromas of herb, vanilla, raspberry, orange peel and black licorice. The palate offers a mouthful of tart fruit flavors that are simultaneously sleek and generous.
Erath 2006 Tuenge Pinot Noir (Chehalem Mountains)	It's pronounced tung-GEE and was formerly known as Chehalem Mountain vineyards. As is often the case with this AVA, it has substantial tannins and a firm, compact structure that bodes well for cellaring. The fruit mixes cherry, pomegranate and hints of banana, while the barrel flavors add some toast and butter.
Dourthe 2013 La Grande Cuvée (Bordeaux Blanc)	Tight and herbaceous, this nervy, crisp wine is full of lemon and lime juice cut with an orange zest. Floral and full of acidity, it leaves a mineral, steely aftertaste.
TintoNegro 2013 Finca La Escuela La Arena Malbec (Mendoza)	Saucy, woody aromas of plum and berry are good but strained. This smacks down with high acidity and a sense of tomato. Salty flavors of tomato, herbs, red plum and currant remain salty on the finish.
I Sodi 2004 Riserva (Chianti Classico)	Here's a chewy, chocolate-driven wine with some lactic notes and a powdery cover of cherry fruit and fresh blueberries. It has a smooth, chewy mouthfeel and bright acidity that is followed by tight, polished tannins.
Albert Bichot 2006 Domaine Long-Depaquit Les Vaillons Premier Cru (Chablis)	Fresh and soft initially, this wine takes a while to reveal some mineral structure. But then, deliciously friendly white fruits surround this taut structure. The acidity is crisp, fresh, leaving orange zest and green fruits for the finish.
Reichsgraf von Kesselstatt 2012 Piesporter Goldtröpfchen GL Spätlese Riesling (Mosel)	Fresh herb and green floral aromas lend a breezy, invigorating character to sweet yellow peach and honey flavors in this medium-sweet Riesling. Rich and fruity, yet bristling with tart tangerine acidity, it's a deeply satisfying sipper.

Note. Each stimulus consisted of the wine name and a professional tasting note. Region and varietal were not separately displayed to participants.



(a) Close-up of shelf talkers.

(b) Shelf talkers in context.

Figure W3: Wine Shelf Talkers in Retail Settings.

Notes: Images from 'The pros and cons of shelf talkers,' by Vicki Denig at SevenFifty Daily, reproduced with permission.

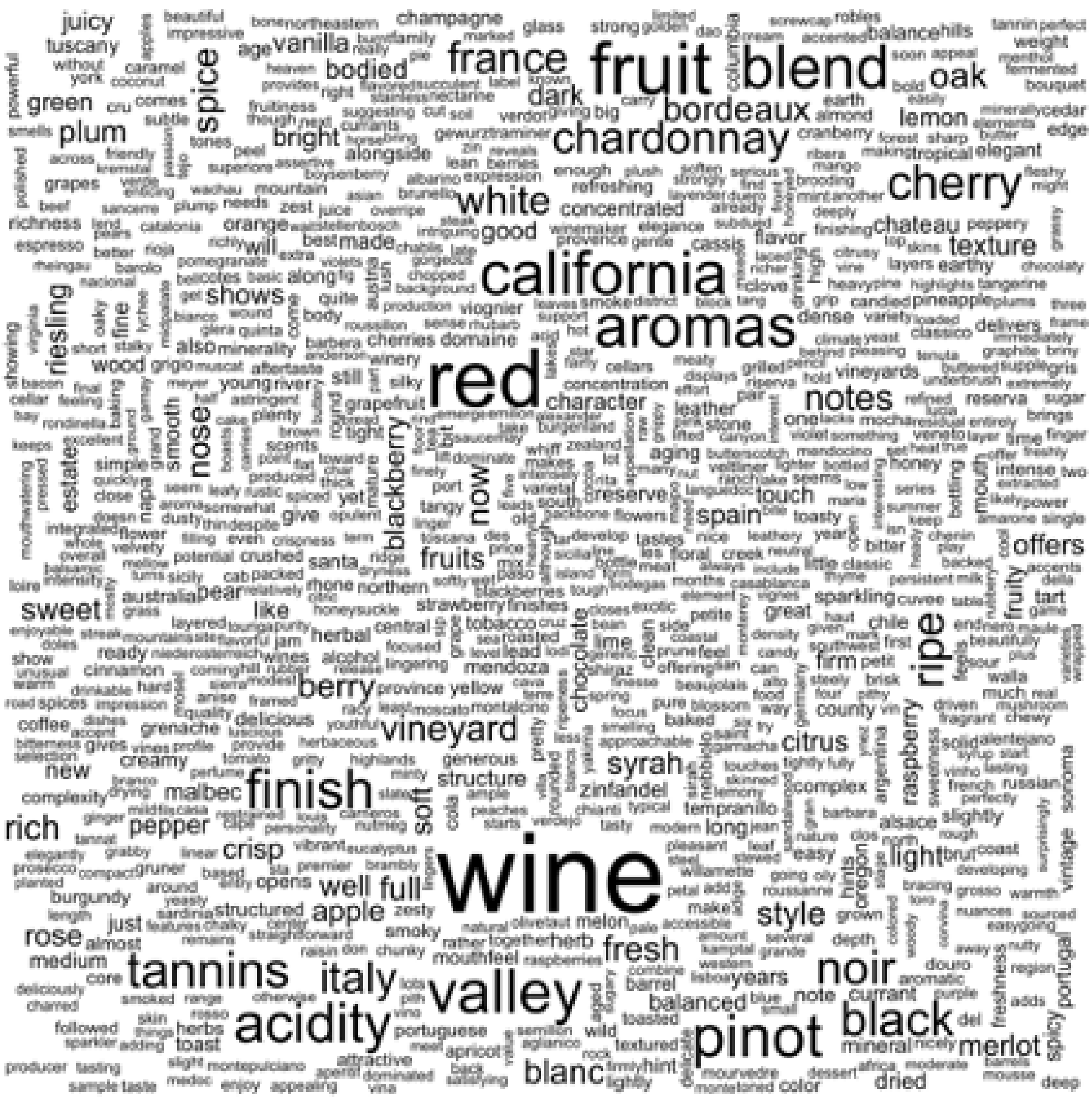


Figure W4: Word Cloud of the Top 1,000 Taste Descriptors.

Participants (N = 1,000) were drawn from wine consumers in Australia, New Zealand, and the United States (see the manuscript for full recruitment details). They reported strong interest in (M = 5.59) and liking for (M = 6.22) wine on 7-point scales (1 = 'not at all'; 7 = 'very'), and high task involvement (M = 5.73; measured using three items, e.g., 'I could relate to the overall situation of evaluating wines'; 1 = 'strongly disagree,' 7 = 'strongly agree'; $\alpha = .74$). They also found the wine descriptions moderately easy to understand (M = 5.43; 7-point scale from 'very difficult' to 'very easy').

F.2 Determining the BRIDGE Architecture Using Optuna

A key aspect of ensuring objectivity and reproducibility in research employing complex models like BRIDGE is the systematic determination of the model architecture. Ad hoc choices for hyperparameters (e.g., number of layers, units per layer, learning rate) can introduce researcher degrees of freedom, potentially impacting results and hindering replication.

To address this, we used Optuna, a Bayesian hyperparameter optimization framework, to identify the optimal neural network architecture for the BRIDGE model in this study. Optuna tests different network configurations to find the best-performing design, replacing manual trial-and-error with a structured, reproducible process. It allows researchers to pre-specify a *search space* and *optimization procedure* rather than a fixed architecture, minimizing subjectivity and supporting transparent, preregistrable research practices (as discussed in §B.3).

We configured Optuna to use Tree-structured Parzen Estimator (TPE), an intelligent Bayesian search strategy that builds a probabilistic model of performance based on previously tested configurations. Unlike simpler methods such as grid search, TPE progressively "learns" which areas of the hyperparameter space are most promising, making the search more efficient as it proceeds.

The model architecture being tuned included components for both classification and contrastive learning. The tuner was configured to find the architecture that minimized the combined validation loss (`val_loss`), which represents a weighted sum of three component losses: sparse categorical cross-entropy for predicting `region`, sparse categorical cross-entropy for predicting `varietal`, and a custom contrastive loss. The contrastive loss component encourages the model to learn representations where embeddings from similar wines (anchor vs. positive, implicitly defined) are closer than embeddings from dissimilar wines (anchor vs. negative samples provided in the input structure). The relative contribution of the contrastive loss was controlled by a fixed weight (`contrastive_weight` = 0.1) and its sensitivity by a fixed temperature parameter (`contrastive_temp` = 0.1).

The evaluation was performed on a held-out validation set representing 10% of the full dataset. We executed 75 trials in total: 16 initial trials explored the hyperparameter space randomly, followed by 59 trials guided by the Bayesian optimization algorithm. Within each trial, the candidate model's training was regulated by callbacks: early stopping halted training if validation loss did not improve for 10 epochs, and the learning rate was reduced if progress plateaued, ensuring efficient and fair evaluation of each configuration.

The specific hyperparameters explored during this search included:

- **Input Embedding Trimming (`mask_size`):** The number of initial dimensions to retain from the input embedding vector (e.g., 3072 for `text-embedding-3-large`). Tested values: {128, 256, 512, 1024, 2048, 3072}. The optimal value found was 2048.
- **Projection Layer Units (`projection_units`):** The number of units in the first dense layer after the optional trimming. Tested range: 64 to 128.
- **Reduction Layer Units (`embedding_units_per_attribute`):** The number of units allocated to each attribute in the second dense layer, before the representation is split into region and varietal components. Tested range: 4 to 16 per attribute (8 to 32 units total across the two attributes).
- **Learning Rate:** The learning rate for the AdamW optimizer. Tested range: 1e-4 to 1e-2.
- **Weight Decay (`weight_decay`):** The L2 regularization strength applied by the AdamW optimizer. Tested range: 1e-4 to 1e-1.

Other architectural choices were fixed: GeLU activation for hidden layers, a dropout rate of 0.125 after projection and reduction layers, the AdamW optimizer, a training batch size of 8, and a mini-batch size of 10 (the group of related descriptions over which the contrastive loss is computed).

The Optuna implementation code used for the hyperparameter search is available in the OSF repository (§A).

F.2.1 Optimal Model Structure

Hyperparameter tuning identified an optimal model configuration balancing performance and complexity. This process determined key architectural parameters, including the dimensionality at various stages and the learning rate used for training. The resulting BRIDGE model processes the initial high-dimensional wine description embeddings through several layers designed to reduce dimensionality while preserving relevant attribute information. A crucial step involves splitting the intermediate representation into distinct components, yielding the final 8-dimensional vectors for region and varietal used in the econometric analysis. The model also incorporates a contrastive learning component during training to further enhance the distinctiveness of these attribute representations. Table W5 summarizes the key layers and transformations in this final architecture.

F.2.2 Classification Performance

The trained BRIDGE model achieved strong classification accuracy, confirming that the learned 8-dimensional representations meaningfully capture the focal attribute dimensions. Table W6 summarizes the classification performance. The reported metrics are computed by scoring the full corpus of 119,955 descriptions with the trained model (training data included); the held-out split described above served hyperparameter selection. Each wine's embedding input comprises the full record — the name, the country and region fields, the varietal, and the tasting note — so the metrics reflect how faithfully the compressed representations preserve the attributes present in the record, not inference of hidden labels from prose.

Table W5: BRIDGE Architecture for Wine Attribute Representation

Layer / Function	Output Shape	Param #
Input (Original Embeddings)	(B, M, 3072)	0
Embedding Trimming	(B, M, 2048)	0
Projection Layer (Dense + Dropout)	(B, M, 128)	262,272
Reduction Layer (Dense + Dropout)	(B, M, 16)	2,064
Representation Splitting	[(B, M, 8), (B, M, 8)]	0
Training-Specific Layers:		
Anchor Selection (Region)	(B, 8)	0
Anchor Selection (Varietal)	(B, 8)	0
Region Classifier Head	(B, 426)	3,834
Varietal Classifier Head	(B, 708)	6,372
Contrastive Loss Calculation	(B)	0
Total params		**274,542**
Trainable params		**274,542**
Non-trainable params		0

Note. Output shapes indicate dimensions as (Batch Size B, Mini-Batch Size M, Features). The 8-dimensional outputs from the "Representation Splitting" layer (one for region, one for varietal) are the final representations used in the main analysis. The layers listed under "Training-Specific Layers" are primarily used during model training for classification and contrastive learning guidance. The number of output units for region (426) and varietal (708) corresponds to the number of unique attribute levels in the dataset.

Table W6: BRIDGE Classification Accuracy

Attribute	Classes	Accuracy	Weighted F1	Top-5 Accuracy	Chance
Region	426	98.2%	97.6%	99.5%	0.23%
Varietal	708	98.0%	97.4%	99.5%	0.14%

Note. Weighted F1 accounts for class imbalance by weighting each class's F1 score by its frequency. Top-5 accuracy measures the proportion of examples for which the true class appeared among the model's five most confident predictions. Chance baselines assume uniform prediction across all classes (i.e., $1/K$ where K is the number of classes).

F.3 Sensitivity to Training Data Size

To assess the robustness of the BRIDGE representation learning process and provide empirical context for data requirements (as discussed in §B.1.1), we conducted a sensitivity analysis examining the impact of training data size on model performance. We systematically evaluated the *fixed optimal BRIDGE neural network architecture* (as determined by Optuna and including the contrastive loss component; see §F.2) when trained on progressively larger subsets of the available labeled wine descriptions. Specifically, we trained distinct instances of the optimal model architecture from scratch using random subsets corresponding to 5%, 10%, 15%, …, 95%, and finally 100% of the full training dataset derived from the original corpus (N $\approx$ 120,000 descriptions, filtered for valid anchor embeddings). For each training run, performance was evaluated based on the validation accuracy achieved for predicting the `region` and `varietal` attributes, as well as the overall weighted validation loss, on a *consistent, held-out validation set* (representing 20% of the filtered dataset, never used during training for any subset size). The accuracy metrics directly reflect how well the model, with its fixed architecture, learns to extract the relevant attribute information from the text embeddings as the amount of training data increases.

The Python code for the training data sensitivity analysis is available in the OSF repository (§A).

F.3.1 Results and Discussion

The sensitivity analysis shows how the BRIDGE model's predictive accuracy on the held-out validation set changes as the amount of training data increases. Figure W5 plots validation accuracy for region and varietal prediction (left axis) and the overall weighted validation loss (right axis) against the percentage of the full training dataset used. The corresponding numerical results are summarized in Table W7.

As illustrated in Figure W5 and Table W7, model performance, measured by validation accuracy, generally improves as the training data size increases, while the validation loss decreases. Region accuracy starts at approximately 87.8% with only 5% of the training data and reaches a plateau

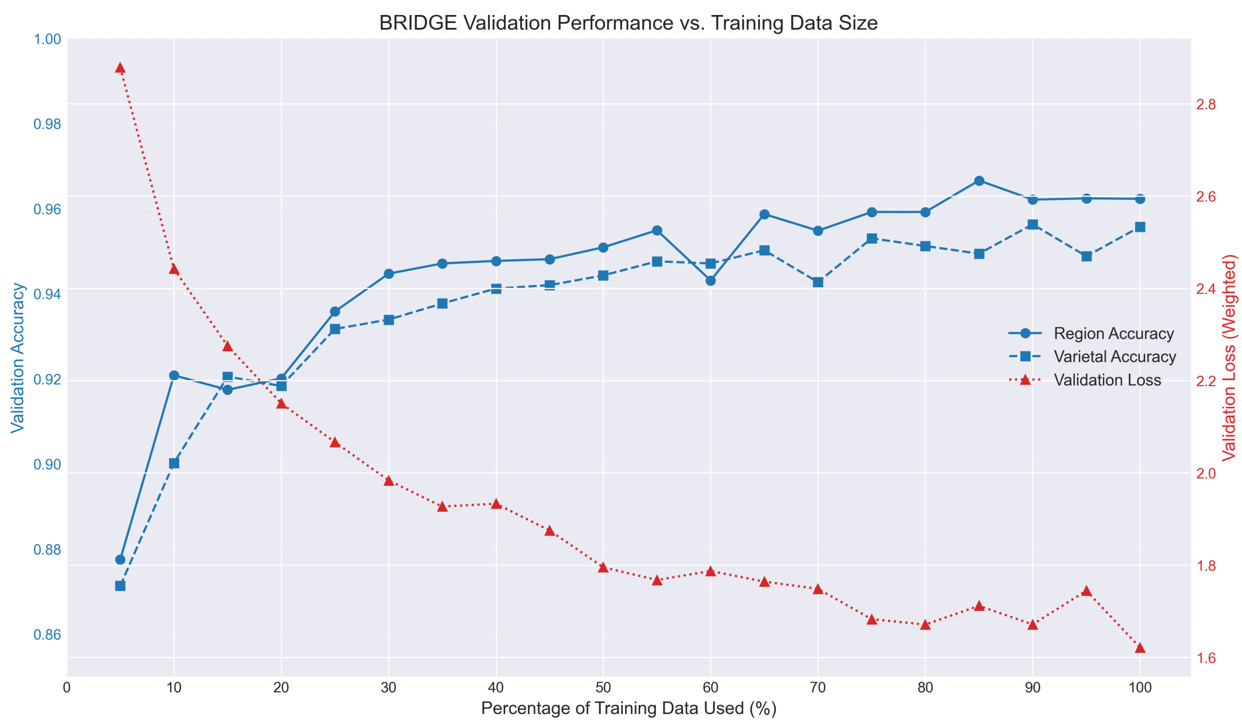


Figure W5: Validation Accuracy and Loss vs. Training Data Percentage

Note: This figure shows the validation accuracy for region and varietal prediction, along with the overall weighted validation loss, achieved by the optimal BRIDGE architecture (including the contrastive loss component) when trained on different percentages of the full training dataset. Performance is evaluated on a fixed, held-out validation set.

Table W7: Validation Performance vs. Training Data Percentage

Train %	Val Loss	Region Acc	Varietal Acc
5%	2.8791	0.8775	0.8713
10%	2.4423	0.9208	0.9002
15%	2.2754	0.9174	0.9205
20%	2.1508	0.9201	0.9183
25%	2.0666	0.9358	0.9317
30%	1.9837	0.9447	0.9339
35%	1.9270	0.9471	0.9377
40%	1.9333	0.9477	0.9412
45%	1.8752	0.9481	0.9420
50%	1.7954	0.9509	0.9443
55%	1.7678	0.9549	0.9476
60%	1.7873	0.9431	0.9471
65%	1.7643	0.9587	0.9502
70%	1.7487	0.9548	0.9427
75%	1.6829	0.9592	0.9530
80%	1.6712	0.9592	0.9512
85%	1.7118	0.9666	0.9494
90%	1.6715	0.9621	0.9563
95%	1.7447	0.9624	0.9488
100%	1.6214	0.9623	0.9557

Note. ‘Val Loss’ is the overall weighted validation loss. ‘Region Acc’ and ‘Varietal Acc’ are the validation accuracies for the respective classification tasks.

around 96.2% when using 90–100% of the data. Varietal accuracy follows a similar trend, starting at 87.1% and reaching approximately 95.6% with the full dataset. The curves exhibit diminishing returns: the largest gains in accuracy occur with the initial increases in data size (e.g., from 5% to 30%), while adding more data beyond roughly 50–60% yields smaller incremental improvements in accuracy, although the validation loss continues to decrease slightly.

This analysis complements the general recommendations provided in the Researcher's Guide (§B.1.1) by demonstrating the model's relative robustness to variations in training set size once a sufficient baseline amount of data (e.g., >20% in this case) is available. Researchers can adapt this sensitivity analysis procedure to their own contexts and datasets to empirically gauge data sufficiency and understand how data limitations might impact the effectiveness of the learned representations.

F.4 Model Fit and Sequence Analysis for Anchoring Effect

To provide statistical support for the anchoring model specification presented in the manuscript (Table 3) and to further investigate the dynamics of the anchoring effect, this section presents detailed model comparison results using the Leave-One-Out Information Criterion (LOOIC) and analyses incorporating task sequence effects.

F.4.1 Single Predictor Model Comparisons

As a baseline analysis, we first compare an intercept-only model (M0) against models including only a single similarity predictor: "Similarity to First Task" (F1), "Similarity to Previous Task" (P1), "Similarity to Next Task" (N1), or "Similarity to Final Task" (L1). Table W8 presents the coefficient estimates and LOOIC values for these models.

The results in Table W8 show that among the single similarity predictors, only "Similarity to First Task" has a coefficient that is credibly different from zero (Estimate = 0.041, 95% CI [0.016, 0.066]). The coefficients for "Similarity to Previous Task" (recency) and the placebo terms ("Similarity to Next Task", "Similarity to Final Task") are statistically nonsignificant, as their

Table W8: Results: Single Predictor Models

Effect	**M0**	**F1**	**P1**	**N1**	**L1**
Intercept	0.083	0.083	0.084	0.083	0.083
	(0.012)	(0.012)	(0.012)	(0.012)	(0.012)
Similarity to First Task	—	0.041	—	—	—
	—	(0.013)	—	—	—
Similarity to Previous Task	—	—	0.023	—	—
	—	—	(0.013)	—	—
Similarity to Next Task	—	—	—	0.004	—
	—	—	—	(0.013)	—
Similarity to Final Task	—	—	—	—	0.006
	—	—	—	—	(0.013)
LOOIC	38770	38761	38768	38772	38771

Note. Bayesian logistic regression results. Estimates are posterior means, standard errors (Est.Error) in parentheses. Dashes (—) indicate the variable is not included. LOOIC (Leave-One-Out Information Criterion) reported for model comparison (lower is better). Based on 28,000 observations. Model names correspond to: M0 (Intercept Only), F1 (Similarity to First Task), P1 (Similarity to Previous Task), N1 (Similarity to Next Task), L1 (Similarity to Final Task).

95% credible intervals include zero. The LOOIC comparison confirms that Model F1 (LOOIC = 38761) provides the best fit among this set, outperforming the intercept-only model (M0; LOOIC = 38770) and models based only on recency (P1; LOOIC = 38768) or placebo similarities (N1, L1). This provides initial evidence supporting the primacy anchoring effect, consistent with the main hypothesis.

F.4.2 Multi-Predictor Model Comparisons

This section presents the detailed results for the sequence of fixed-effects models (M1–M4) estimated to test the anchoring hypothesis while controlling for recency, placebo effects, and nuisance variables. Model F1 (Similarity to First Task only) from §F.4.1 is included for comparison.

The coefficient estimates in Table W9 reinforce the findings from the single predictor models: "Similarity to First Task" remains the only consistently significant similarity predictor across all fixed-effects specifications (M1–M4). Its posterior mean estimate is positive and its 95% credible interval excludes zero in all models. The coefficients for recency, placebo similarities ($\mathrm{Sim}^{\mathrm{Next}}$,

Table W9: Results: Fixed-Effects Multi-Predictor Models

Effect	**F1**	**Model 1**	**Model 2**	**Model 3**	**Model 4**
Intercept	0.083	0.084	0.084	0.083	0.083
	(0.012)	(0.012)	(0.012)	(0.012)	(0.012)
Similarity to First Task	0.041	0.039	0.041	0.038	0.039
	(0.013)	(0.014)	(0.014)	(0.014)	(0.014)
Similarity to Previous Task	—	0.017	0.017	0.015	0.016
	—	(0.014)	(0.014)	(0.013)	(0.014)
Similarity to Next Task	—	-0.007	-0.007	-0.009	-0.009
	—	(0.014)	(0.014)	(0.014)	(0.014)
Similarity to Final Task	—	-0.003	-0.003	-0.005	-0.004
	—	(0.014)	(0.014)	(0.014)	(0.014)
Similarity to Second Task	—	—	0.001	—	-0.001
	—	—	(0.014)	—	(0.014)
Similarity to Two Tasks Before	—	—	-0.011	—	-0.012
	—	—	(0.014)	—	(0.014)
Similarity to Two Tasks After	—	—	-0.009	—	-0.009
	—	—	(0.014)	—	(0.013)
Similarity to Second-to-Final Task	—	—	0.013	—	0.012
	—	—	(0.014)	—	(0.014)
Nuisance Controls:					
Nuisance Control 1	—	—	—	-0.037	-0.040
	—	—	—	(0.035)	(0.035)
Nuisance Control 2	—	—	—	-0.389	-0.389
	—	—	—	(0.071)	(0.068)
Nuisance Control 3	—	—	—	-0.566	-0.565
	—	—	—	(0.082)	(0.082)
Nuisance Control 4	—	—	—	-0.130	-0.128
	—	—	—	(0.090)	(0.091)
Nuisance Control 5	—	—	—	0.089	0.088
	—	—	—	(0.096)	(0.095)
LOOIC	38761	38766	38772	38694	38700

Note. Bayesian logistic regression results. Estimates are posterior means, standard errors (Est.Error) in parentheses. Dashes (—) indicate the variable is not included. Nuisance controls (1–5) are derived from BRIDGE. LOOIC (Leave-One-Out Information Criterion) reported for model comparison (lower is better).

$\text{Sim}^{\text{Final}}$), and the additional similarity terms ($\text{Sim}^{\text{Second}}$, etc.) are generally nonsignificant.

The LOOIC comparison shows that Model 3 (LOOIC = 38694), which includes the four similarity anchors along with the BRIDGE-derived nuisance controls, provides the best fit among these fixed-effects models. Adding the nuisance controls (comparing M3 to M1) results in a substantial improvement in fit (ΔLOOIC $\approx$ 72). However, adding the additional similarity terms (comparing M4 to M3, or M2 to M1) does not improve fit relative to complexity; Model 4 performs slightly worse than Model 3 (LOOIC = 38700 vs. 38694), and Model 2 performs worse than Model 1 (LOOIC = 38772 vs. 38766). This highlights the importance of both the primacy anchoring effect and the nuisance controls, while indicating that the additional similarity terms do not add significant explanatory power.

F.4.3 Heterogeneity Model Comparisons

This section compares the fixed-effects models that include nuisance controls (M3, M4) with their corresponding random-effects counterparts (M5, M6). These latter models account for participant-level heterogeneity by allowing coefficients to vary across individuals. Specifically, Model 5 adds heterogeneity to the specification of Model 3, while Model 6 adds heterogeneity to the specification of Model 4. Table W10 presents the full results for these four models, including fixed-effect estimates, the standard deviations of the random effects for M5 and M6, and LOOIC values for model comparison.

Table W10: Fixed-Effects vs. Heterogeneity Models

Effect	**Model 3** (Fixed Eff.)	**Model 4** (Fixed Eff.)	**Model 5** (Het. on M3)	**Model 6** (Het. on M4)
Fixed Effects:				
Intercept	0.083	0.083	0.089	0.089
	(0.012)	(0.012)	(0.012)	(0.013)
Similarity to First Task	0.038	0.039	0.038	0.042
	(0.014)	(0.014)	(0.015)	(0.016)
Similarity to Previous Task	0.015	0.016	0.015	0.016
	(0.013)	(0.014)	(0.015)	(0.015)

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Table W10: Fixed-Effects vs. Het. Models (Contd.)

Effect	**Model 3** (Fixed Eff.)	**Model 4** (Fixed Eff.)	**Model 5** (Het. on M3)	**Model 6** (Het. on M4)
Similarity to Next Task	-0.009	-0.009	-0.010	-0.009
	(0.014)	(0.014)	(0.014)	(0.014)
Similarity to Final Task	-0.005	-0.004	-0.004	-0.002
	(0.014)	(0.014)	(0.015)	(0.015)
Similarity to Second Task	—	-0.001	—	-0.003
	—	(0.014)	—	(0.016)
Similarity to Two Tasks Before	—	-0.012	—	-0.011
	—	(0.014)	—	(0.015)
Similarity to Two Tasks After	—	-0.009	—	-0.013
	—	(0.013)	—	(0.015)
Similarity to Second-to-Final Task	—	0.012	—	0.010
	—	(0.014)	—	(0.016)
Nuisance Control 1 (Nuisance_1)	-0.037	-0.040	-0.039	-0.046
	(0.035)	(0.035)	(0.037)	(0.037)
Nuisance Control 2 (Nuisance_2)	-0.389	-0.389	-0.425	-0.431
	(0.071)	(0.068)	(0.101)	(0.103)
Nuisance Control 3 (Nuisance_3)	-0.566	-0.565	-0.593	-0.605
	(0.082)	(0.082)	(0.092)	(0.091)
Nuisance Control 4 (Nuisance_4)	-0.130	-0.128	-0.151	-0.151
	(0.090)	(0.091)	(0.100)	(0.100)
Nuisance Control 5 (Nuisance_5)	0.089	0.088	0.104	0.110
	(0.096)	(0.095)	(0.108)	(0.110)
Random Effects SDs (σ_β):				
sd(Intercept)	—	—	0.048	0.050
	—	—	(0.032)	(0.033)
sd(Sim First)	—	—	0.149	0.112
	—	—	(0.033)	(0.044)
sd(Sim Prev)	—	—	0.073	0.067
	—	—	(0.041)	(0.039)
sd(Sim Next)	—	—	0.038	0.038
	—	—	(0.028)	(0.026)
sd(Sim Final)	—	—	0.128	0.092
	—	—	(0.040)	(0.046)
sd(Sim Second)	—	—	—	0.185
	—	—	—	(0.030)
sd(Sim Two Before)	—	—	—	0.053
	—	—	—	(0.036)
sd(Sim Two After)	—	—	—	0.073
	—	—	—	(0.041)

Continued on next page

Table W10: Fixed-Effects vs. Het. Models (Contd.)

Effect	**Model 3** (Fixed Eff.)	**Model 4** (Fixed Eff.)	**Model 5** (Het. on M3)	**Model 6** (Het. on M4)
sd(Sim Second-to-Final)	—	—	—	0.117
	—	—	—	(0.043)
sd(Nuisance 1)	—	—	0.156	0.152
	—	—	(0.097)	(0.097)
sd(Nuisance 2)	—	—	2.125	2.131
	—	—	(0.116)	(0.118)
sd(Nuisance 3)	—	—	0.867	0.870
	—	—	(0.248)	(0.252)
sd(Nuisance 4)	—	—	0.940	0.957
	—	—	(0.279)	(0.287)
sd(Nuisance 5)	—	—	1.078	1.055
	—	—	(0.273)	(0.311)
LOOIC	38694	38700	38320	38293

Note. Bayesian hierarchical logistic regression results. Estimates are posterior means, standard errors (Est.Error) in parentheses. Dashes (—) indicate the term is not included or not estimated (e.g., random effects for fixed-effects models). Similarity to First Task = $\text{Sim}^{\text{First}}$; Similarity to Previous Task = Sim^{Prev}; Similarity to Next Task = Sim^{Next}; Similarity to Final Task = $\text{Sim}^{\text{Final}}$; Similarity to Second Task = $\text{Sim}^{\text{Second}}$; Similarity to Two Tasks Before = $\text{Sim}^{\text{Two Before}}$; Similarity to Two Tasks After = $\text{Sim}^{\text{Two After}}$; Similarity to Second-to-Final Task = $\text{Sim}^{\text{Second-to-Final}}$. Random effects SDs ($\sigma_\beta$) represent the estimated standard deviation of the participant-level coefficients around the fixed-effect mean. LOOIC (Leave-One-Out Information Criterion) reported for model comparison (lower is better).

Table W10 shows that our primary finding—a positive and significant fixed effect for "Similarity to First Task" ($\text{Sim}^{\text{First}}$)—holds even when accounting for participant-level heterogeneity in Models 5 and 6. The fixed-effect estimate remains credibly different from zero in both M5 and M6. Other similarity terms remain nonsignificant on average across participants.

The random effects standard deviations (σ_β) indicate the extent of heterogeneity across participants for each coefficient. For $\text{Sim}^{\text{First}}$, the estimated SD is credibly positive in both M5 ($\sigma_\beta = 0.149$) and M6 ($\sigma_\beta = 0.112$), suggesting meaningful variation in the strength of the anchoring effect across individuals. Several nuisance controls, particularly Nuisance 2, 3, 4, and 5, also exhibit substantial heterogeneity (e.g., $\sigma_\beta \approx 2.13$ for Nuisance 2 in M5), indicating participants varied considerably in their sensitivity to the underlying textual features captured by these controls.

Incorporating participant-level heterogeneity leads to a dramatic improvement in model fit, as assessed by the LOOIC. Comparing M5 to M3, the LOOIC drops from 38694 to 38320 (ΔLOOIC $\approx$ 374). Similarly, comparing M6 to M4, the LOOIC drops from 38700 to 38293 (ΔLOOIC $\approx$ 407). Model 6 achieves the best LOOIC (38293), improving upon Model 5 (38320) by ΔLOOIC $\approx$ 27, indicating that the additional similarity terms in M6 provide meaningful improvement when individual heterogeneity is accounted for. This demonstrates the importance of accounting for individual differences when modeling these preference dynamics. Despite the presence of significant heterogeneity, the average anchoring effect remains robust.

F.4.4 Sequence Interaction Models

To test whether the anchoring effect wanes over the course of the experiment (e.g., due to fatigue), we estimated models including an interaction term between "Similarity to First Task" ($\text{Sim}^{\text{First}}$) and the "Sequence" of each task, defined as its position in the 32-task sequence (the models are estimated on tasks 3–30). Table W11 presents the results for the key terms in these sequence interaction models. The models correspond to the fixed-effects specifications M1, M2, M3, and M4, but with the addition of the "Sequence" and its interaction with $\text{Sim}^{\text{First}}$.

Table W11: Results: Sequence Interaction Models

Effect	**M1_seq**	**M2_seq**	**M3_seq**	**M4_seq**
Intercept	0.016	0.016	0.017	0.018
	(0.027)	(0.028)	(0.027)	(0.028)
Similarity to First Task ($\text{Sim}^{\text{First}}$)	0.067	0.070	0.067	0.070
	(0.030)	(0.030)	(0.030)	(0.029)
$\text{Sim}^{\text{First}} \times$ Sequence	-0.002	-0.002	-0.002	-0.002
	(0.002)	(0.002)	(0.002)	(0.002)
Sequence	0.004	0.004	0.004	0.004
	(0.001)	(0.001)	(0.001)	(0.002)
Other Sim. Terms Included?	Yes (M1)	Yes (M2)	Yes (M1)	Yes (M2)
Nuisance Controls Included?	No	No	Yes	Yes

Note. Bayesian logistic regression results showing key coefficients for sequence interaction models. Estimates are posterior means, standard errors (Est.Error) in parentheses. "Other Sim. Terms Included?" indicates whether the model included the same set of similarity terms as the corresponding model (e.g., M1_seq includes Sim^{Prev}, Sim^{Next}, $\text{Sim}^{\text{Final}}$ like M1).

Across all four sequence interaction models (M1_seq to M4_seq), the interaction term "$\text{Sim}^{\text{First}}$ × Task Sequence" is consistently indistinguishable from zero (posterior mean ≈ -0.002), with the 95% credible interval including zero in every specification. This indicates that the anchoring effect remains stable as participants progress through the tasks.

There is a small, positive main effect of Task Sequence, suggesting participants show a slight, general drift toward selecting the right-hand option as the session progresses—a pattern that may reflect decision fatigue—but this drift neither weakens nor strengthens the anchoring effect. The influence of the first set of options remains essentially constant from the opening trial to the end of the experiment. Such temporal stability is consistent with comparison-based accounts of anchoring, in which the accessibility changes a comparison induces are long-lasting rather than transient (Mussweiler 2003).

F.4.5 Decomposition of the Anchoring Effect

To distinguish between exogenous anchoring and the revelation of stable preferences, we decomposed the $\text{Sim}^{\text{First}}$ variable into two components: similarity to the wine the participant *chose* in the first task ($\text{Sim}^{\text{Chosen First}}$) and similarity to the wine the participant *did not choose* ($\text{Sim}^{\text{Unchosen First}}$). We re-estimated each of the six model specifications (M1–M6) with the two components in place of $\text{Sim}^{\text{First}}$; all other terms are as in the original specifications, so Models 2, 4, and 6 retain the second-task and lag-2 similarity terms, Models 3–6 retain the nuisance controls, and Models 5 and 6 retain participant random slopes on all predictors. If the observed effect were driven by stable preferences (i.e., a "consistent chooser" mechanism), we would expect $\text{Sim}^{\text{Chosen First}}$ to strongly predict subsequent choices, as participants would continue to select wines similar to the one they initially preferred. Table W12 presents the results.

Across all six specifications, the coefficient for similarity to the *chosen* wine is null (-0.002 to 0.006, all $p > .76$), whereas the coefficient for similarity to the *unchosen* wine is positive and significant (0.072 to 0.085, all $p < .001$); the difference between the two coefficients is itself significant in every specification (posterior mean difference 0.067 to 0.087, $p \leq .019$). This pattern

Table W12: Results: Decomposition of the Anchoring Effect Across All Model Specifications

Effect	**Model 1**	**Model 2**	**Model 3**	**Model 4**	**Model 5**	**Model 6**
Intercept	0.083	0.083	0.084	0.083	0.089	0.090
	(0.012)	(0.012)	(0.012)	(0.012)	(0.013)	(0.013)
Similarity to *Chosen* First Task Wine	0.004	0.005	0.004	0.006	−0.002	−0.001
	(0.020)	(0.020)	(0.020)	(0.020)	(0.022)	(0.023)
Similarity to *Unchosen* First Task Wine	0.075	0.077	0.072	0.073	0.080	0.085
	(0.020)	(0.020)	(0.020)	(0.020)	(0.023)	(0.023)
Similarity to Previous Task	0.017	0.018	0.015	0.016	0.015	0.016
	(0.013)	(0.014)	(0.013)	(0.014)	(0.014)	(0.015)
Similarity to Next Task	−0.007	−0.007	−0.009	−0.009	−0.010	−0.008
	(0.013)	(0.014)	(0.014)	(0.014)	(0.015)	(0.015)
Similarity to Final Task	−0.004	−0.003	−0.006	−0.004	−0.005	−0.003
	(0.014)	(0.014)	(0.013)	(0.014)	(0.015)	(0.016)
Similarity to Second Task	—	0.000	—	−0.001	—	−0.002
	—	(0.014)	—	(0.014)	—	(0.017)
Similarity to Two Tasks Before	—	−0.011	—	−0.012	—	−0.011
	—	(0.014)	—	(0.014)	—	(0.015)
Similarity to Two Tasks After	—	−0.008	—	−0.008	—	−0.013
	—	(0.014)	—	(0.014)	—	(0.015)
Similarity to Second-to-Final Task	—	0.014	—	0.012	—	0.011
	—	(0.014)	—	(0.014)	—	(0.016)
Nuisance Controls 1–5	—	—	Incl.	Incl.	Incl.	Incl.
Participant Random Slopes	—	—	—	—	Incl.	Incl.

Note. Bayesian logistic regression. Estimates are posterior means, standard errors (Est.Error) in parentheses. Each column re-estimates the corresponding model specification (M1–M6) with $\text{Sim}^{\text{First}}$ replaced by its chosen and unchosen components; all other terms are as in the original specification. Dashes (—) indicate the variable is not included. Models 5 and 6 include participant random slopes on all predictors; reported estimates are population-level effects. Based on 28,000 observations.

indicates that the anchoring effect is not driven by the revelation of pre-existing preferences: similarity to the chosen wine—the component a stable-preference account predicts—does not predict later choice, whereas similarity to the unchosen wine does, supporting an exogenous anchoring account. The conceptual account underlying our framework suggests that comparisons are directional: a target is evaluated against a standard, and the standard organizes the judgment (Mussweiler 2003). The asymmetry is consistent with the possibility that the wine a participant came to favor in the first task served as the target, whereas the unchosen wine served as the comparison standard, carrying the initial reference frame into later tasks.

F.5 Benchmark Comparisons

To evaluate the precision and validity of BRIDGE relative to alternative approaches, we estimated the anchoring effect using several benchmark specifications. Table W13 presents the complete results.

Table W13: Benchmark Comparison: Anchoring Effect Estimates Across Alternative Specifications

	Specification	**Input**	**Dims**	**First_1 $\hat{\beta}$**	**95% CI**	**CI Width**
Panel A: BRIDGE and Controls						
(F)	BRIDGE (with nuisance)	OpenAI → BRIDGE	8	0.038	[0.008, 0.068]	0.059
(A)	BRIDGE (no nuisance)	OpenAI → BRIDGE	8	0.039	[0.013, 0.065]	0.052
(B)	BRIDGE + Empath controls	OpenAI → BRIDGE	8	0.040	[0.015, 0.067]	0.052
Panel B: Raw Embedding Approaches						
(C)	Full OpenAI embedding	OpenAI raw	3,072	0.215	[0.030, 0.398]	0.368
	Nuisance-only	OpenAI nuisance	5	−0.007	[−0.030, 0.015]	0.045
Panel C: Non-Embedding Approaches						
(D)	Jaccard (attribute-only)	Binary matching	2	0.065	[−0.002, 0.128]	0.130
(E)	Linear Projection	OpenAI → Ridge+SVD	8	0.135	[−0.000, 0.269]	0.269

Note. Most models use Bayesian logistic regression with random intercepts for participants; rows (C) and nuisance-only are fixed-effects only, and row (F) includes random slopes for all predictors. Estimates are posterior means; 95% credible intervals in brackets. CI Width is the width of the 95% credible interval. "Dims" refers to the dimensionality of the similarity measure. Letters (A)–(F) label the benchmark specifications. BRIDGE models use cosine similarity on 8-dimensional learned representations. Raw embedding models use cosine similarity on the full embedding vectors. Jaccard uses binary attribute matching: $(I(\text{same region}) + I(\text{same varietal}))/2$. Linear Projection applies Ridge regression of embeddings on attribute labels, followed by SVD on residuals.

Several patterns emerge from Table W13. First, BRIDGE provides the tightest inference: its

CI width (0.059) is approximately 6 times narrower than raw OpenAI embeddings (0.368) and approximately 5 times narrower than Linear Projection (0.269). Second, the full OpenAI embedding yields a significant but inflated estimate ($\hat{\beta} = 0.215$) with a substantially wider credible interval, because the raw embedding similarity incorporates both attribute variation—which is theoretically and empirically central to the anchoring effect—and nuisance variation (tone, style, length) that is irrelevant to the effect. This conflation inflates both the point estimate and the uncertainty. Without nuisance controls, the same full-embedding similarity fails to reach significance ($\hat{\beta} = 0.177$, 95% CI [-0.005, 0.354]). Third, the nuisance-only model yields a null result ($\hat{\beta} = -0.007$), confirming that the anchoring effect is driven by attribute similarity, not by superficial textual resemblance. Fourth, Jaccard attribute matching cannot detect the effect ($\hat{\beta} = 0.065$, $p = .056$) because only 12.8% of trial pairs share any attribute (9.9% region, 4.2% varietal, 1.3% both), and 67% of trial-level Jaccard similarity scores are exactly zero, creating a near-degenerate similarity distribution. Fifth, Linear Projection detects a marginal effect ($\hat{\beta} = 0.135$, $p = .051$) but with a CI width approximately 5 times wider than BRIDGE, consistent with the Monte Carlo simulation findings: with 426 regions and 708 varietals, the wine descriptions plausibly exhibit the kind of interaction effect confounding that requires BRIDGE's learned representations.

The Empath controls model (Benchmark B) shows that adding 10 SVD-reduced Empath psycholinguistic features to the BRIDGE similarity measures does not alter the anchoring estimate (0.040 vs. 0.039 without controls). Five of ten Empath dimensions are independently significant predictors of choice, confirming that textual features matter for wine preferences, but they are orthogonal to the anchoring mechanism isolated by BRIDGE.

F.6 Embedding Backend Sensitivity: BERT vs. OpenAI

BRIDGE is embedding-agnostic: any semantic embedding can serve as input to the compression network. To assess how the choice of embedding backend affects downstream inference, we retrained BRIDGE using BERT embeddings (bert-base-nli-mean-tokens, 768 dimensions) as input—the same BRIDGE architecture and training procedure, but with a different embedding backend—

and re-estimated Models 1, 3, and 5. These three models span the range from the most parsimonious specification (four similarity terms only) to the fully specified model with random slopes. All models use the same estimation sample ($N = 28{,}000$) as the main analysis. Table W14 presents the results.

Table W14: Embedding Backend Sensitivity: OpenAI vs. BERT Inputs to BRIDGE

	Model 1		**Model 3**		**Model 5**	
	OpenAI	BERT	OpenAI	BERT	OpenAI	BERT
Sim First	0.039 (0.014)	0.015 (0.010)	0.038 (0.014)	0.011 (0.010)	0.038 (0.015)	0.012 (0.013)
Sim Previous	0.017 (0.014)	0.013 (0.010)	0.015 (0.013)	0.010 (0.010)	0.015 (0.015)	0.011 (0.011)
Sim Next	−0.007 (0.014)	−0.001 (0.010)	−0.009 (0.014)	−0.002 (0.010)	−0.010 (0.014)	−0.005 (0.011)
Sim Final	−0.003 (0.014)	0.003 (0.010)	−0.005 (0.014)	0.001 (0.011)	−0.004 (0.015)	0.003 (0.012)

Note. Posterior means with standard errors in parentheses. M1 = four similarity terms (fixed effects); M3 = M1 + nuisance controls (fixed effects); M5 = M3 + random slopes. OpenAI uses text-embedding-3-large (3,072-dim); BERT uses bert-base-nli-mean-tokens (768-dim). $N = 28{,}000$. OpenAI Sim First: $p = .002$ (M1), $p = .008$ (M3), $p = .014$ (M5); all BERT Sim First coefficients are nonsignificant.

The BERT→BRIDGE anchoring coefficient is directionally consistent ($\hat{\beta} = 0.011$–0.015) but attenuated by roughly two-thirds relative to OpenAI→BRIDGE ($\hat{\beta} = 0.038$–0.039) and is not statistically significant in any specification. All other similarity coefficients remain nonsignificant in both backends, preserving the qualitative pattern. The attenuation is consistent with classical measurement error: the BERT input—an older, lower-dimensional embedding—produces less precise attribute representations, attenuating the coefficient toward zero.

These results confirm that BRIDGE is embedding-agnostic in design but that the quality of the input embeddings matters for the precision of the extracted attribute representations. We therefore present this comparison as a boundary condition as much as a robustness check: BRIDGE's ability to control confounds depends on what the underlying embedding encodes — the legacy BERT embedding supports weaker, less precise inference than the current-generation embedding, and present embeddings may likewise have blind spots that future ones close — so confound-control claims are embedding-specific and should be revalidated as embeddings change. We recommend modern LLM embeddings (e.g., OpenAI text-embedding-3-large or comparable models) for best results; the `bridge` package also supports local alternatives (e.g., Gemma, 768-dim, free) for researchers without API access.

F.7 Name-Cue Robustness

The displayed wine names often contain the wines' focal attributes. This section documents how often, shows that classification performance is not an artifact of the attribute being visible in the name, and tests directly whether the anchoring effect depends on the attributes being shown. Throughout, a name is counted as containing the varietal or region when the wine's own varietal or region string appears verbatim in its title (case-insensitive exact substring match; empty strings never match); the focal region label is the wine's province, and finer appellation strings are not counted. Synonyms and translated attribute names are not matched, so the counts are conservative; varietal strings embedded in proprietary wine names do count. Most titles (83.7%) end in a parenthetical appellation, which is counted only when it contains the focal province string.

Prevalence. About two-thirds of displayed names contain the varietal and about a quarter the region (Table W15); because displayed wines were sampled at random, the corpus fractions and the fractions among the wines actually shown to participants are near-identical.

Table W15: Displayed Wine Names Containing the Wine's Own Varietal and/or Region (Counts)

Cue in displayed name	**Full corpus** (N = 119,955)	**Wines shown to participants** (N = 49,548)
Contains varietal	77,664 (64.7%)	32,011 (64.6%)
Contains region	31,635 (26.4%)	12,985 (26.2%)
Contains both	20,895 (17.4%)	8,516 (17.2%)
Contains either	88,404 (73.7%)	36,480 (73.6%)
Contains neither	31,551 (26.3%)	13,068 (26.4%)

Note. A displayed name is counted as containing the varietal (resp. region) if the wine's own varietal (resp. region) string appears verbatim in its title (case-insensitive exact substring match; empty strings never match). The focal region label is the wine's province; finer appellation strings are not counted. "Wines shown to participants" are the 49,548 unique wines appearing as options across the displayed choice tasks; percentages are of the column total.

Classification by attribute visibility. Table W16 reports the same classification metrics used in the main analysis (accuracy, weighted F1, top-5 accuracy), overall and separately for wines whose displayed names show the varietal, the region, both, or neither. Performance is uniformly high — every subgroup far above the 0.23%/0.14% chance baselines — and it is not higher where the attribute is visible: region accuracy is lowest precisely where the region is visible in the name (93.2% when the name shows the region; 92.1% when it shows both attributes) and highest for wines showing neither (99.96%). Because each wine's embedding input includes its region and varietal fields, these metrics reflect how faithfully the compressed representations preserve the focal attributes present in the record, not inference of hidden labels from prose (see §F.2).

Table W16: Focal-Attribute Classification Validation by Attribute Visibility in the Wine Name

Attribute	**Wines**	***N***	**Accuracy**	**Weighted F1**	**Top-5 Acc.**
Region	Overall	119,955	98.18%	97.64%	99.48%
	Name shows varietal	77,664	97.83%	97.21%	99.41%
	Name shows region	31,635	93.24%	91.49%	98.05%
	Name shows both	20,895	92.14%	90.12%	97.85%
	Name shows neither	31,551	99.96%	99.97%	99.99%
Varietal	Overall	119,955	97.99%	97.44%	99.49%
	Name shows varietal	77,664	97.46%	96.79%	99.31%
	Name shows region	31,635	97.41%	96.77%	99.26%
	Name shows both	20,895	96.87%	96.10%	99.01%
	Name shows neither	31,551	99.13%	98.99%	99.85%

Note. Region (426 classes) and varietal (708 classes) classification for the trained BRIDGE model, overall and by which attribute the displayed name shows. Subgroups overlap ("both" wines also appear in "shows varietal" and "shows region"). Chance baselines are 0.23% (region) and 0.14% (varietal). Each wine's embedding input includes its region and varietal fields, so these figures reflect how faithfully the representations preserve the focal attributes; accuracy is not higher where the name displays the attribute.

Does anchoring depend on whether the attributes are shown? Restricting the anchoring analysis to cue-free subsamples would be both underpowered and uninformative: requiring that neither the anchor task nor the current task display either attribute leaves 115 observations, and the varietal-specific restriction leaves 350 — so a null there would mostly reflect low power — while splitting on the absence of a cue discards the variation that is most informative (the region-specific restriction retains 8,383 observations, but the interactions below test that question directly on the full sample). We therefore retain the full sample (1,000 participants; 28,000 observations) and interact the anchoring term (similarity to the first task) with four indicators for whether the varietal or region was visible in the first (anchor) task and in the current task. Because the design randomizes the wines shown in every task, the four indicators are nearly orthogonal (all six off-diagonal correlations below 0.03 in magnitude), so each interaction is separately identified. Table W17 reports the results across Models 1–6: the average anchoring slope remains 0.035–0.039 (exact $p = .005$–$.020$), no interaction is significant in any specification, and a joint test cannot reject that all four are simultaneously zero ($p = .49$–$.76$). The only term approaching significance is a *positive* coefficient on the current task's varietal visibility ($p = .07$–$.08$ in Models 5 and 6; $p \geq .13$ otherwise), indicating that anchoring is, if anything, modestly stronger — not weaker — when the current wine's name shows its varietal. The anchoring effect thus does not depend on whether the structured attributes appear in the free-form names.

Table W17: Anchoring Does Not Vary with the Presence of Structured Attributes: Interaction Specifications

Anchoring (First Task) $\times$	**M1**	**M2**	**M3**	**M4**	**M5**	**M6**
Varietal in first task	0.004	0.004	−0.007	−0.005	−0.012	−0.018
	(.93)	(.93)	(.85)	(.89)	(.79)	(.68)
Region in first task	−0.008	−0.007	−0.007	−0.007	−0.009	−0.009
	(.76)	(.79)	(.80)	(.78)	(.73)	(.76)
Varietal in current task	0.059	0.060	0.066	0.067	0.082	0.083
	(.17)	(.18)	(.13)	(.14)	(.08)	(.07)
Region in current task	−0.001	−0.001	−0.001	−0.000	−0.004	−0.004
	(.98)	(.99)	(.97)	(.98)	(.89)	(.88)
Avg. anchoring slope	0.038	0.039	0.035	0.037	0.035	0.039
(exact *p*)	(.005)	(.009)	(.008)	(.009)	(.020)	(.011)
Joint test, all four $= 0$ (*p*)	.75	.76	.68	.68	.52	.49

Note. Posterior means of the four interaction coefficients (exact posterior *p* in parentheses) from Bayesian logistic regressions on the full sample (1,000 participants; 28,000 observations), Models 1–6. "Varietal/Region in first [current] task" equals one if at least one wine displayed in the first (anchor) [current] task shows that attribute in its name. "Avg. anchoring slope" is the coefficient on similarity to the first task with the four indicators mean-centered (the population-average slope). The joint test is a posterior Wald test (χ^2_4) that the four interactions are simultaneously zero. No interaction is significant in any specification, and the joint test is nonsignificant throughout.

Web Appendix G: Validation Study with Controlled Stimuli (Coffee)

To enhance the credibility of our main findings and to extend them, we conducted a validation study using carefully controlled mock descriptions. While our primary investigation relied on complex, real-world wine descriptions and an AI-based model (BRIDGE) to handle nuanced text, this follow-up study shifted to the context of coffee—a product category that, like wine, offers rich sensory experiences—and employed conventional analysis techniques. This approach was chosen both (a) to verify the generalizability of the anchoring effects observed in the primary wine study to a different category and (b) to provide support for the efficacy of BRIDGE. By establishing similar results using conventional analytical techniques and simpler, less realistic stimuli, we demonstrate that the observed preference dynamics are a robust phenomenon and not an artifact of our proposed method.

Specifically, we employed the following process to construct controlled yet realistic stimuli. We focused on two key coffee attributes, each with six levels, yielding 36 unique attribute combinations. Building on these structured profiles, we created standardized single-sentence verbal descriptions using a generative AI model. By maintaining a strict definition of each aroma and taste level within these descriptions, we introduced enough complexity to simulate a realistic coffee scenario but avoided confounds that could arise from varying sentence structure (style or tone). This allows us to isolate the core attributes and use straightforward measures of attribute overlap to quantify product similarity, offering a clear point of comparison to the AI-driven methodology employed in our main wine study.

G.1 Data

A total of 611 undergraduate students (69.2% women, average age = 21.7) participated in this study in exchange for course credit. Participants were presented with 18 pairs of coffee descriptions and asked to indicate their relative preference between the coffees in each pair on a 7-point scale (1 = strongly prefer Option A, 7 = strongly prefer Option B). Each coffee in each pair was randomly selected, without replacement, from the total set of 36 coffees. Although all participants were exposed to the same set of 36 coffees, the specific pairings and the sequence of presentation were randomly varied across participants. After completing the evaluation tasks, participants provided basic demographic information (gender, age).

Each coffee description was defined by two attributes—aroma and taste—each with six possible levels (aroma: floral, fruity, nutty, chocolatey, spicy, earthy; taste: bitter, sweet, acidic, sour, umami, balanced). To generate 36 distinct profiles (representing all possible attribute combinations), we enumerated every aroma–taste pairing. For each pairing, we used the autocomplete feature of ChatGPT to help create a short, one-sentence descriptor in a standardized format.

Specifically, we prompted ChatGPT with the following stem: *"This coffee's [adjective] [aroma level] and [adjective] [taste level]…"* to ensure that each description (1) referenced the correct aroma and taste attributes, (2) adopted a similarly neutral and concise tone, and (3) adhered to a single-sentence structure. On average, these descriptions were 24 words long (median = 24 words), which helped maintain a uniform level of detail and complexity across all coffee profiles. By relying on a single-sentence format and a standardized generation process, we aimed to provide enough descriptive richness to simulate a real product context while standardizing extraneous elements. This approach minimized potential confounds related to length or writing style, such that the observed preference dynamics could be attributed to the systematically varied aroma and taste attributes. Table W18 presents example descriptions.

G.2 Variable Operationalization

To test whether participants' preferences in subsequent tasks demonstrated anchoring to the options presented in the initial task, we defined a similarity measure based on the extent of attribute match between the options in each task and those in the initial task. Specifically, for each subsequent choice task, we calculated the sum of matching attribute levels (aroma and taste) between the options in the current task and the options presented in the initial task. This served as our key independent measure of similarity, where a larger value indicated greater alignment with the options shown in the initial task. This operationalization allowed us to test whether

Table W18: Examples of Coffee Description Stimuli

Coffee Description	Length (words)
This coffee's enchanting floral aroma and distinct bitterness weave together, creating a robust depth that invigorates the senses.	18
This coffee's enchanting floral aroma and perfectly balanced profile create a harmonious experience where delicate flavors and smooth depth come together seamlessly.	22
This coffee's exuberant fruity aroma and bright acidity come together to offer a crisp, tart profile that refreshes and delights with every sip.	23
This coffee's soothing nutty aroma and subtle sweetness come together to deliver a smooth, pleasantly sweet sensation that gently unfolds with each sip.	23
This coffee's luxurious chocolatey aroma and savory umami taste create a decadent, full-bodied experience that indulges the senses with layers of richness and complexity.	24
This coffee's aromatic spicy notes and savory umami profile come together to create a rich, full-bodied experience that captivates the senses with its depth and complexity.	26
This coffee's rich earthy aroma and sharp sourness combine to present a bold, tangy profile that intrigues and stimulates the palate with its adventurous complexity.	25
This coffee's vibrant fruity aroma and well-balanced taste deliver a delightful interplay of bright and smooth flavors, resulting in a satisfying and rounded experience.	24

Note. Eight examples from the 36 standardized single-sentence coffee descriptions (one per aroma–taste combination) generated with ChatGPT.

participants' subsequent choices were influenced by the attributes of the options they encountered initially, consistent with the anchoring effects observed in the wine study.

To rule out alternative explanations, particularly those related to unobserved preference heterogeneity, we implemented two key controls within the experimental design and analysis. First, the specific options shown in the initial task were determined randomly for each participant (thus entirely incidental), ensuring that any observed similarity effects were not driven by systematic differences in the initial options encountered across the sample. Second, we computed additional similarity measures to test for recency effects and serve as placebo tests. Specifically, we calculated similarity to the options shown in the final task, the next task, and the previous task, mirroring the measures employed in our wine study. Since the options shown in the next and final tasks were unknown to participants at the time of their current choices, we expected nonsignificant coefficients for these variables. If participants' preferences were primarily driven by primacy effects (anchoring on the first task), we would expect similarity to the first task to significantly explain their choices, whereas similarity to the next and final tasks (our placebo measures) should not.

To operationalize similarity for the analysis, we defined the following variables based on attribute overlap:

1. **Similarity to the First Task (similarity_first)**: This variable measures the degree of attribute match between the options in the current task (t) and those presented in the first task ($t = 1$). For each option (A and B) in the current task, we counted the number of matching attributes (aroma and taste; possible values: 0, 1, or 2) with each option (A and B) in the first task, summing these matches to obtain a total similarity score for each option. The final variable is the difference between the total similarity score for Option B and that for Option A in the current task. A positive value indicates that Option B is more similar to the options in the first task than Option A is, and vice versa. This variable tests for primacy (anchoring) effects.

2. **Similarity to the Previous Task (similarity_previous)**: This variable measures the

degree of attribute match between the options in the current task (t) and those shown in the immediately preceding task ($t-1$). The calculation follows the same procedure as for similarity_first, but uses the options from the previous task as the anchor. This variable tests for immediate recency effects.

3. **Similarity to the Next Task (similarity_next)**: This variable measures the degree of attribute match between the options in the current task (t) and those shown in the next task ($t+1$). The calculation follows the same procedure as above, but uses the options from the next task as the anchor. Since participants could not know the options in the next task at the time of their current choice, this variable serves as a placebo test. A nonsignificant coefficient would confirm that observed effects are not due to spurious correlations or unobserved factors.

4. **Similarity to the Final Task (similarity_last)**: This variable measures the degree of attribute match between the options in the current task (t) and those shown in the final task ($t = 18$). The calculation follows the same procedure as for similarity_first, but uses the options from the final task as the anchor. Like similarity_next, this variable serves as a placebo test, as participants could not anticipate the options in the final task.

In addition to these four main similarity variables, we also created four analogous variables to capture potential anchoring effects at a "two-step" distance from the current task. Specifically, for each task t, we calculated similarity to the second task (similarity_second), similarity to the task two steps before the current task (similarity_two_before), similarity to the task two steps after the current task (similarity_two_after), and similarity to the second-to-final task (similarity_second_last). These variables were computed using the same attribute overlap procedure as the main similarity variables, comparing the options in the current task to those in the specified anchor task. Each variable was defined only when the corresponding anchor task fell within the valid task range (e.g., similarity_two_before was defined only for tasks $t \geq 3$, and similarity_two_after only for tasks $t \leq 16$). Including these additional variables allows for a more comprehensive test of whether the anchoring effect is specific to the first task or extends to other positions in the sequence.

G.3 Model

To test our core hypothesis and evaluate the robustness of the observed anchoring effects, we fit two Bayesian models. Both models used participants' relative preference ratings (on the 1–7 scale, where higher values indicate stronger preference for Option B) as the dependent variable and included the calculated similarity measures as predictors. The models were designed to test whether participants' preferences in subsequent tasks were influenced by the similarity of the current options to the options presented in specific anchor tasks. The two specifications were estimated sequentially to first isolate the primary effects of interest (primacy, recency, placebo) and then to test the robustness and specificity of these effects to the inclusion of the additional similarity measures.

G.3.0.1 Specification 1: Single-Anchor Model The first model focused on the similarity of the current task's options to the options presented in four key anchor tasks: the first task (primacy), the previous task (immediate recency), the next task (placebo), and the final task (placebo). The relative preference rating of participant i in task t was modeled as:

$$\begin{aligned} y_{it} = \beta_0 + \beta_1 \text{similarity first}_{it} + \beta_2 \text{similarity previous}_{it} \\ + \beta_3 \text{similarity next}_{it} + \beta_4 \text{similarity final}_{it} + \varepsilon_{it}. \end{aligned} \tag{W2}$$

Here:

- y_{it} represents the relative preference rating of participant i in task t.
- similarity first_{it}, similarity previous_{it}, similarity next_{it}, and similarity final_{it} are the similarity difference scores (Option B minus Option A) for task t, as defined above.
- β_1, β_2, β_3, and β_4 are the coefficients capturing the influence of each similarity measure on participants' relative preferences.

- ε_{it} is an i.i.d. Gaussian error term.

This model allowed us to test the primacy hypothesis by examining the coefficient for similarity to the first task (β_1). A significant positive coefficient for β_1 would indicate anchoring on the initial options. We tested for immediate recency effects using the coefficient for similarity to the previous task (β_2). Finally, we included two placebo tests: similarity to the next task (β_3) and similarity to the final task (β_4). Since participants could not know the options in the next or final tasks at the time of their current choice, nonsignificant coefficients for β_3 and β_4 would support the validity of the anchoring interpretation and help rule out alternative explanations like unobserved heterogeneity.

G.3.0.2 Specification 2: Two-Anchor Model The second model extended the first by including additional similarity measures that captured anchoring effects at a "two-step" distance from the current task. Specifically, we added similarity to the second task (similarity second), two tasks before the current task (similarity two before), two tasks after the current task (similarity two after), and the second-to-final task (similarity second final). The model for this specification was:

$$\begin{aligned} y_{it} = \beta_0 &+ \beta_1 \text{similarity first}_{it} + \beta_2 \text{similarity previous}_{it} \\ &+ \beta_3 \text{similarity next}_{it} + \beta_4 \text{similarity final}_{it} \\ &+ \beta_5 \text{similarity second}_{it} + \beta_6 \text{similarity two before}_{it} \\ &+ \beta_7 \text{similarity two after}_{it} + \beta_8 \text{similarity second final}_{it} + \varepsilon_{it}. \end{aligned} \tag{W3}$$

This specification allowed us to test whether anchoring effects extended beyond the immediate task context. A significant positive coefficient for β_1 (similarity to the first task) in this extended model would further support the primacy hypothesis, while nonsignificant coefficients for the additional variables (β_5 and β_6) would suggest that anchoring effects were specific to the first task and did not generalize to other anchor tasks or time lags. The coefficients β_7 and β_8 serve as further placebo tests.

G.4 Results

The estimation results, summarized in Table W19, indicate that participants' subsequent preferences were significantly influenced by the coffee options encountered in the very first task. This finding replicates the anchoring phenomenon observed in our primary wine study, providing evidence that initial exposures serve as crucial reference points shaping later decisions, even within a different product category and using controlled stimuli analyzed via conventional methods.

Table W19: Results: Coffee Validation Study

	Single-Anchor Model		**Two-Anchor Model**	
Effect	**Estimate**	**Std. Error**	**Estimate**	**Std. Error**
Intercept	4.04	0.02	4.04	0.02
Similarity to First Task	0.07	0.02	0.07	0.03
Similarity to Previous Task	-0.00	0.02	0.00	0.03
Similarity to Next Task	0.01	0.02	-0.00	0.03
Similarity to Final Task	0.01	0.02	0.01	0.03
Similarity to Second Task	—	—	0.00	0.03
Similarity to Two Tasks Before	—	—	0.02	0.03
Similarity to Two Tasks After	—	—	0.01	0.02
Similarity to Second-to-Final Task	—	—	0.03	0.02

Note. Entries are posterior means and standard errors from Bayesian linear regressions of the 7-point relative preference rating (higher values indicate stronger preference for Option B). Dashes (—) indicate that the variable is not included in the Single-Anchor Model. Single-Anchor $N = 9{,}776$ and Two-Anchor $N = 8{,}554$ task-level observations (of $611 \times 18 = 10{,}998$; tasks without a defined anchor at the sequence boundaries are excluded).

Specifically, the coefficient for `Similarity to First Task` was positive and statistically significant in both the Single-Anchor Model ($\beta_1 = 0.07$, SE = 0.02, $p = .002$) and the more comprehensive Two-Anchor Model ($\beta_1 = 0.07$, SE = 0.03, $p = .008$). This robust positive effect indicates that participants consistently preferred options in later tasks that shared more attributes (aroma and taste) with the options they saw initially, strongly supporting the primacy (anchoring) hypothesis.

Conversely, the coefficients for similarity to the previous task, the next task, and the final task were all nonsignificant in both models. The nonsignificance for `Similarity to Previous`

`Task` indicates that recency was not a driver of preferences. The nonsignificance of `Similarity to Next Task` and `Similarity to Final Task` suggests that the findings are not merely an artifact of spurious correlation: if the results were a function of unobserved heterogeneity, then we would expect all similarity measures to show a significant effect. That is, given that the sequence and pairing of options were randomized across participants, if stable underlying preferences were the main factor, we would expect spurious correlations with similarity to various tasks. The fact that only similarity to the *first* task emerged as significant strongly suggests a genuine anchoring effect tied to the initial exposure.

Furthermore, in the Two-Anchor Model, all the additional similarity measures—`Similarity to Second Task`, `Similarity to Two Tasks Before`, `Similarity to Two Tasks After`, and `Similarity to Second-to-Final Task`—were also nonsignificant. This pattern indicates that the observed anchoring effect was specific to the *first* task and did not extend to other positions in the sequence, further supporting our findings.

G.5 Discussion

The results from this validation study, conducted using controlled mock descriptions and conventional analysis, confirm the anchoring effects observed in our primary study, which used complex, real-world wine descriptions analyzed via BRIDGE. Across both model specifications, participants consistently demonstrated a preference for coffee options in later tasks that were more similar in attributes (aroma and taste) to the options presented in their initial task.

These findings provide strong converging evidence for our main conclusions regarding preference dynamics. They demonstrate that the anchoring phenomenon—whereby initial product exposures significantly shape subsequent preferences—is not an artifact confined to the specific context of complex, unstructured wine descriptions, nor is it dependent on the AI-based analytical methodology (BRIDGE) employed in the main paper. Instead, the anchoring effect emerges consistently across different experiential product domains (wine and coffee) and across varying levels of stimulus complexity and analytical approaches. This consistency suggests that anchoring on initial options represents a robust psychological mechanism influencing sequential consumer decision-making.

Furthermore, by successfully replicating the core finding using controlled stimuli and standard statistical techniques, this study serves to validate the methodological utility of BRIDGE. It shows that BRIDGE can effectively uncover theoretically meaningful behavioral patterns within complex, ecologically valid settings, yielding insights consistent with those obtainable through more traditional, controlled experimental paradigms when applied to simpler contexts. This validation bolsters confidence in both the substantive findings regarding preference anchoring and the broader applicability of the BRIDGE approach for investigating consumer behavior with rich, real-world stimuli.

References

Akiba, Takuya, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama (2019), "Optuna: A next-generation hyperparameter optimization framework," in *The 25th ACM SIGKDD international conference on knowledge discovery & data mining*, 2623–31.

Alba, Joseph W and Elanor F Williams (2013), "Pleasure principles: A review of research on hedonic consumption," *Journal of consumer psychology*, 23 (1), 2–18.

Amir, On and Jonathan Levav (2008), "Choice construction versus preference construction: The instability of preferences learned in context," *Journal of Marketing Research*, 45 (2), 145–58.

Ariely, Dan, George Loewenstein, and Drazen Prelec (2003), "'Coherent arbitrariness': Stable demand curves without stable preferences," *The Quarterly journal of economics*, 118 (1), 73–106.

Baribault, Beth, Chris Donkin, Daniel R Little, Jennifer S Trueblood, Zita Oravecz, Don Van Ravenzwaaij, Corey N White, Paul De Boeck, and Joachim Vandekerckhove (2018), "Metastudies for robust tests of theory," *Proceedings of the National Academy of Sciences*, 115 (11), 2607–12.

Bishop, Christopher M (1995), *Neural networks for pattern recognition*, Oxford university press.

Blackwell, Matthew and Michael P Olson (2022), "Reducing model misspecification and bias in the estimation of interactions," *Political Analysis*, 30 (4), 495–514.

Calder, Bobby J, Lynn W Phillips, and Alice M Tybout (1981), "Designing research for application," *Journal of consumer research*, 8 (2), 197–207.

Camerer, Colin (1997), *Rules for experimenting in psychology and economics, and why they differ*, Springer.

Caruana, Rich (1997), "Multitask learning," *Machine learning*, 28, 41–75.

Carvalho, Diogo V, Eduardo M Pereira, and Jaime S Cardoso (2019), "Machine learning interpretability: A survey on methods and metrics," *Electronics*, 8 (8), 832.

Chang, Hannah H, Anirban Mukherjee, and Amitava Chattopadhyay (2023), "More voices persuade: The attentional benefits of voice numerosity," *Journal of Marketing Research*, 60 (4), 687–706.

Chapman, Gretchen B and Eric J Johnson (1999), "Anchoring, activation, and the construction of values," *Organizational behavior and human decision processes*, 79 (2), 115–53.

Chernev, Alexander (2011), "Semantic anchoring in sequential evaluations of vices and virtues," *Journal of Consumer Research*, 37 (5), 761–74.

Clark, Herbert H (1973), "The language-as-fixed-effect fallacy: A critique of language statistics in psychological research," *Journal of verbal learning and verbal behavior*, 12 (4), 335–59.

Cook, Thomas D and Donald T Campbell (1979), *Quasi-experimentation: Design & analysis issues for field settings*, Chicago, IL: Rand McNally.

Donkers, Bas, Benedict GC Dellaert, Rory M Waisman, and Gerald Häubl (2020), "Preference dynamics in sequential consumer choice with defaults," *Journal of Marketing Research*, 57 (6), 1096–1112.

Ebbesen, Ebbe B and Vladimir J Konecni (1980), "On the external validity of decision-making research: What do we know about decisions in the real world," *Cognitive processes in choice and decision behavior*, 21–45.

Einhorn, Hillel J and Robin M Hogarth (1981), "Behavioral decision theory: Processes of judgement and choice," *Annual review of psychology*, 32, 53–88.

Fast, Ethan, Binbin Chen, and Michael S Bernstein (2016), "Empath: Understanding topic signals in large-scale text," in *Proceedings of the 2016 CHI conference on human factors in computing systems*, ACM, 4647–57.

Frederick, Shane, Leonard Lee, and Ernest Baskin (2014), "The limits of attraction," *Journal of Marketing Research*, 51 (4), 487–507.

Gandhi, Natasha, Wanling Zou, Caroline Meyer, Sudeep Bhatia, and Lukasz Walasek (2022), "Computational methods for predicting and understanding food judgment," *Psychological Science*, 33 (4), 579–94.

Gelman, Andrew and Eric Loken (2013), "The garden of forking paths: Why multiple comparisons can be a problem, even when there is no 'fishing expedition' or 'p-hacking' and the research hypothesis was posited ahead of time," *Department of Statistics, Columbia University*, 348, 1–17.

Gigerenzer, Gerd (1991), "How to make cognitive illusions disappear: Beyond 'heuristics and biases'," *European review of social psychology*, 2 (1), 83–115.

Goldstein, Ariel, Haocheng Wang, Leonard Niekerken, Mariano Schain, Zaid Zada, Bobbi Aubrey, Tom Sheffer, Samuel A Nastase, Harshvardhan Gazula, Aditi Singh, and others (2025), "A unified acoustic-to-speech-to-language embedding space captures the neural basis of natural language processing in everyday conversations," *Nature Human Behaviour*, 1–15.

Hoeffler, Steve and Dan Ariely (1999), "Constructing stable preferences: A look into dimensions of experience and their impact on preference stability," *Journal of consumer psychology*, 8 (2), 113–39.

Hornik, Kurt, Maxwell Stinchcombe, and Halbert White (1989), "Multilayer feedforward networks are universal approximators," *Neural networks*, 2 (5), 359–66.

Johnson, William B. and Joram Lindenstrauss (1984), "Extensions of Lipschitz mappings into a Hilbert space," in *Conference in Modern Analysis and Probability (Contemporary Mathematics, vol. 26)*, 189–206.

Judd, Charles M, Jacob Westfall, and David A Kenny (2012), "Treating stimuli as a random factor in social psychology: A new and comprehensive solution to a pervasive but largely ignored problem." *Journal of personality and social psychology*, 103 (1), 54–69.

Kirk, Roger E (2013), *Experimental design: Procedures for the behavioral sciences*, Thousand Oaks, CA: Sage Publications.

Kusupati, Aditya, Gantavya Bhatt, Aniket Rege, Matthew Wallingford, Aditya Sinha, Vivek Ramanujan, William Howard-Snyder, Kaifeng Chen, Sham Kakade, Prateek Jain, and Ali Farhadi (2022), "Matryoshka representation learning," in *Advances in neural information processing systems*, 30233–49.

Latour, Kathryn A and Michael S Latour (2010), "Bridging aficionados' perceptual and conceptual knowledge to enhance how they learn from experience," *Journal of Consumer Research*, 37 (4), 688–97.

Levesque, Hector J (1986), "Knowledge representation and reasoning," *Annual review of computer science*, 1 (1), 255–87.

MacNeil, Karen (2015), *The wine bible*, Workman Publishing.

Martin, Leonard L, John J Seta, and Rick A Crelia (1990), "Assimilation and contrast as a function of people's willingness and ability to expend effort in forming an impression." *Journal of personality and social psychology*, 59 (1), 27–37.

Morales, Andrea C, On Amir, and Leonard Lee (2017), "Keeping it real in experimental research—understanding when, where, and how to enhance realism and measure consumer behavior," *Journal of Consumer Research*, 44 (2), 465–76.

Mussweiler, Thomas (2003), “Comparison processes in social judgment: Mechanisms and consequences.” *Psychological review*, 110 (3), 472–89.

Payne, John W, James R Bettman, David A Schkade, Norbert Schwarz, and Robin Gregory (2000), “Measuring constructed preferences: Towards a building code,” *Elicitation of preferences*, 243–75.

Pham, Michel Tuan (2013), “The seven sins of consumer psychology,” *Journal of Consumer Psychology*, 23 (4), 411–23.

Rabin, Matthew (1998), “Psychology and economics,” *Journal of economic literature*, 36 (1), 11–46.

Rocklage, Matthew D, Derek D Rucker, and Loran F Nordgren (2021), “Emotionally numb: Expertise dulls consumer experience,” *Journal of Consumer Research*, 48 (3), 355–73.

Rosenthal, Robert (1979), “The file drawer problem and tolerance for null results.” *Psychological bulletin*, 86 (3), 638–41.

Shadish, William R, Thomas D Cook, and Donald T Campbell (2002), *Experimental and quasi-experimental designs for generalized causal inference*, Boston, MA: Houghton Mifflin.

Sharot, Tali, Cristina M Velasquez, and Raymond J Dolan (2010), “Do decisions shape preference? Evidence from blind choice,” *Psychological science*, 21 (9), 1231–35.

Simonsohn, Uri, Andres Montealegre, and Ioannis Evangelidis (2025), “Stimulus sampling reimagined: Designing experiments with mix-and-match, analyzing results with stimulus plots,” *Journal of Personality and Social Psychology*, 129 (1), 71–90.

Slovic, Paul (1995), “The construction of preference.” *American Psychologist*, 50 (5), 364–71.

Spicer, Jake, Jian-Qiao Zhu, Nick Chater, and Adam N Sanborn (2022), “Perceptual and cognitive judgments show both anchoring and repulsion,” *Psychological Science*, 33 (9), 1395–1407.

Stone, Dan N and David A Schkade (1991), “Numeric and linguistic information representation in multiattribute choice,” *Organizational Behavior and Human Decision Processes*, 49 (1), 42–59.

Strack, Fritz and Thomas Mussweiler (1997), “Explaining the enigmatic anchoring effect: Mechanisms of selective accessibility.” *Journal of personality and social psychology*, 73 (3), 437–46.

Tenenbaum, Joshua B, Charles Kemp, Thomas L Griffiths, and Noah D Goodman (2011), “How to grow a mind: Statistics, structure, and abstraction,” *science*, 331 (6022), 1279–85.

Wells, Gary L and Paul D Windschitl (1999), “Stimulus sampling and social psychological experimentation,” *Personality and Social Psychology Bulletin*, 25 (9), 1115–25.

Westfall, Jacob, Charles M Judd, and David A Kenny (2015), “Replicating studies in which samples of participants respond to samples of stimuli,” *Perspectives on Psychological Science*, 10 (3), 390–99.

Wickens, Thomas D and Geoffrey Keppel (1983), “On the choice of design and of test statistic in the analysis of experiments with sampled materials,” *Journal of Verbal Learning and Verbal Behavior*, 22 (3), 296–309.

Wilson, Timothy D, Elliot Aronson, and Kevin Carlsmith (2010), “The art of laboratory experimentation,” *Handbook of social psychology*, 1, 51–81.